Maximum Principle based deep Learning for Optimal Control of a Two-Link Revolute Manipulator using Neural ODEs

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1. Modellierung eines Roboters

Wir halten uns an [8] und [7]

$$\underbrace{f - mg}_{\text{Kräftebilanz}} = \underbrace{m\ddot{x}}_{\text{Newtons 2nd Law}} = \frac{\mathrm{d}}{\mathrm{d}t}\left[m\dot{x}\right]$$

$$f - \frac{\partial}{\partial x}\underbrace{\left[mgx(t)\right]}_{\text{Potenzielle Energie } V} = \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{x}}\underbrace{\left[m\frac{1}{2}\dot{x}(t)^2\right]}_{\text{kinetische Energie } T}$$

$$f - \frac{\partial V}{\partial x} = \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial T}{\partial \dot{x}} \Rightarrow f = \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{x}} - \frac{\partial L}{\partial x}$$

Wobei $L = T - V = \frac{1}{2}m\dot{x}(t)^2 - mgx(t)$ gilt. Für generalisierte Koordinaten $\mathbf{q} = \theta \in \mathbb{R}^n$

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{\mathbf{q}}_i}\right) - \frac{\partial T}{\partial \mathbf{q}_i} + \frac{\partial V}{\partial \mathbf{q}_i} = \tau_i \quad i = 1, \cdots, n$$

Für die kinetische Energie eines starren Körpers gilt:

$$T = \frac{1}{2}m\mathbf{v}^T\mathbf{v} + \frac{1}{2}\boldsymbol{\omega}^T I\boldsymbol{\omega}$$

Wobei $\mathbf{v}$ die Geschwindigkeit des Schwerpunkts, $\boldsymbol{\omega}$ die Winkelgeschwindigkeit und I der Trägheitstensor bezüglich des Schwerpunkts ist. Für den Trägheitstensor gilt:

$$I = \int_V \rho(\mathbf{r})[(\mathbf{r}\cdot\mathbf{r})\mathbf{1} - \mathbf{r}\otimes\mathbf{r}]\mathrm{d}V \tag{1}$$

Für ein Mehrkörpersystem mit generalisierten Koordinaten $\mathbf{q}$ lässt sich die Gesamtenergie in kompakter Form schreiben als:

$$T = \frac{1}{2}\dot{\mathbf{q}}^T M(\mathbf{q})\dot{\mathbf{q}}$$

wobei $M(\mathbf{q})$ die konfigurationsabhängige Massenmatrix ist, die alle Massen und Trägheiten der einzelnen Glieder erfasst. Für die Geschwindigkeiten gilt mittels der Kettenregel:

$$\mathbf{v} = J_v(\mathbf{q})\dot{\mathbf{q}} \quad \text{und} \quad \boldsymbol{\omega} = J_\omega(\mathbf{q})\dot{\mathbf{q}} \quad J_v = \frac{\partial \mathbf{r}}{\partial \mathbf{q}} \quad \text{und} \quad J_\omega = \frac{\partial \boldsymbol{\omega}}{\partial \dot{\mathbf{q}}}$$

Einsetzen in die kinetische Energie liefert:

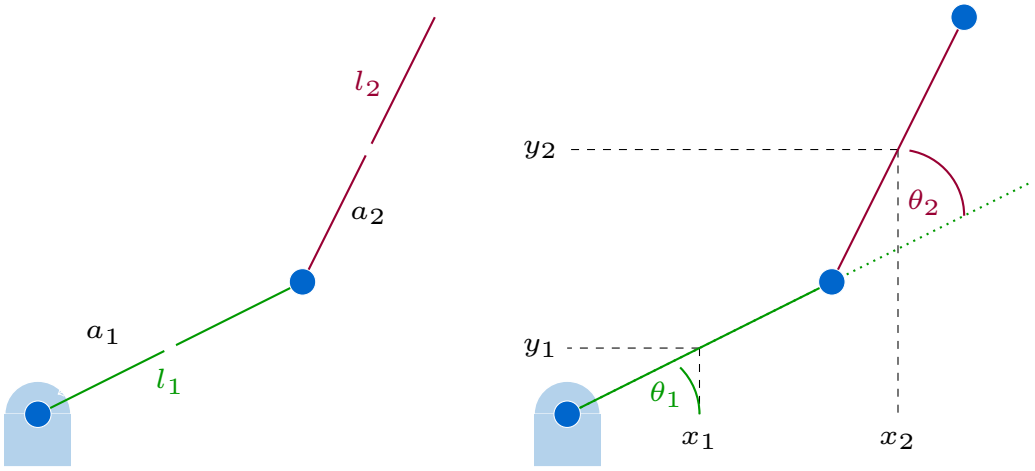
$$\begin{aligned} T &= \frac{1}{2}m(J_v\dot{\mathbf{q}})^T(J_v\dot{\mathbf{q}}) + \frac{1}{2}(J_\omega\dot{\mathbf{q}})^T I(J_\omega\dot{\mathbf{q}}) \\ &= \frac{1}{2}\dot{\mathbf{q}}^T(mJ_v^T J_v)\dot{\mathbf{q}} + \frac{1}{2}\dot{\mathbf{q}}^T(J_\omega^T I J_\omega)\dot{\mathbf{q}} \\ &= \frac{1}{2}\dot{\mathbf{q}}^T \underbrace{\left(mJ_v^T J_v + J_\omega^T I J_\omega\right)}_{M(\mathbf{q})}\dot{\mathbf{q}} \end{aligned}$$

Die Gesamtenergie setzt sich zusammen aus der Summe der Energien aller n Glieder:

$$M(\mathbf{q}) = \sum_{i=1}^{n} m_i J_{v_i}^T J_{v_i} + J_{\omega_i}^T I_i J_{\omega_i}$$

1.1. Two Link Revolute Manipulator

Wir bezeichnen $\mathbf{q} = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}$ als die generalisierten Koordinaten des Systems. Die Schwerpunkte der beiden Links bezeichnen wir mit (x_1, y_1) und (x_2, y_2). Die Abstände der Schwerpunkte von den Gelenken werden mit a_1 und a_2 bezeichnet.



Für die Koordinaten der Schwerpunkte gilt:

$$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \begin{pmatrix} a_1 \cdot \cos(\theta_1) \\ a_1 \cdot \sin(\theta_1) \\ 0 \end{pmatrix} \quad \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix} = \begin{pmatrix} l_1 \cdot \cos(\theta_1) + a_2 \cos(\theta_1 + \theta_2) \\ l_1 \cdot \sin(\theta_1) + a_2 \sin(\theta_1 + \theta_2) \\ 0 \end{pmatrix}$$

1.1.1. Formulierung der Massenmatrix

Für die Massenmatrix gilt hier:

$$M(\mathbf{q}) = m_1 J_{v_1}^T J_{v_1} + J_{\omega_1}^T I_1 J_{\omega_1} + m_2 J_{v_2}^T J_{v_2} + J_{\omega_2}^T I_2 J_{\omega_2}$$

Hierbei bezeichnet J die Jacobi-Matrix, I_i das Trägheitstensor und m_i die Masse des i-ten Links.

$$J_{v_1} = \frac{\partial \mathbf{r}_1}{\partial \mathbf{q}} = \frac{\partial (x_1, y_1, z_1)}{\partial (\theta_1, \theta_2)} = \begin{pmatrix} \frac{\partial x_1}{\partial \theta_1} & \frac{\partial x_1}{\partial \theta_2} \\ \frac{\partial y_1}{\partial \theta_1} & \frac{\partial y_1}{\partial \theta_2} \\ \frac{\partial z_1}{\partial \theta_1} & \frac{\partial z_1}{\partial \theta_2} \end{pmatrix} = \begin{pmatrix} -a_1 \sin(\theta_1) & 0 \\ a_1 \cos(\theta_1) & 0 \\ 0 & 0 \end{pmatrix}$$

$$J_{v_2} = \frac{\partial \mathbf{r}_2}{\partial \mathbf{q}} = \frac{\partial (x_2, y_2, z_2)}{\partial (\theta_1, \theta_2)} = \begin{pmatrix} \frac{\partial x_2}{\partial \theta_1} & \frac{\partial x_2}{\partial \theta_2} \\ \frac{\partial y_2}{\partial \theta_1} & \frac{\partial y_2}{\partial \theta_2} \\ \frac{\partial z_2}{\partial \theta_1} & \frac{\partial z_2}{\partial \theta_2} \end{pmatrix} = \begin{pmatrix} -l_1 \sin(\theta_1) - a_2 \sin(\theta_1 + \theta_2) & -a_2 \sin(\theta_1 + \theta_2) \\ l_1 \cos(\theta_1) + a_2 \cos(\theta_1 + \theta_2) & a_2 \cos(\theta_1 + \theta_2) \\ 0 & 0 \end{pmatrix}$$

Für das Matrizenprodukt $J_{v_1}^T J_{v_1}$ gilt:

$$\begin{aligned} J_{v_1}^T J_{v_1} &= \begin{pmatrix} -a_1 \sin(\theta_1) & a_1 \cos(\theta_1) & 0 \\ 0 & 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} -a_1 \sin(\theta_1) & 0 \\ a_1 \cos(\theta_1) & 0 \\ 0 & 0 \end{pmatrix} \\ &= \begin{pmatrix} a_1^2 \sin^2(\theta_1) + a_1^2 \cos^2(\theta_1) & 0 \\ 0 & 0 \end{pmatrix} \\ &= \begin{pmatrix} a_1^2 & 0 \\ 0 & 0 \end{pmatrix} \end{aligned}$$

Für das Matrizenprodukt $J_{v_2}^T J_{v_2}$ gilt:

$$\begin{aligned} J_{v_2}^T J_{v_2} &= \begin{pmatrix} -l_1\sin(\theta_1) - a_2\sin(\theta_1+\theta_2) & l_1\cos(\theta_1) + a_2\cos(\theta_1+\theta_2) & 0 \\ -a_2\sin(\theta_1+\theta_2) & a_2\cos(\theta_1+\theta_2) & 0 \end{pmatrix} \\ &\quad \cdot \begin{pmatrix} -l_1\sin(\theta_1) - a_2\sin(\theta_1+\theta_2) & -a_2\sin(\theta_1+\theta_2) \\ l_1\cos(\theta_1) + a_2\cos(\theta_1+\theta_2) & a_2\cos(\theta_1+\theta_2) \\ 0 & 0 \end{pmatrix} \\ &= \begin{pmatrix} (l_1^2 + a_2^2 + 2l_1a_2\cos(\theta_2)) & (a_2^2 + l_1a_2\cos(\theta_2)) \\ (a_2^2 + l_1a_2\cos(\theta_2)) & a_2^2 \end{pmatrix} \end{aligned}$$

Für die Winkelgeschwindigkeiten setzen wir $\boldsymbol{\omega}_1 = \begin{pmatrix} 0 & 0 & \theta_1' \end{pmatrix}^T$ und $\boldsymbol{\omega}_2 = \begin{pmatrix} 0 & 0 & \theta_1' + \theta_2' \end{pmatrix}^T$.

$$J_{\omega_1} = \frac{\partial \boldsymbol{\omega}_1}{\partial \dot{\mathbf{q}}} = \frac{\partial(\omega_1)}{\partial(\theta_1', \theta_2')} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ \frac{\partial\omega_1}{\partial\theta_1'} & \frac{\partial\omega_1}{\partial\theta_2'} \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \end{pmatrix}$$

$$J_{\omega_2} = \frac{\partial \boldsymbol{\omega}_2}{\partial \dot{\mathbf{q}}} = \frac{\partial(\omega_2)}{\partial(\theta_1', \theta_2')} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ \frac{\partial\omega_2}{\partial\theta_1'} & \frac{\partial\omega_2}{\partial\theta_2'} \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 1 \end{pmatrix}$$

Für den Trägheitstensor I_i gilt:

$$I = \begin{pmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{pmatrix} = \begin{pmatrix} \int (y^2+z^2)\mathrm{d}m & -\int xy\mathrm{d}m & -\int xz\mathrm{d}m \\ -\int yx\mathrm{d}m & \int (x^2+z^2)\mathrm{d}m & -\int yz\mathrm{d}m \\ -\int zx\mathrm{d}m & -\int zy\mathrm{d}m & \int (x^2+y^2)\mathrm{d}m \end{pmatrix}$$

Für symmetrische Körper (z.B. Zylinder, Kugel, Quader) gilt: $I_{xy} = I_{xz} = I_{yz} = 0$ also erhalten wir eine Diagonalmatrix.

$$I_i = \begin{pmatrix} \int_y^2 \mathrm{d}m & 0 & 0 \\ 0 & \int x^2\mathrm{d}m & 0 \\ 0 & 0 & \int (x^2+y^2)\mathrm{d}m \end{pmatrix} = \begin{pmatrix} \int_V \rho y^2 \mathrm{d}V & 0 & 0 \\ 0 & \int_V \rho x^2 \mathrm{d}V & 0 \\ 0 & 0 & \int_V \rho(x^2+y^2)\mathrm{d}V \end{pmatrix}$$

Für die Massenmatrix $M(\mathbf{q})$ gilt somit:

$$\begin{aligned} M(\mathbf{q}) &= m_1 J_{v_1}^T J_{v_1} + J_{\omega_1}^T I_1 J_{\omega_1} + m_2 J_{v_2}^T J_{v_2} + J_{\omega_2}^T I_2 J_{\omega_2} \\ &= m_1 \begin{pmatrix} a_1^2 & 0 \\ 0 & 0 \end{pmatrix} + \underbrace{\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} I_{1,xx} & 0 & 0 \\ 0 & I_{1,yy} & 0 \\ 0 & 0 & I_{1,zz} \end{pmatrix}}_{= \begin{pmatrix} 0 & 0 & I_{1,zz} \\ 0 & 0 & 0 \end{pmatrix}} \cdot \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \end{pmatrix} \\ &\quad + m_2 \begin{pmatrix} (l_1^2 + a_2^2 + 2l_1a_2\cos(\theta_2)) & (a_2^2 + l_1a_2\cos(\theta_2)) \\ (a_2^2 + l_1a_2\cos(\theta_2)) & a_2^2 \end{pmatrix} \\ &\quad + \underbrace{\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} I_{2,xx} & 0 & 0 \\ 0 & I_{2,yy} & 0 \\ 0 & 0 & I_{2,zz} \end{pmatrix}}_{= \begin{pmatrix} 0 & 0 & I_{2,zz} \\ 0 & 0 & I_{2,zz} \end{pmatrix}} \cdot \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 1 \end{pmatrix} \\ &= \begin{pmatrix} m_1 a_1^2 + I_{1,zz} + m_2 \cdot (l_1^2 + a_2^2 + 2l_1a_2\cos(\theta_2)) + I_{2,zz} & m_2(a_2^2 + l_1a_2\cos(\theta_2)) + I_{2,zz} \\ m_2(a_2^2 + l_1a_2\cos(\theta_2)) + I_{2,zz} & m_2 a_2^2 + I_{2,zz} \end{pmatrix} \end{aligned}$$

Die Trägheitselemente $I_{i,xx}$ und $I_{i,yy}$ sind nicht relevant. Für unser Model verwenden wir den Standardansatz eines dünnen Stabes der Länge l_i und Masse m_i. Somit gilt $z = y \approx 0$ und wir erhalten: $I_{i,xx} = I_{i,yy} = 0$

$$I_{i,zz} \approx \int_{-\frac{l_i}{2}}^{\frac{l_i}{2}} \underbrace{\rho}_{\frac{m_i}{l_i}} x^2 \mathrm{d}x = \frac{1}{3} \cdot \frac{m_i}{l_i} \cdot \left[\left(\frac{l_i}{2}\right)^3 - \left(-\frac{l_i}{2}\right)^3 \right] = \frac{1}{12} m_i l_i^2 \Rightarrow I_i = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{1}{12} m_i l_i^2 \end{pmatrix}$$

1.1.2. Aufstellen der Coriolis-Matrix

Für die partiellen Ableitungen der kinetischen Energie $T(\mathbf{q}, \dot{\mathbf{q}}) = \frac{1}{2} \dot{\mathbf{q}}^T M(\mathbf{q}) \dot{\mathbf{q}}$ gilt:

$$\begin{aligned}
\frac{\partial T}{\partial \theta_1} &= \frac{1}{2} \frac{\partial M_{11}}{\partial \theta_1} {\theta'_1}^2 + \frac{\partial M_{12}}{\partial \theta_1} \theta'_1 \theta'_2 + \frac{1}{2} \frac{\partial M_{22}}{\partial \theta_1} \theta'_2(t)^2 \\
\frac{\partial T}{\partial \theta_2} &= \frac{1}{2} \frac{\partial M_{11}}{\partial \theta_2} {\theta'_1}^2 + \frac{\partial M_{12}}{\partial \theta_2} \theta'_1 \theta'_2 + \frac{1}{2} \frac{\partial M_{22}}{\partial \theta_2} \theta'_2(t)^2 \\
\frac{\partial T}{\partial \theta'_1} &= M_{11} \theta'_1 + M_{12} \theta'_2 \\
\frac{\partial T}{\partial \theta'_2} &= M_{12} \theta'_1 + M_{22} \theta'_2
\end{aligned}$$

Für die zeitlichen Ableitungen der Größen gilt

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial T}{\partial \theta'_1} &= \underbrace{\frac{\mathrm{d}}{\mathrm{d}t} M_{11}}_{\frac{\partial M_{11}}{\partial \theta_1} \theta'_1 + \frac{\partial M_{11}}{\partial \theta_2} \theta'_2} \cdot \theta'_1 + M_{11} \theta''_1 + \underbrace{\frac{\mathrm{d}}{\mathrm{d}t} M_{12}}_{\frac{\partial M_{12}}{\partial \theta_1} \theta'_1 + \frac{\partial M_{12}}{\partial \theta_2} \theta'_2} \cdot \theta'_2 + M_{12} \theta''_2 \\
&= \frac{\partial M_{11}}{\partial \theta_1} {\theta'_1}^2 + \frac{\partial M_{11}}{\partial \theta_2} \theta'_1 \theta'_2 + M_{11} \theta''_1 + \frac{\partial M_{12}}{\partial \theta_1} \theta'_1 \theta'_2 + \frac{\partial M_{12}}{\partial \theta_2} {\theta'_2}^2 + M_{12} \theta''_2 \\
&= \frac{\partial M_{11}}{\partial \theta_1} {\theta'_1}^2 + \left[\frac{\partial M_{11}}{\partial \theta_2} + \frac{\partial M_{12}}{\partial \theta_1} \right] \theta'_1 \theta'_2 + \frac{\partial M_{12}}{\partial \theta_2} {\theta'_2}^2 + M_{11} \theta''_1 + M_{12} \theta''_2
\end{aligned}$$

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial T}{\partial \theta'_2} &= \underbrace{\frac{\mathrm{d}}{\mathrm{d}t} M_{12}}_{\frac{\partial M_{12}}{\partial \theta_1} \theta'_1 + \frac{\partial M_{12}}{\partial \theta_2} \theta'_2} \cdot \theta'_1 + M_{12} \theta''_1 + \underbrace{\frac{\mathrm{d}}{\mathrm{d}t} M_{22}}_{\frac{\partial M_{22}}{\partial \theta_1} \theta'_1 + \frac{\partial M_{22}}{\partial \theta_2} \theta'_2} \cdot \theta'_2 + M_{22} \theta''_2 \\
&= \frac{\partial M_{12}}{\partial \theta_1} {\theta'_1}^2 + \frac{\partial M_{12}}{\partial \theta_2} \theta'_1 \theta'_2 + M_{12} \theta''_1 + \frac{\partial M_{22}}{\partial \theta_1} \theta'_1 \theta'_2 + \frac{\partial M_{22}}{\partial \theta_2} {\theta'_2}^2 + M_{22} \theta''_2
\end{aligned}$$

Es gilt

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial T}{\partial \theta'_1} - \frac{\partial T}{\partial \theta_1} &= \left[\frac{\partial M_{11}}{\partial \theta_1} - \frac{1}{2} \frac{\partial M_{11}}{\partial \theta_1} \right] {\theta'_1}^2 + \left[\frac{\partial M_{11}}{\partial \theta_2} + \frac{\partial M_{12}}{\partial \theta_1} - \frac{\partial M_{12}}{\partial \theta_1} \right] \theta'_1 \theta'_2 + \left[\frac{\partial M_{12}}{\partial \theta_2} - \frac{1}{2} \frac{\partial M_{22}}{\partial \theta_1} \right] {\theta'_2}^2 \\
&\quad + M_{11} \theta''_1 + M_{12} \theta''_2
\end{aligned}$$

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial T}{\partial \theta'_2} - \frac{\partial T}{\partial \theta_2} &= \left[\frac{\partial M_{12}}{\partial \theta_1} - \frac{1}{2} \frac{\partial M_{11}}{\partial \theta_2} \right] {\theta'_1}^2 + \left[\frac{\partial M_{12}}{\partial \theta_2} + \frac{\partial M_{22}}{\partial \theta_1} - \frac{\partial M_{12}}{\partial \theta_2} \right] \theta'_1 \theta'_2 + \left[\frac{\partial M_{22}}{\partial \theta_2} - \frac{1}{2} \frac{\partial M_{22}}{\partial \theta_2} \right] {\theta'_2}^2 \\
&\quad + M_{12} \theta''_1 + D_{22} \theta''_2 \\
&= \underbrace{\left[\frac{\partial M_{12}}{\partial \theta_1} \theta'_1 - \frac{1}{2} \frac{\partial M_{11}}{\partial \theta_2} \theta'_1 \right]}_{C_{21}} \theta'_1 + \underbrace{\left[\frac{1}{2} \frac{\partial M_{22}}{\partial \theta_2} \theta'_2 + \frac{\partial M_{22}}{\partial \theta_1} \theta'_1 \right]}_{C_{22}} \theta'_2 + M_{12} \theta''_1 + M_{22} \theta''_2
\end{aligned}$$

Für die Einträge der Matrix C gilt

$$
\begin{aligned}
C_{11} &= \frac{1}{2}\underbrace{\frac{\partial M_{11}}{\partial \theta_1}}_{=0}\theta_1' + \underbrace{\frac{\partial M_{11}}{\partial \theta_2}}_{=-2m_2 l_1 a_2 \sin(\theta_2)}\theta_2' = -2m_2 l_1 a_2 \sin(\theta_2)\theta_2' \\
C_{12} &= \underbrace{\frac{\partial M_{12}}{\partial \theta_2}}_{=-m_2 l_1 a_2 \sin(\theta_2)}\theta_2' - \frac{1}{2}\underbrace{\frac{\partial M_{22}}{\partial \theta_1}}_{=0}\theta_2' = -m_2 l_1 a_2 \sin(\theta_2)\theta_2' \\
C_{21} &= \underbrace{\frac{\partial M_{12}}{\partial \theta_1}}_{=0}\theta_1' - \frac{1}{2}\underbrace{\frac{\partial M_{11}}{\partial \theta_2}}_{=-2m_2 l_1 a_2 \sin(\theta_2)}\theta_1' = m_2 l_1 a_2 \sin(\theta_2)\theta_1' \\
C_{22} &= \frac{1}{2}\underbrace{\frac{\partial M_{22}}{\partial \theta_2}}_{=0}\theta_2' + \underbrace{\frac{\partial M_{22}}{\partial \theta_1}}_{=0}\theta_1' = 0
\end{aligned}
$$

Es gilt somit

$$
C(\mathbf{q}, \dot{\mathbf{q}}) \cdot \dot{\mathbf{q}} = \begin{pmatrix} -2m_2 l_1 a_2 \sin(\theta_2)\theta_2' & -m_2 l_1 a_2 \sin(\theta_2)\theta_2' \\ m_2 l_1 a_2 \sin(\theta_2)\theta_1' & 0 \end{pmatrix} \begin{pmatrix} \theta_1' \\ \theta_2' \end{pmatrix}
$$

1.1.3. Gravitationsterme

Für die potenzielle Energie V gilt:

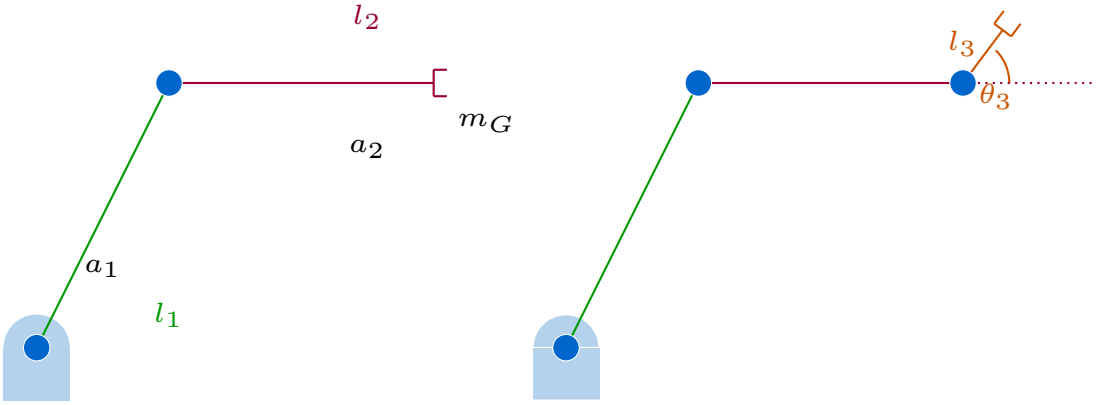
$$
V = m_1 \cdot g \cdot y_1 + m_2 \cdot g \cdot y_2 = m_1 g a_1 \sin(\theta_1) + m_2 g \left(l_1 \sin(\theta_1) + a_2 \sin(\theta_1 + \theta_2)\right)
$$

Für die partiellen Ableitungen erhalten wir

$$
G(\mathbf{q}) = \frac{\partial V}{\partial \mathbf{q}} = \begin{pmatrix} \frac{\partial V}{\partial \theta_1} \\ \frac{\partial V}{\partial \theta_2} \end{pmatrix} = \begin{pmatrix} m_1 g a_1 \cos(\theta_1) + m_2 g \left(l_1 \cos(\theta_1) + a_2 \cos(\theta_1 + \theta_2)\right) \\ m_2 g \left(l_1 \cos(\theta_1) + a_2 \cos(\theta_1 + \theta_2)\right) \end{pmatrix}
$$

1.2. Einfaches Greifobjekt

Wir modellieren ein Variables Greifobjekt, welches am Ende von Link 2 gehalten wird. Wir betrachten nachfolgend das Greifobjekt als Punktmasse mit Masse m_G.



Für die Masse von Link 2 inklusive Greifobjekt gilt:

$$m_i' = m_i + m_G$$

Für den Schwerpunkt von Link 2 inklusive Greifobjekt gilt:

$$\mathbf{a}_2' = \frac{m_i \cdot \mathbf{a}_2 + m_G \cdot \mathbf{d}}{m_i + m_G} \quad \mathbf{d}(\theta_3) = \begin{pmatrix} l_2 + r \cdot \cos(\theta_3) \\ r \cdot \sin(\theta_3) \\ 0 \end{pmatrix}$$

Für den Trägheitstensor von Link 2 inklusive Greifobjekt gilt:

$$I_i' = I_i + m_i \cdot S(\mathbf{a}_i - \mathbf{a}_i') + m_G \cdot S(\mathbf{d} - \mathbf{a}_i') \quad S(\mathbf{v}) = \begin{pmatrix} v_y^2 + v_z^2 & -v_x v_y & -v_x v_z \\ -v_x v_y & v_x^2 + v_z^2 & -v_y v_z \\ -v_x v_z & -v_y v_z & v_x^2 + v_y^2 \end{pmatrix}$$

Wobei $S(\mathbf{v})$ die Steiner-Matrix ist. Wir erhalten die Matrix direkt aus der Definition des Trägheitstensors (1) einer Punktmasse mit $\rho(\mathbf{r}) = m \cdot \delta(\mathrm{r} - \mathbf{d})$. Für die Verschiebungsvektoren gilt:

$$\mathbf{d} = \begin{pmatrix} l_2 \\ 0 \\ 0 \end{pmatrix} \mathbf{a}_2 = \begin{pmatrix} \frac{l_2}{2} \\ 0 \\ 0 \end{pmatrix} \Rightarrow \mathbf{a}_2' = \frac{l_2}{m_2 + m_G} \begin{pmatrix} \frac{m_2}{2} + m_G \\ 0 \\ 0 \end{pmatrix}$$

Für die Steiner-Matrizen gilt entsprechend

$$S(\mathbf{a}_2 - \mathbf{a}_2') = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \left[\frac{l_2}{2} - a_2'\right]^2 & 0 \\ 0 & 0 & \left[\frac{l_2}{2} - a_2'\right]^2 \end{pmatrix} \quad S(\mathbf{d} - \mathbf{a}_2') = \begin{pmatrix} 0 & 0 & 0 \\ 0 & [l_2^2 - a_2']^2 & 0 \\ 0 & 0 & [l_2^2 - a_2']^2 \end{pmatrix}$$

Somit erhalten wir für den Trägheitstensor von Link 2 inklusive Greifobjekt:

$$I_2' = \begin{pmatrix} I_{2,xx} & 0 & 0 \\ 0 & I_{2,yy} + m_2 \cdot \left[\frac{l_2^2}{2} - a_2'\right]^2 + m_G \cdot \left[\frac{l_2^2}{2} - a_2'\right]^2 & 0 \\ 0 & 0 & I_{2,zz} + m_2 \cdot \left[\frac{l_2^2}{2} - a_2'\right]^2 + m_G \cdot \left[\frac{l_2^2}{2} - a_2'\right]^2 \end{pmatrix}$$

Für die Koordinaten des Schwerpunktes von Link 3 gilt:

$$\mathbf{r}_3 = \begin{pmatrix} l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2) + l_3 \cos(\theta_1 + \theta_2 + \theta_3) \\ l_1 \sin(\theta_1) + l_2 \sin(\theta_1 + \theta_2) + l_3 \sin(\theta_1 + \theta_2 + \theta_3) \\ 0 \end{pmatrix}$$

Berechnung der Jacobi-Matrizen für Link 3 bezüglich der Geschwindigkeiten:

$$J_{v_3} = \frac{\partial \mathbf{r}_3}{\partial \mathbf{q}}$$

Für die Winkelgeschwindigkeit des Links 3 setzen wir $\boldsymbol{\omega_3} = \begin{pmatrix} 0 & 0 & \theta_1' + \theta_2' + \theta_3' \end{pmatrix}$ und erhalten für die Jacobi-Matrix bezüglich der Winkelgeschwindigkeit:

$$J_{\omega_3} = \frac{\partial \boldsymbol{\omega_3}}{\partial \dot{\mathbf{q}}} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

2. Theorie optimaler Steuerungungsprobleme

Wir halten uns an die Definitionen in [3]

Definition 2.1. Seien $t_0 < t_f$ feste Zeiten und

$$\begin{aligned}\Phi &: \mathbb{R}^{n_{\boldsymbol{x}}} \to \mathbb{R},\\ L &: [t_0, t_f] \times \mathbb{R}^{n_{\boldsymbol{x}}} \to \mathbb{R},\\ f &: [t_0, t_f] \times \mathbb{R}^{n_{\boldsymbol{x}}} \times \mathbb{R}^{n_{\boldsymbol{u}}} \to \mathbb{R}^{n_{\boldsymbol{x}}}\end{aligned}$$

hinreichend glatte Funktionen und $\mathcal{U} \subset \mathbb{R}^{n_{\boldsymbol{u}}}$ eine abgeschlossene konvexe nichtleere Menge.

$$\min_{\boldsymbol{u}} \int_{t_0}^{t_f} L(t, \boldsymbol{x}(t), \boldsymbol{u}(t))\,\mathrm{d}t + \Phi(\boldsymbol{x}(t_f))$$

mit $\boldsymbol{x} \in W_{1,\infty}^{n_{\boldsymbol{x}}}([t_0, t_f]), \boldsymbol{u} \in L_{\infty}^{n_{\boldsymbol{u}}}([t_0, t_f])$ und

$$\dot{\boldsymbol{x}}(t) = f(t, \boldsymbol{x}(t), \boldsymbol{u}(t)), \quad \boldsymbol{x}(t_0) = \boldsymbol{x}_0, \quad \boldsymbol{u}(t) \in \mathcal{U} \text{ f.ü. } t \in [t_0, t_f]$$

Wir betrachten ein Steuerungsproblem und definieren die Value-Function V mittels

$$V(x(t), t) := \min_{u} \left\{ \int_{t}^{t_f} L(s, x(s), u(s))\,\mathrm{d}s + \Phi(x(t_f)) \right\}$$

Satz 2.2. Die Value-Function erfüllt

$$0 = \frac{\partial V(x,t)}{\partial t} + \min_{u} \left[L(t,x,u) + \frac{\partial V(x,t)}{\partial x} \cdot f(t,x,u) \right], \quad x \in \mathbb{R}^n,\ t_0 \le t \le t_f$$

mit den Randbedingungen

$$V(x, t_f) = \Phi(x(t_f)) \quad V(x, t_0) = \min_{u} \int_{t_0}^{t_f} L(t, x(t), u(t))\,\mathrm{d}t + \Phi(x(t_f))$$

Beweis: Die Value-Function $V(x,t)$ beschreibt die minimalen Kosten von Zustand x zum Endzeitpunkt t_f. Für $s \in [t, t_f]$ gilt

- Systemdynamik

$$x'(s) = f(x(s), u(s)), \quad x(t) = x$$

- Kostenfunktional

$$J(u) = \int_{t}^{t_f} L(s, x(s), u(s))\,\mathrm{d}s + \Phi(x(t_f))$$

Angenommen, wir wechseln zum Zeitpunkt $t + h$ zu einer optimalen Steuerung, dann gilt für unser Zielfunktional

$$\begin{aligned}J(u) &= \int_{t}^{t+h} L(s, x(s), u(s))\,\mathrm{d}s + \int_{t+h}^{t_f} L(s, x(s), u(s))\,\mathrm{d}s + \Phi(x(t_f))\\ &= \int_{t}^{t+h} L(s, x(s), u(s))\,\mathrm{d}s + V(x(t+h), t+h)\end{aligned}$$

Da $V(x(t),t)$ die minimalen Kosten vom Zustand x zur Zeit t angibt, gilt:

$$\begin{aligned} V(x(t),t) &\leq \int_t^{t+h} L(s,x(s),u(s))\,\mathrm{d}s + V(x(t+h),t+h) \\ \Rightarrow 0 &\leq \int_t^{t+h} L(s,x(s),u(s))\,\mathrm{d}s + V(x(t+h),t+h) - V(x(t),t) \end{aligned}$$

Im Grenzübergang $h \to 0$ erhalten wir

$$\begin{aligned} 0 &\leq \lim_{h\to 0} \frac{1}{h} \int_t^{t+h} L(s,x(s),u(s))\,\mathrm{d}s + \lim_{h\to 0} \frac{V(x(t+h),t+h) - V(x,t)}{h} \\ &= L(t,x(t),u(t)) + \frac{\mathrm{d}}{\mathrm{d}t} V(x(t),t) \\ &= L(t,x(t),u(t)) + \frac{\partial V}{\partial t} + \frac{\partial V}{\partial x} \cdot x'(t) \\ &= L(t,x(t),u(t)) + \frac{\partial V}{\partial t} + \frac{\partial V}{\partial x} \cdot f(t,x(t),u(t)) \end{aligned}$$

Durch Minimierung bezüglich u erhalten wir die Gleichheit und somit

$$\begin{aligned} 0 &= \frac{\partial V}{\partial t} + \min_u \left[L(t,x(t),u(t)) + \frac{\partial V}{\partial x} \cdot f(t,x(t),u(t)) \right] \\ &= \frac{\partial V}{\partial t} + \mathsf{H}\left(t, x(t), \frac{\partial V}{\partial x}(t,x(t)) \right) \end{aligned}$$

□

2.1. Herleitung des Minimumprinzips

Berechnung der totalen Ableitung der Value-Function V entlang der Charakteristik $x(t)$:

$$\frac{dV}{dt} = \frac{\partial V}{\partial x}\frac{dx}{dt} + \frac{\partial V}{\partial t} = p(x,t) \cdot x'(t) - H\left(t, x, \frac{\partial V}{\partial x}\right)$$

Berechnung der totalen Ableitung von $p = \frac{\partial V}{\partial x}$:

$$\frac{dp}{dt} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial V}{\partial x}\right) = \frac{\partial^2 V}{\partial x^2} \cdot x'(t) + \frac{\partial V}{\partial x \partial t}$$

Differenzieren der HJB bezüglich x liefert

$$0 = \frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{\partial V}{\partial t} + H\left(t, x, \frac{\partial V}{\partial x}\right)\right] = \frac{\partial V}{\partial t \partial x} + \frac{\partial H}{\partial x} + \frac{\partial H}{\partial p} \cdot \frac{\partial^2 V}{\partial x^2}$$

Einsetzen der Gleichung liefert:

$$\frac{dp}{dt} = \frac{\partial^2 V}{\partial x^2} \cdot x'(t) + \left(-\frac{\partial H}{\partial x} - \frac{\partial H}{\partial p} \cdot \frac{\partial^2 V}{\partial x^2}\right) = \frac{\partial^2 V}{\partial x^2} \cdot \left(x'(t) - \frac{\partial H}{\partial p}\right) - \frac{\partial H}{\partial x}$$

Wir erhalten somit die bekannten, notwendigen Optimalitätsbedingungen:

$$p'(t) = -H_x, \quad x'(t) = f(t,x,u^*), \quad u^* = \mathsf{argmin}\,[L(t,x,u) + p(t,x)f(t,x,u)]$$

2.2. Anwendung auf Linear-Quadratische Probleme

2.2.1. Finite time problem

Gegeben sei das Steuerungsproblem

$$\min_u \int_{t_0}^{t_f} x^T Q x + u^T R u \, \mathrm{d}t + \underbrace{(x(t_f) - x_f)^T S (x(t_f) - x_f)}_{x(t_f)^T S x(t_f) - 2x_f^T S x(t_f) + x_f^T S x_f} = V(t_0, x(t_0)) + x_f^T S x_f$$

unter der Nebenbedingung $x'(t) = Ax + Bu$ und $x(t_0) = x_0$. Der Term $x_f^T S x_f$ ist unabhängig von u und kann bei der Minimierung weggelassen werden.

$$V(x,t) = x^T K(t) x + 2s(t)^T x + r(t) \quad V(x,t_f) = x(t_f)^T K(t_f) x(t_f) + 2s(t_f)^T x(t_f) + r(t_f)$$

Wir erhalten für die Endwerte

$$K(t_f) = S, \quad s(t_f) = -Sx_f \quad r(t_f) = 0$$

$$V_t = x^T K'(t) x + 2s'(t)^T x + r'(t) \quad V_x = 2K(t)x + 2s(t)$$

$$\frac{\partial}{\partial u}\left[x^T Q x + u^T R u + V_x^T \cdot (Ax + Bu)\right] = 2Ru + B^T \cdot V_x \Rightarrow u^*(t) = -\frac{1}{2} R^{-1} B^T V_x$$

$$\begin{aligned}
(u^*)^T R u^* &= \left(-\frac{1}{2} R^{-1} B^T V_x\right)^T R \left(-\frac{1}{2} R^{-1} B^T V_x\right) \\
&= \frac{1}{4} \cdot \left(R^{-1} B^T V_x\right)^T R \left(R^{-1} B^T V_x\right) \\
&= \frac{1}{4} \cdot \left(2R^{-1} B^T (K(t)x + s(t))\right)^T R \left(2R^{-1} B^T (K(t)x + s(t))\right) \\
&= (K(t)x + s(t))^T B R^{-1} R R^{-1} B^T (K(t)x + s(t)) \\
&= < K(t)x + s(t), P(2K(t)x + s(t)) > \\
&= < K(t)x, PK(t)x > + \underbrace{< K(t)x, Ps(t) >}_{< s(t), P^T K(t)x >} + < s(t), PK(t)x > + < s(t), Ps(t) > \\
&= x^T K(t)^T P K(t) x + 2 \cdot s(t)^T P K(t) x + s(t)^T P s(t)
\end{aligned}$$

mit $P = BR^{-1}B^T$ und $P^T = P$.

$$\begin{aligned}
V_x^T \left(-\frac{1}{2} B R^{-1} B^T V_x\right) &= (2K(t)x + 2s(t))^T \left(-\frac{1}{2} B R^{-1} B^T (2K(t)x + 2s(t))\right) \\
&= -2 \cdot (K(t)x + s(t))^T P (K(t)x + s(t)) \\
&= -2 \cdot x^T K(t)^T P K(t) x - 4 \cdot s(t)^T P K(t) x - 2 \cdot s(t)^T P s(t)
\end{aligned}$$

$$V_x^T A x = 2(K(t)x + s(t))^T A x = 2x^T K(t)^T A x + 2s(t)^T A x = x^T (K(t)^T A + A^T K(t)) x + 2s(t)^T A x$$

$$-V_t = \min_u \left[x^T Q x + u^T R u + V_x^T \cdot (Ax + Bu)\right]$$

$$-V_t = x^T Q x + (u^*)^T R u^* + V_x^T \cdot \left(Ax - \frac{1}{2} B R^{-1} B^T V_x\right)$$
$$-x^T K'(t) x - 2 s'(t)^T x - r'(t) = x^T (Q - KPK + KA + A^T K) x - 2 \cdot ((PK - A)^T s(t))^T x - s(t)^T P s(t)$$

$$\begin{aligned} K'(t) &= -Q + K(t) B R^{-1} B^T K(t) - K(t) A - A^T K(t) \quad K(t_f) = S \\ s'(t) &= (K(t) B R^{-1} B^T - A^T) s(t) \quad s(t_f) = -S x_f \\ r'(t) &= s(t)^T B R^{-1} B^T s(t) \quad r(t_f) = 0 \end{aligned}$$

2.2.2. infinite time problem

Im Grenzfall $t \to \infty$ betrachten wir das Steuerungsproblem

$$\min_u \int_{t_0}^{\infty} (x - x_f)^T Q (x - x_f) + u^T R u \, \mathrm{d}t = V(x(t_0))$$

unter der Nebenbedingung $x'(t) = Ax + Bu$ und $x(t_0) = x_0$. Als Ansatz für die Value Funktion wählen wir erneut

$$V(x) = x^T K x + 2 s^T x + r \Rightarrow V_x = 2Kx + 2s$$

Es gilt erneut

$$\frac{\partial}{\partial u} \left((x - x_f)^T Q (x - x_f) + u^T R u + V_x^T (Ax + Bu)\right) = 2Ru + B^T V_x = 0 \Rightarrow u^* = -\frac{1}{2} R^{-1} B^T V_x$$

$$(x - x_f)^T Q (x - x_f) = x^T Q x - 2 x_f^T Q x + x_f^T Q x_f$$

$$0 = x^T (Q - KPK + KA + A^T K) x - 2((PK - A)^T s + Q x_f)^T x - s^T(P) s + x_f^T Q x_f$$

2.3. Optimality Condition

Steuerung $\boldsymbol{u} \in C^1(\mathbb{R}; \mathbb{R}^{n_u})$: Wir betrachten die Störung

$$\boldsymbol{u}_\epsilon(t) := \boldsymbol{u}(t) + \epsilon \cdot \boldsymbol{h}(t) \quad \boldsymbol{x}_\epsilon(t) := \boldsymbol{x}(t) + \epsilon \cdot \boldsymbol{\eta}(t)$$

Einsetzen in die Systemdynamik $\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u})$ liefert:

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}\epsilon} \dot{\boldsymbol{x}}(t) + \epsilon \cdot \dot{\boldsymbol{\eta}}(t) \Big|_{\epsilon=0} &= \frac{\mathrm{d}}{\mathrm{d}\epsilon} \boldsymbol{f}\left(\boldsymbol{x}(t) + \epsilon \cdot \boldsymbol{\eta}(t), \boldsymbol{u}(t) + \epsilon \cdot \boldsymbol{h}(t)\right) \Big|_{\epsilon=0} \\ \dot{\boldsymbol{\eta}}(t) &= \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}} \cdot \boldsymbol{\eta}(t) + \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{u}} \cdot \boldsymbol{h}(t) \end{aligned}$$

Aus $\boldsymbol{x}_\epsilon(t_0) = \boldsymbol{x}_0$ für alle ϵ folgt $\boldsymbol{\eta}(t_0) = 0$.

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}\epsilon} J(\boldsymbol{u} + \epsilon \boldsymbol{h})\bigg|_{\epsilon=0} &= \frac{\mathrm{d}}{\mathrm{d}\epsilon} \int_{t_0}^{t_f} L(\boldsymbol{x}_\epsilon, \boldsymbol{u}_\epsilon)\,\mathrm{d}t + \Phi(\boldsymbol{x}_\epsilon(t_f))\bigg|_{\epsilon=0} \\ &= \int_{t_0}^{t_f} L_{\boldsymbol{x}}(\boldsymbol{x}, \boldsymbol{u}) \cdot \boldsymbol{\eta}(t) + L_{\boldsymbol{u}}(\boldsymbol{x}, \boldsymbol{u}) \cdot \boldsymbol{h}(t)\,\mathrm{d}t + \Phi(\boldsymbol{x}(t_f)) \cdot \boldsymbol{\eta}(t_f)\end{aligned}$$

Multiplikation des adjungierten Zustandes $\boldsymbol{p}$ [1] mit der Sensitivitätsgleichung für $\boldsymbol{\eta}$ liefert:

$$\boldsymbol{p}(t)^T \left(\dot{\boldsymbol{\eta}}(t) - \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}} \cdot \boldsymbol{\eta}(t) - \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{u}} \cdot \boldsymbol{h}(t) \right) = 0$$

Einsetzen in die Ableitung des Zielfunktionals liefert:[2]

$$\begin{aligned}D(\boldsymbol{u}, \boldsymbol{h}) &:= \frac{\mathrm{d}}{\mathrm{d}\epsilon} J(\boldsymbol{u} + \epsilon \boldsymbol{h})\bigg|_{\epsilon=0} \\ &= \frac{\mathrm{d}}{\mathrm{d}\epsilon} \int_{t_0}^{t_f} L(\boldsymbol{x}_\epsilon, \boldsymbol{u}_\epsilon)\,\mathrm{d}t + \Phi(\boldsymbol{x}_\epsilon(t_f))\bigg|_{\epsilon=0} \\ &= \int_{t_0}^{t_f} L_{\boldsymbol{x}} \cdot \boldsymbol{\eta} + L_{\boldsymbol{u}} \cdot \boldsymbol{h} - \boldsymbol{p}^T \left[\dot{\boldsymbol{\eta}} - \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}} \cdot \boldsymbol{\eta} - \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{u}} \cdot \boldsymbol{h} \right] \mathrm{d}t + \frac{\partial \Phi(\boldsymbol{x}(t_f))}{\partial \boldsymbol{x}} \cdot \boldsymbol{\eta}(t_f) \\ &= \int_{t_0}^{t_f} \left(L_{\boldsymbol{u}} + \boldsymbol{p}^T \boldsymbol{f}_{\boldsymbol{u}} \right) \cdot \boldsymbol{h}(t)\,\mathrm{d}t\end{aligned}$$

für jede beliebige Testfunktion $\boldsymbol{\eta}$. Somit gilt $\nabla_{\boldsymbol{u}} J(\boldsymbol{u}) = L_{\boldsymbol{u}}^T + \boldsymbol{f}_{\boldsymbol{u}}^T \boldsymbol{p}(t)$. Sei $\boldsymbol{u} \in \mathbb{R}^N$ ein endlichdimensionaler Vektor, dann gilt für die Variation von $\boldsymbol{u}$:

$$\boldsymbol{u}_\epsilon = \boldsymbol{u} + \epsilon \boldsymbol{h}$$

da $\boldsymbol{h}$ nicht zeitabhängig ist, gilt

$$D(\boldsymbol{u}, \boldsymbol{h}) = \boldsymbol{h} \cdot \int_{t_0}^{t_f} \left(L_{\boldsymbol{u}} + \boldsymbol{p}^T \boldsymbol{f}_{\boldsymbol{u}} \right) \mathrm{d}t \Rightarrow \nabla_{\boldsymbol{u}} J(\boldsymbol{u}) = \int_{t_0}^{t_f} \left(L_{\boldsymbol{u}}^T + \boldsymbol{f}_{\boldsymbol{u}}^T \cdot \boldsymbol{p} \right) \mathrm{d}t$$

Wobei $L_{\boldsymbol{u}}^T \in \mathbb{R}^{N \times 1}$, $\boldsymbol{f}_{\boldsymbol{u}}^T \in \mathbb{R}^{N \times n_x}$ gilt.

2.3.1. Schrittweitenbestimmung

Ziel ist es nun eine Schrittweite $\alpha > 0$ zu bestimmen, die J entlang des negativen Gradienten verringert, d.h. $J\left[\boldsymbol{u} - \alpha \cdot \nabla_{\boldsymbol{u}} J(\boldsymbol{u})\right] < J\left[\boldsymbol{u}\right]$. Wir linearisieren J lokal um $\boldsymbol{u}$:

$$J\left[\boldsymbol{u} - \alpha \cdot \nabla_{\boldsymbol{u}} J(\boldsymbol{u})\right] \approx J\left[\boldsymbol{u}\right] - \alpha \cdot \underbrace{\nabla_{\boldsymbol{u}} J(\boldsymbol{u})^T \cdot \nabla_{\boldsymbol{u}} J(\boldsymbol{u})}_{\|\nabla_{\boldsymbol{u}} J(\boldsymbol{u})\|^2}$$

[1] The adjoint state $\boldsymbol{p} : \mathbb{R} \to \mathbb{R}^{n_x}$ satisfies

$$\dot{\boldsymbol{p}}(t) = -\left(\frac{\partial H}{\partial \boldsymbol{x}}\right)^T = -\left(\frac{\partial L}{\partial \boldsymbol{x}}\right)^T - \left(\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right)^T \cdot \boldsymbol{p}(t) \quad \boldsymbol{p}(t_f) = \left(\frac{\partial \Phi(\boldsymbol{x}(t_f))}{\partial \boldsymbol{x}}\right)^T$$

[2] Integration by parts leads to

$$-\int_{t_0}^{t_f} \boldsymbol{p}(t)^T \dot{\boldsymbol{\eta}}(t)\,\mathrm{d}t = \int_{t_0}^{t_f} \dot{\boldsymbol{p}}(t)^T \boldsymbol{\eta}(t)\,\mathrm{d}t - \left[\boldsymbol{p}(t)^T \boldsymbol{\eta}(t)\right]_{t_0}^{t_f} = \int_{t_0}^{t_f} \left(-\frac{\partial L}{\partial \boldsymbol{x}} - \boldsymbol{p}(t)^T \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}} \right) \boldsymbol{\eta}(t)\,\mathrm{d}t - \frac{\partial \Phi(\boldsymbol{x}(t_f))}{\partial \boldsymbol{x}} \cdot \boldsymbol{\eta}(t_f)$$

Die Existenz eines $c > 0$ ist garantiert und es gilt die Armijo-Bedingung für die Schrittweite α [5]:

$$J\left[\boldsymbol{u} - \alpha \cdot \nabla_{\boldsymbol{u}} J(\boldsymbol{u})\right] \leq J(\boldsymbol{u}) - c \cdot \alpha \left\|\nabla_{\boldsymbol{u}} J(\boldsymbol{u})\right\|^2$$

Nun ersetzen wir den globalen Parameter α für einen Schritt:

$$\boldsymbol{m}_t := \beta_1 \cdot \boldsymbol{m}_{t-1} + (1 - \beta_1) \cdot \nabla_{\boldsymbol{u}} J(\boldsymbol{u}) \quad \boldsymbol{m}_0 = \mathbf{0}$$

Wir erhalten also iterativ:

$$\begin{aligned}
\boldsymbol{m}_1 &= (1 - \beta_1) \cdot \nabla_{\boldsymbol{u}} J(\boldsymbol{u}) \\
\boldsymbol{m}_2 &= \beta_1 \cdot (1 - \beta_1) \cdot \nabla_{\boldsymbol{u}} J(\boldsymbol{u}) + (1 - \beta_1) \cdot \nabla_{\boldsymbol{u}} J(\boldsymbol{u}) \\
\boldsymbol{m}_3 &= \beta_1^2 \cdot (1 - \beta_1) \cdot \nabla_{\boldsymbol{u}} J(\boldsymbol{u}) + \beta_1 \cdot (1 - \beta_1) \cdot \nabla_{\boldsymbol{u}} J(\boldsymbol{u}) + (1 - \beta_1) \cdot \nabla_{\boldsymbol{u}} J(\boldsymbol{u}) \\
&= (1 - \beta_1) \cdot \boldsymbol{m}_K \qquad\qquad = (1 - \beta_1)
\end{aligned}$$

3. Linearisierung der Dynamik

Wir gehen analog zu [4] vor. Wir definieren den Zustand $\mathbf{x} = \begin{bmatrix}\mathbf{q}\\ \dot{\mathbf{q}}\end{bmatrix}$ und schreiben unsere Dynamikgleichung wir folgt um

$$M(\mathbf{q})\ddot{\mathbf{q}} + C(\mathbf{q},\dot{\mathbf{q}})\dot{\mathbf{q}} + G(\mathbf{q}) = \boldsymbol{u} \Rightarrow \underbrace{\begin{bmatrix}\dot{\mathbf{q}}\\ \ddot{\mathbf{q}}\end{bmatrix}}_{\dot{\mathbf{x}}} = \underbrace{\begin{bmatrix}\dot{\mathbf{q}}\\ M^{-1}(\mathbf{q})\left[\boldsymbol{u} - C(\mathbf{q},\dot{\mathbf{q}})\dot{\mathbf{q}} - G(\mathbf{q})\right]\end{bmatrix}}_{f(\mathbf{x},\boldsymbol{u})}$$

Wir linearisieren um den Arbeitspunkt $(\boldsymbol{x}^*, \boldsymbol{u}^*)$

$$\begin{aligned}\dot{\boldsymbol{x}} &= f(\boldsymbol{x},\boldsymbol{u})\\ &\approx \underbrace{f(\boldsymbol{x}^*,\boldsymbol{u}^*)}_{=0} + \underbrace{\frac{\partial f}{\partial x}(\boldsymbol{x}^*,\boldsymbol{u}^*)}_{A(\boldsymbol{x}^*,\boldsymbol{u}^*)}(\boldsymbol{x}-\boldsymbol{x}^*) + \underbrace{\frac{\partial f}{\partial u}(\boldsymbol{x}^*,\boldsymbol{u}^*)}_{B(\boldsymbol{x}^*,\boldsymbol{u}^*)}(\boldsymbol{u}-\boldsymbol{u}^*)\\ &= A(\boldsymbol{x}^*,\boldsymbol{u}^*)\bar{\boldsymbol{x}} + B(\boldsymbol{x}^*,\boldsymbol{u}^*)\bar{\boldsymbol{u}}\end{aligned}$$

Wobei $\bar{\boldsymbol{x}}(t) = \boldsymbol{x}(t) - \boldsymbol{x}^*$ und $\bar{\boldsymbol{u}}(t) = \boldsymbol{u}(t) - \boldsymbol{u}^*$ gilt. Für die Ableitung der Größen gilt

$$\frac{\mathrm{d}\bar{\boldsymbol{x}}}{\mathrm{d}t} = \dot{\boldsymbol{x}} - \underbrace{\frac{\mathrm{d}\boldsymbol{x}^*}{\mathrm{d}t}}_{=0} = \dot{\boldsymbol{x}}(t) \Rightarrow \bar{\boldsymbol{x}}(t) = A\bar{\boldsymbol{x}}(t) + B\bar{\boldsymbol{u}}(t)$$

Somit ist die Rückführung von $\bar{\boldsymbol{x}}$ auf $\mathbf{0}$ äquivalten zu der Rückführung von $\boldsymbol{x}$ auf den Arbeitspunkt $\boldsymbol{x}^*$. Wir definieren $h(\boldsymbol{q},\dot{\boldsymbol{q}},\boldsymbol{u}) := \boldsymbol{u} - C(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} - G(\boldsymbol{q})$. und wenden die Produktregel auf den Term $M(\boldsymbol{q})\ddot{\boldsymbol{q}}$ an, um die Ableitung von f nach $\boldsymbol{x}$ zu bestimmen.

$$\frac{\partial}{\partial \boldsymbol{x}}\left[M(\boldsymbol{q})\ddot{\boldsymbol{q}}\right] = \frac{\partial}{\partial \boldsymbol{x}}\left[h(\boldsymbol{q},\dot{\boldsymbol{q}},\boldsymbol{u})\right] \Rightarrow \frac{\partial M}{\partial \boldsymbol{x}}\ddot{\boldsymbol{q}} + M(\boldsymbol{q})\frac{\partial \ddot{\boldsymbol{q}}}{\partial \boldsymbol{x}} = \frac{\partial h}{\partial \boldsymbol{x}}$$

Werten wir h im Arbeitspunkt $(\boldsymbol{x}^*,\boldsymbol{u}^*)$ aus, so gilt $\ddot{\boldsymbol{q}} = \dot{\boldsymbol{q}} = \mathbf{0}$, somit fällt der erste Term weg und wir erhalten

$$\frac{\partial h}{\partial \boldsymbol{x}}(\boldsymbol{x}^*,\boldsymbol{u}^*) = \underbrace{\frac{\partial M}{\partial \boldsymbol{x}}\ddot{\boldsymbol{q}}}_{=0} + M(\boldsymbol{x}^*)\frac{\partial \ddot{\boldsymbol{q}}}{\partial \boldsymbol{x}} \Rightarrow \frac{\partial \ddot{\boldsymbol{q}}}{\partial \boldsymbol{x}} = M^{-1}(\boldsymbol{q}^*)\frac{\partial h}{\partial \boldsymbol{x}}(\boldsymbol{x}^*,\boldsymbol{u}^*)$$

Für die Ableitung der Funktion $h = \boldsymbol{u} - C(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} - G(\boldsymbol{q})$ in $(\boldsymbol{x}^*,\boldsymbol{u}^*)$ gilt

$$\begin{aligned}\frac{\partial h}{\partial \boldsymbol{x}}(\boldsymbol{x}^*,\boldsymbol{u}^*) = \begin{bmatrix}\frac{\partial h}{\partial \boldsymbol{q}} & \frac{\partial h}{\partial \dot{\boldsymbol{q}}}\end{bmatrix} &= \begin{bmatrix}-\frac{\partial C}{\partial \boldsymbol{q}}\dot{\boldsymbol{q}} - \frac{\partial G}{\partial \boldsymbol{q}} & -C(\boldsymbol{q},\dot{\boldsymbol{q}}) - \frac{\partial C}{\partial \dot{\boldsymbol{q}}}\dot{\boldsymbol{q}}\end{bmatrix}(\boldsymbol{q}^*, \dot{\boldsymbol{q}}^* = \mathbf{0})\\ &= \begin{bmatrix}-\frac{\partial G}{\partial \boldsymbol{q}} & -C(\boldsymbol{q}^*,\mathbf{0})\end{bmatrix}\end{aligned}$$

Fassen wir die Ergebnisse zusammen, so gilt für die Ableitung von f nach $\boldsymbol{x}$ im Arbeitspunkt $(\boldsymbol{x}^*,\boldsymbol{u}^*)$

$$A(\boldsymbol{x}^*,\boldsymbol{u}^*) = \begin{bmatrix}\frac{\partial \dot{\boldsymbol{q}}}{\partial \boldsymbol{q}} & \frac{\partial \dot{\boldsymbol{q}}}{\partial \dot{\boldsymbol{q}}}\\ \frac{\partial \ddot{\boldsymbol{q}}}{\partial \boldsymbol{q}} & \frac{\partial \ddot{\boldsymbol{q}}}{\partial \dot{\boldsymbol{q}}}\end{bmatrix}(\boldsymbol{x}^*,\boldsymbol{u}^*) = \begin{bmatrix}0 & I\\ M^{-1}(\boldsymbol{q}^*)\left[-\frac{\partial G}{\partial \boldsymbol{q}}(\boldsymbol{q}^*)\right] & M^{-1}(\boldsymbol{q}^*)\left[-C(\boldsymbol{q}^*,\mathbf{0})\right]\end{bmatrix}$$

Für die Ableitung von f nach $\boldsymbol{u}$ gilt

$$B(\boldsymbol{x}^*,\boldsymbol{u}^*) = \begin{bmatrix}\frac{\partial \dot{\boldsymbol{q}}}{\partial \boldsymbol{u}}\\ \frac{\partial \ddot{\boldsymbol{q}}}{\partial \boldsymbol{u}}\end{bmatrix}(\boldsymbol{x}^*,\boldsymbol{u}^*) = \begin{bmatrix}0\\ M^{-1}(\boldsymbol{q}^*)\end{bmatrix}$$

Satz 3.1. Gegeben sei die Dynamikgleichung

$$M(\boldsymbol{q})\ddot{\boldsymbol{q}} + C(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + G(\boldsymbol{q}) = \boldsymbol{u} \Rightarrow \dot{\boldsymbol{x}} = f(\boldsymbol{x}, \boldsymbol{u})$$

und ein Arbeitspunkt $(\boldsymbol{x}^*, \boldsymbol{u}^*)$ mit $f(\boldsymbol{x}^*, \boldsymbol{u}^*) = \boldsymbol{0}$. Dann gilt für die Linearisierung:

$$\begin{aligned}\dot{\boldsymbol{x}} &= A(\boldsymbol{x}^*, \boldsymbol{u}^*)\boldsymbol{x} + B(\boldsymbol{x}^*, \boldsymbol{u}^*)\boldsymbol{u} \\ &= \begin{bmatrix} 0 & I \\ M^{-1}(\boldsymbol{q}^*)\left[-\frac{\partial G}{\partial \boldsymbol{q}}(\boldsymbol{q}^*)\right] & M^{-1}(\boldsymbol{q}^*)\left[-C(\boldsymbol{q}^*, \boldsymbol{0})\right] \end{bmatrix} \boldsymbol{x} + \begin{bmatrix} 0 \\ M^{-1}(\boldsymbol{q}^*) \end{bmatrix} \boldsymbol{u}\end{aligned}$$

3.1. Linearisierung des two link revolute manipulators

Wir definieren $x_1 = q_1$, $x_2 = q_2$, $x_3 = \dot{q}_1$, $x_4 = \dot{q}_2$ und erhalten $\dot{\boldsymbol{x}} = f(\boldsymbol{x}, \boldsymbol{u})$ mit

$$\begin{bmatrix} f_1(\boldsymbol{x}, \boldsymbol{u}) \\ f_2(\boldsymbol{x}, \boldsymbol{u}) \end{bmatrix} = \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \end{bmatrix} \quad \begin{bmatrix} f_3(\boldsymbol{x}, \boldsymbol{u}) \\ f_4(\boldsymbol{x}, \boldsymbol{u}) \end{bmatrix} = M(\boldsymbol{q})^{-1}\left[\boldsymbol{u} - C(\boldsymbol{x})\dot{\boldsymbol{q}} - G(\boldsymbol{q})\right]$$

Für die Corioliskraft im Punkt $(\boldsymbol{x}^*, \boldsymbol{u}^*)$, bzw. für die Matrix $C(\boldsymbol{q}, \dot{\boldsymbol{q}} = \boldsymbol{0})$ gilt:

$$C(\boldsymbol{q}^*, \boldsymbol{0}) = \begin{bmatrix} 0 & -m_2 l_1 a_2 \sin(q_2^*) \cdot 0 \\ m_2 l_1 a_2 \sin(q_2^*) \cdot 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Wir berechnen zunächst die Ableitung von h bzgl. $\boldsymbol{x}$ im Punkt $(\boldsymbol{x}^*, \boldsymbol{u}^*)$ und erhalten

$$\frac{\partial h}{\partial \boldsymbol{x}}(\boldsymbol{x}^*, \boldsymbol{u}^*) = \begin{bmatrix} -\frac{\partial G}{\partial \boldsymbol{q}} & -\underbrace{C(\boldsymbol{q}^*, \boldsymbol{0})}_{=\boldsymbol{0}} \end{bmatrix} = \begin{bmatrix} -\frac{\partial G_1}{\partial q_1}(\boldsymbol{q}^*) & -\frac{\partial G_1}{\partial q_2}(\boldsymbol{q}^*) & 0 & 0 \\ -\frac{\partial G_2}{\partial q_1}(\boldsymbol{q}^*) & -\frac{\partial G_2}{\partial q_2}(\boldsymbol{q}^*) & 0 & 0 \end{bmatrix}$$

Für die konkrete Matrix $A = f_{\boldsymbol{x}}(\boldsymbol{x}^*, \boldsymbol{u}^*)$ gilt

$$A(\boldsymbol{x}^*, \boldsymbol{u}^*) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \frac{\partial f_1}{\partial x_3} & \frac{\partial f_1}{\partial x_4} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \frac{\partial f_2}{\partial x_3} & \frac{\partial f_2}{\partial x_4} \\ \frac{\partial f_3}{\partial x_1} & \frac{\partial f_3}{\partial x_2} & \frac{\partial f_3}{\partial x_3} & \frac{\partial f_3}{\partial x_4} \\ \frac{\partial f_4}{\partial x_1} & \frac{\partial f_4}{\partial x_2} & \frac{\partial f_4}{\partial x_3} & \frac{\partial f_4}{\partial x_4} \end{bmatrix}(\boldsymbol{x}^*, \boldsymbol{u}^*) = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -M^{-1}(\boldsymbol{q}^*)\frac{\partial G}{\partial \boldsymbol{q}}(\boldsymbol{q}^*) & & 0 & 0 \\ & & 0 & 0 \end{bmatrix}$$

Für die konkrete Matrix $B = f_{\boldsymbol{u}}(\boldsymbol{x}^*, \boldsymbol{u}^*)$ gilt

$$B(\boldsymbol{x}^*, \boldsymbol{u}^*) = \begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \frac{\partial f_1}{\partial u_2} \\ \frac{\partial f_2}{\partial u_1} & \frac{\partial f_2}{\partial u_2} \\ \frac{\partial f_3}{\partial u_1} & \frac{\partial f_3}{\partial u_2} \\ \frac{\partial f_4}{\partial u_1} & \frac{\partial f_4}{\partial u_2} \end{bmatrix}(\boldsymbol{x}^*, \boldsymbol{u}^*) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ M^{-1}(\boldsymbol{q}^*) \end{bmatrix}$$

Für die konkret Berechnung der partiellen Ableitungen der Gravitationskraft gilt

$$\begin{aligned} G(\boldsymbol{q}) &= \begin{bmatrix} m_1 g a_1 \cos(q_1) + m_2 g\left(l_1 \cos(q_1) + a_2 \cos(q_1 + q_2)\right) \\ m_2 g\left(l_1 \cos(q_1) + a_2 \cos(q_1 + q_2)\right) \end{bmatrix} \\ \frac{\partial G}{\partial \boldsymbol{q}} &= \begin{bmatrix} \frac{\partial G_1}{\partial q_1} & \frac{\partial G_1}{\partial q_2} \\ \frac{\partial G_2}{\partial q_1} & \frac{\partial G_2}{\partial q_2} \end{bmatrix} \\ &= \begin{bmatrix} -m_1 g a_1 \sin(q_1) - m_2 g\left(l_1 \sin(q_1) + a_2 \sin(q_1 + q_2)\right) & -m_2 g a_2 \sin(q_1 + q_2) \\ -m_2 g\left(l_1 \sin(q_1) + a_2 \sin(q_1 + q_2)\right) & -m_2 g a_2 \sin(q_1 + q_2) \end{bmatrix} \end{aligned}$$

Die Massenmatrix $M(\boldsymbol{q})$ ist gegeben durch den Ausdruck:

$$M(\boldsymbol{q}) = \begin{bmatrix} m_1 a_1^2 + I_{1,zz} + m_2 \cdot (l_1^2 + a_2^2 + 2 l_1 a_2 \cos(q_2)) + I_{2,zz} & m_2 \cdot (a_2^2 + l_1 a_2 \cos(q_2)) + I_{2,zz} \\ m_2 \cdot (a_2^2 + l_1 a_2 \cos(q_2)) + I_{2,zz} & m_2 a_2^2 + I_{2,zz} \end{bmatrix}$$

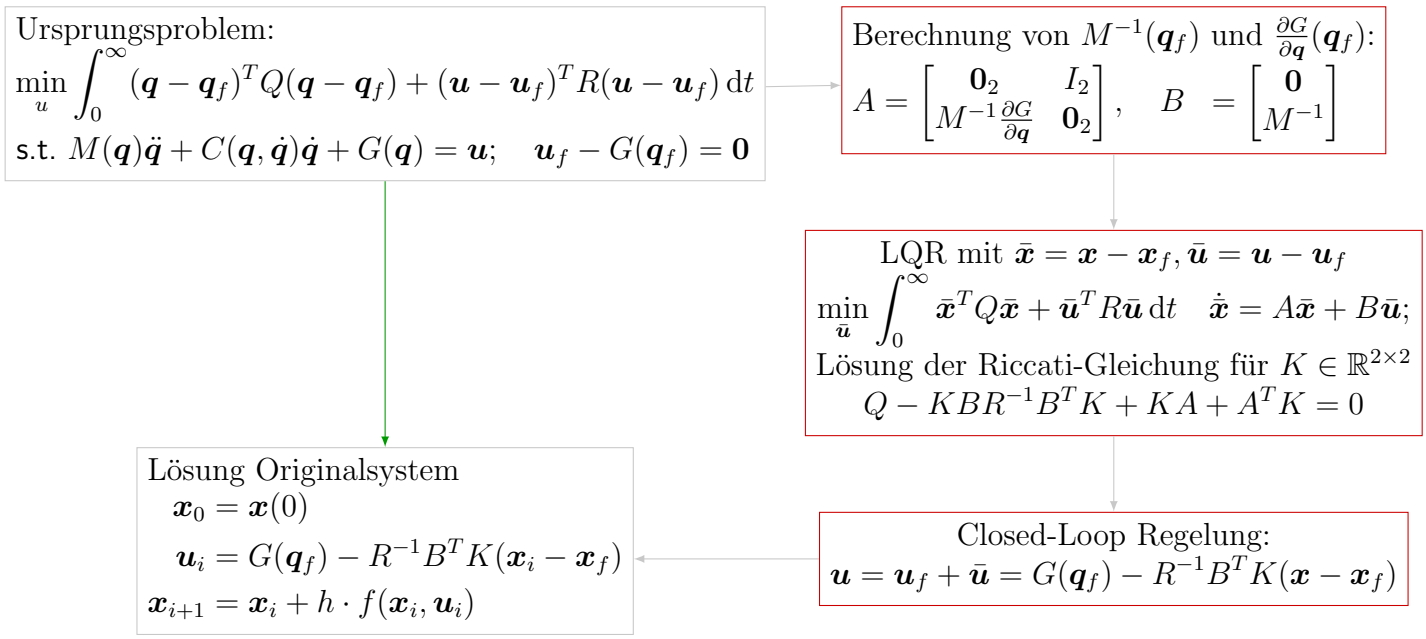
Für die Inverse der Massenmatrix M^{-1} gilt analytisch mittels $F_{q_2} := m_2 l_1 a_2 \cos(q_2)$

$$\begin{aligned}
\det(M) &= M_{11} M_{22} - M_{12}^2 \\
&= \left[m_1 a_1^2 + I_{1,zz} + m_2 \cdot (l_1^2 + a_2^2 + 2 l_1 a_2 \cos(q_2)) + I_{2,zz} \right] \cdot (m_2 a_2^2 + I_{2,zz}) \\
&\quad - (m_2 \cdot (a_2^2 + l_1 a_2 \cos(q_2)) + I_{2,zz})^2 \\
&= \left[m_1 a_1^2 + I_{1,zz} + I_{2,zz} + m_2 l_1^2 + m_2 a_2^2 + 2 \underbrace{m_2 l_1 a_2 \cos(q_2)}_{F_{q_2}} \right] (m_2 a_2^2 + I_{2,zz}) \\
&\quad - (m_2 a_2^2 + I_{2,zz})^2 - 2 \underbrace{m_2 l_1 a_2 \cos(q_2)}_{F_{q_2}} \cdot (m_2 a_2^2 + I_{2,zz}) - F_{q_2}^2 \\
&= (m_2 a_2^2 + I_{2,zz}) \cdot \left[m_1 a_1^2 + I_{1,zz} + m_2 l_1^2 + \underbrace{I_{2,zz} + m_2 a_2^2 - m_2 a_2^2 - I_{2,zz}}_{=0} \right] - \underbrace{(m_2 l_1 a_2)^2 (1 - \sin(q_2)^2)}_{F_{q_2}^2} \\
&= (m_2 a_2^2 + I_{2,zz}) \cdot (m_1 a_1^2 + I_1 + m_2 l_1^2) - (m_2 l_1 a_2)^2 + (m_2 l_1 a_2)^2 \sin(q_2)^2 \\
&= (m_2 a_2^2 + I_{2,zz}) \cdot (m_1 a_1^2 + I_1) + \underbrace{(m_2 a_2^2 + I_{2,zz}) \cdot m_2 l_1^2}_{m_2^2 l_1^2 a_2^2 + m_2 l_1^2 I_{2,zz}} - (m_2 l_1 a_2)^2 + (m_2 l_1 a_2)^2 \sin(q_2)^2 \\
&= (m_1 a_1^2 + I_1 + m_2 l_1^2) \cdot (m_2 a_2^2 + I_2) + m_2 l_1^2 I_{2,zz} + m_2^2 l_1^2 a_2^2 \sin(q_2)^2
\end{aligned}$$

Für die Inverse der Massenmatrix gilt somit

$$\begin{aligned}
M^{-1} &= \frac{1}{\det(M)} \begin{pmatrix} M_{22} & -M_{12} \\ -M_{21} & M_{11} \end{pmatrix} \\
&= \frac{1}{\det(M)} \begin{pmatrix} m_2 a_2^2 + I_2 & -(m_2 \cdot (a_2^2 + l_1 a_2 \cos(x_2)) + I_2) \\ -(m_2 \cdot (a_2^2 + l_1 a_2 \cos(x_2)) + I_2) & m_1 a_1^2 + I_1 + m_2 \cdot (l_1^2 + a_2^2 + 2 l_1 a_2 \cos(x_2)) + I_2 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
M_{11} &= m_1 a_1^2 + I_1 + m_2 \cdot (l_1^2 + a_2^2 + 2 l_1 a_2 \cos(x_2)) + I_2 \\
M_{12} &= m_2 \cdot (a_2^2 + l_1 a_2 \cos(x_2)) + I_2 \\
M_{21} &= M_{12} \\
M_{22} &= m_2 a_2^2 + I_2
\end{aligned}$$



3.2. Riccati Regler

Es gilt

$$F(K) := Q - KPK + KA + A^TK \Rightarrow F(K) = 0$$

Definition 3.2. Sei $X \in \mathbb{R}^{n\times n}$ dann definieren wir

$$\mathsf{vec}(X) = \begin{pmatrix} X_{11} & X_{21} & \cdots & X_{n1} & X_{12} & X_{22} & \cdots X_{nn}\end{pmatrix}^T \in \mathbb{R}^{n^2}$$

Seien $A \in \mathbb{R}^{n\times m}, B \in \mathbb{R}^{p\times r}$, dann definieren wir

$$A \otimes B = \begin{pmatrix} A_{11}B & \cdots & A_{1m}B \\ A_{12}B & \dots & A_{2m}B \\ \vdots & \ddots & \vdots \\ A_{n1}B & \cdots & A_{nm}B \end{pmatrix} \in \mathbb{R}^{mp\times nr}$$

Es gilt die Formel

$$\mathsf{vec}(MXN) = (N^T \otimes M)\mathsf{vec}(X)$$

Betrachten wir den Fall $M = 1$ und $N = A$ bzw. $N = 1$ und $M = A^T$ erhalten wir mit $X = K$

$$\mathsf{vec}(KA) = (A^T \otimes 1)\mathsf{vec}(K) \quad \mathsf{vec}(A^TK) = (1 \otimes A^T)\mathsf{vec}(K)$$

Anwendung auf unser Problem liefert

$$\underbrace{\mathsf{vec}(F(K))}_{f(x=\mathsf{vec}(K))} = \underbrace{\mathsf{vec}(Q)}_{q} - \underbrace{\mathsf{vec}(KPK)}_{g(x=\mathsf{vec}(K))} + \underbrace{\mathsf{vec}(KA)}_{(A^T\otimes 1)\mathsf{vec}(K)} + \underbrace{\mathsf{vec}(A^TK)}_{(1\otimes A^T)\mathsf{vec}(K)}$$

Durch Einführung von $x = \mathsf{vec}(K) \in \mathbb{R}^{n^2}$ erhalten wir

$$f(x) = q - g(x) + (A^T \otimes 1)x + (1 \otimes A^T)x$$

Da $(KP)^T = KP$ gilt

$$g(\mathsf{vec}(K)) = \mathsf{vec}(KPK) = (1 \otimes KP)\mathsf{vec}(K)$$

Wir betrachten die Variation $K(\epsilon) = K + \epsilon \cdot \bar{K}$ und die Funktion $h : \mathbb{R} \to \mathbb{R}^{n^2}$ mit

$$\begin{aligned} h(\epsilon) := g(\mathsf{vec}(K(\epsilon))) &= \mathsf{vec}(K(\epsilon)PK(\epsilon)) \\ &= \mathsf{vec}((K + \epsilon \cdot \bar{K})P(K + \epsilon \cdot \bar{K})) \\ &= \mathsf{vec}(KPK + \epsilon KP\bar{K} + \epsilon \bar{K}PK + \epsilon^2 \bar{K}P\bar{K}) \end{aligned}$$

Ableitung nach ϵ und Auswertung in $\epsilon = 0$ ergibt

$$h'(0) = \frac{d}{d\epsilon} g(\mathsf{vec}(K + \epsilon\bar{K}))\Big|_{\epsilon=0} = \mathsf{vec}(KP\bar{K} + \bar{K}PK) = \mathsf{vec}(KP\bar{K}) + \mathsf{vec}(\bar{K}PK)$$

Umschreiben als Matrix-Vektor-Produkt

$$\mathsf{vec}(KP \cdot \bar{K} \cdot 1) = (1 \otimes KP)\mathsf{vec}(\bar{K}) \quad \mathsf{vec}(1 \cdot \bar{K} \cdot PK) = ((PK)^T \otimes 1)\mathsf{vec}(\bar{K})$$

Wir erhalten als Ableitung

$$g'(x) = (1 \otimes KP) + ((PK)^T \otimes 1) \Rightarrow f'(x) = -(1 \otimes KP) - ((PK)^T \otimes 1) + (A^T \otimes 1) + (1 \otimes A^T)$$

4. Gradientenverfahren

4.1. Gradientenverfahren für den two link revolute Manipulator

Wir betrachten das Steuerungsproblem

$$\begin{aligned}
\min_{\boldsymbol{u}} \int_{t_0}^{t_f} L(\boldsymbol{x},\boldsymbol{u})\,\mathrm{d}t + \Phi(\boldsymbol{x}(t_f)) \\
L(\boldsymbol{x},\boldsymbol{u}) &= (\boldsymbol{u}-\boldsymbol{u}_f)^T \cdot R \cdot (\boldsymbol{u}-\boldsymbol{u}_f) \\
\Phi(\boldsymbol{x}(t_f)) &= (\boldsymbol{q}(t_f)-\boldsymbol{q}_f)^T \cdot Q \cdot (\boldsymbol{q}(t_f)-\boldsymbol{q}_f) \\
\dot{\boldsymbol{x}} = f(\boldsymbol{x},\boldsymbol{u}) &= \begin{bmatrix} \dot{\boldsymbol{q}} \\ M^{-1}(\boldsymbol{q}) \cdot (-C(\boldsymbol{q},\dot{\boldsymbol{q}}) \cdot \dot{\boldsymbol{q}} - G(\boldsymbol{q}) + \boldsymbol{u}) \end{bmatrix} \\
\boldsymbol{x} = \begin{bmatrix} \boldsymbol{q} \\ \dot{\boldsymbol{q}} \end{bmatrix} \quad \boldsymbol{x}(t_0) &= \begin{bmatrix} \boldsymbol{q}(t_0) \\ \dot{\boldsymbol{q}}(t_0) \end{bmatrix}
\end{aligned}$$

Für die adjungierte Gleichung gilt

$$\dot{\boldsymbol{p}} = -H_{\boldsymbol{x}} = -\begin{bmatrix} \frac{\partial H}{\partial \boldsymbol{q}} \\ \frac{\partial H}{\partial \dot{\boldsymbol{q}}} \end{bmatrix}, \quad \boldsymbol{p}(t_f) = \Phi_{\boldsymbol{x}}(\boldsymbol{x}(t_f)) = \begin{bmatrix} 2Q(\boldsymbol{q}(t_f)-\boldsymbol{q}_f) \\ \boldsymbol{0} \end{bmatrix} \quad H[t] = L(\boldsymbol{u}) + \boldsymbol{p}^T \cdot f(\boldsymbol{x},\boldsymbol{u})$$

Für den Gradienten gilt

$$\nabla J(\boldsymbol{u}) = \frac{\partial H}{\partial \boldsymbol{u}} = 2R \cdot (\boldsymbol{u}-\boldsymbol{u}_f) + \frac{\partial f}{\partial \boldsymbol{u}}^T \cdot \boldsymbol{p}$$

Wir nutzen die spezielle Struktur unseres Problems

$$\frac{\partial f}{\partial \boldsymbol{u}} = \begin{bmatrix} \boldsymbol{0}_2 \\ M^{-1}(\boldsymbol{q}) \end{bmatrix} \Rightarrow \frac{\partial f}{\partial \boldsymbol{u}}^T \cdot \begin{bmatrix} \boldsymbol{p}_{\boldsymbol{q}} \\ \boldsymbol{p}_{\dot{\boldsymbol{q}}} \end{bmatrix} = M^{-1}(\boldsymbol{q}) \cdot \boldsymbol{p}_{\dot{\boldsymbol{q}}} \Rightarrow \nabla J(\boldsymbol{u}) = 2R \cdot (\boldsymbol{u}-\boldsymbol{u}_f) + M^{-1}(\boldsymbol{q}) \cdot \boldsymbol{p}_{\dot{\boldsymbol{q}}}$$

1) Foward Integration der State Equation: $\dot{\boldsymbol{x}} = f(\boldsymbol{x},\boldsymbol{u})$

$$\begin{aligned}
\boldsymbol{x}_0 &= \boldsymbol{x}(t_0) \\
\boldsymbol{x}_{i+1} &= \boldsymbol{x}_i + h \cdot \begin{bmatrix} \dot{\boldsymbol{q}}_i \\ M^{-1}(\boldsymbol{q}_i) \cdot (-C(\boldsymbol{q}_i,\dot{\boldsymbol{q}}_i) \cdot \dot{\boldsymbol{q}}_i - G(\boldsymbol{q}_i) + \boldsymbol{u}_i) \end{bmatrix} \quad i = 0,\ldots,N-1
\end{aligned}$$

Abspeichern Array $\boldsymbol{X} = [\boldsymbol{x}_0, \boldsymbol{x}_1, \ldots, \boldsymbol{x}_N]$

2) Backward Integration der Adjungierten Gleichung: $\dot{\boldsymbol{p}} = -H_{\boldsymbol{x}} = -\frac{\partial f}{\partial \boldsymbol{x}}$

$$\begin{aligned}
\boldsymbol{p}_N = \boldsymbol{p}(t_f) &= \begin{bmatrix} 2Q(\boldsymbol{q}(t_f)-\boldsymbol{q}_f) \\ \boldsymbol{0} \end{bmatrix} \\
\boldsymbol{p}_{i-1} &= \boldsymbol{p}_i + h \cdot \begin{bmatrix} \boldsymbol{0}_2 & I_2 \\ \frac{\partial \ddot{\boldsymbol{q}}}{\partial \boldsymbol{q}} & \frac{\partial \ddot{\boldsymbol{q}}}{\partial \dot{\boldsymbol{q}}} \end{bmatrix} \cdot \boldsymbol{p}_i \\
\frac{\partial \ddot{\boldsymbol{q}}}{\partial \boldsymbol{q}} &= \frac{\partial}{\partial \boldsymbol{q}} \left(M^{-1}(\boldsymbol{q}_i) \cdot (-C(\boldsymbol{q}_i,\dot{\boldsymbol{q}}_i) \cdot \dot{\boldsymbol{q}}_i - G(\boldsymbol{q}_i) + \boldsymbol{u}_i) \right) \\
\frac{\partial \ddot{\boldsymbol{q}}}{\partial \boldsymbol{q}} &= \frac{\partial}{\partial \dot{\boldsymbol{q}}} \left(M^{-1}(\boldsymbol{q}_i) \cdot (-C(\boldsymbol{q}_i,\dot{\boldsymbol{q}}_i) \cdot \dot{\boldsymbol{q}}_i - G(\boldsymbol{q}_i) + \boldsymbol{u}_i) \right)
\end{aligned}$$

2.1) Finite Differenzen für $\frac{\partial \ddot{\boldsymbol{q}}}{\partial x_j}$ $j = 1,2,3,4$

$$\begin{aligned}
\boldsymbol{x}_{plus} &= \boldsymbol{x} + \epsilon \cdot \boldsymbol{e}_j \quad \boldsymbol{x}_{minus} = \boldsymbol{x} - \epsilon \cdot \boldsymbol{e}_j \\
\ddot{\boldsymbol{q}}(\boldsymbol{x}) &= M^{-1}(\boldsymbol{q}) \cdot (-C(\boldsymbol{q},\dot{\boldsymbol{q}}) \cdot \dot{\boldsymbol{q}} - G(\boldsymbol{q}) + \boldsymbol{u}) \\
\frac{\partial \ddot{\boldsymbol{q}}}{\partial x_j} &\approx \frac{\ddot{\boldsymbol{q}}(\boldsymbol{x}_{plus}) - \ddot{\boldsymbol{q}}(\boldsymbol{x}_{minus})}{2\epsilon}
\end{aligned}$$

Abspeichern Array $\boldsymbol{P} = [\boldsymbol{p}_0, \boldsymbol{p}_1, \ldots, \boldsymbol{p}_N]$

3) Gradientenverfahren $\boldsymbol{u}_{i+1} = \boldsymbol{u}_i - \alpha_j \cdot \nabla J(\boldsymbol{u}_i)$

$$\boldsymbol{u}_{i+1} = \boldsymbol{u}_i - \alpha_j \cdot \left[2R \cdot (\boldsymbol{u}_i - \boldsymbol{u}_f) + M^{-1}(\boldsymbol{q}_i) \cdot \boldsymbol{P}_{\dot{\boldsymbol{q}},i}\right] \quad i = 0, \ldots, N-1$$

3.1) Berechnung der Schrittweite α_j über Linesearch

$$\begin{aligned}
\alpha_{test} &= \beta^m \cdot \alpha_0 \\
\boldsymbol{u}_{test} &= \boldsymbol{u}_k - \alpha_{test} \cdot \nabla J(\boldsymbol{u}_k) \\
J_{test} &= J(\boldsymbol{u}_{test}) \\
\text{IF } J_{test} &\leq J(\boldsymbol{u}_k) - c \cdot \alpha_{test} \cdot \|\nabla J(\boldsymbol{u}_k)\|^2 : \\
\alpha_k &= \alpha_{test} \\
&\Rightarrow \text{Break}
\end{aligned}$$

5. Reinforcement Learning in Optimal Control

Wir folgen den Notationen aus [6] Für eine gegebene Trajektorie τ des MDPs gilt:

$$\tau_t^n := \{\boldsymbol{s}_t, a_t, r_t, \boldsymbol{s}_{t+1}, \cdots, \boldsymbol{s}_{t+n}, a_{t+n}, r_{t+n}, \boldsymbol{s}_{t+n+1}\} = \left\{\boldsymbol{s}_{t+i}, a_{t+i}, r_{t+i}, \boldsymbol{s}_{t+i+1}\right\}_{i=0,\dots,n}$$

In einem unendlichen Zeithorizont notieren wir $\tau_t^\infty = \tau_t$. Für die Verteilung der Trajektorien gilt:

$$p(\tau_0 \mid \pi) = p(\boldsymbol{s}_0) \cdot \prod_{t=0}^{\infty} \pi(a_t|\boldsymbol{s}_t) \cdot T_{a_t}(\boldsymbol{s}_t, \boldsymbol{s}_{t+1})$$

Für das Zielfunktional definieren wir:

Definition 5.1: Objective Function. Wir definieren die Objective Function als

$$J(\pi) := V^\pi(\boldsymbol{s}_0) = \mathsf{E}_{\tau_0 \sim p(\tau_0|\pi)}\left[R(\tau_0)\right] \Rightarrow \pi^*(a|\boldsymbol{s}) = \arg\max_\pi V^\pi(\boldsymbol{s}_0)$$

5.1. Policy-Based Algorithms

Wir betrachten nun nachfolgend eine Familie von parametrisierten Policies $\pi_\theta(a|\boldsymbol{s})$ mit den Parametern $\theta \in \mathbb{R}^d$.

	Unendlichdimensionales Problem	Parametrisiertes Problem
Policy	$\pi : \mathcal{S} \to p(\mathcal{A})$	$\theta \to \pi_\theta$ mit $\pi_\theta(a\|\boldsymbol{s})$
$J(\pi)$	$J(\pi) = \mathsf{E}_{\tau\sim p(\tau\|\pi)}\left[R(\tau)\right]$	$J(\theta) = \mathsf{E}_{\tau\sim p(\tau\|\theta)}\left[R(\tau)\right]$
$p(\tau)$	$p(\tau \mid \pi) = p(\boldsymbol{s}_0) \cdot \prod_{t=0}^{T-1} \pi(a_t\|\boldsymbol{s}_t) \cdot T_{a_t}(\boldsymbol{s}_t, \boldsymbol{s}_{t+1})$	$p(\tau \mid \theta) = p(\boldsymbol{s}_0) \cdot \prod_{t=0}^{T-1} \pi_\theta(a_t\|\boldsymbol{s}_t) \cdot T_{a_t}(\boldsymbol{s}_t, \boldsymbol{s}_{t+1})$
∇J	$\nabla J(\pi) = \sum_{t=0}^{T-1} p(\boldsymbol{s}_t = \boldsymbol{s}) \cdot Q^\pi(\boldsymbol{s}, a)$	$\nabla J(\theta) = \mathsf{E}_{\tau\sim p(\tau\|\theta)}\left[\sum_{t=0}^{T-1} Q^{\pi_\theta}(\boldsymbol{s}_t, a_t) \cdot \nabla_\theta \log \pi_\theta(a_t\|\boldsymbol{s}_t)\right]$

Satz 5.2. Policy Gradient Theorem Für die Objective Function $J(\theta)$ gilt:

$$\nabla J(\theta) = \mathsf{E}_{\tau\sim p(\tau|\theta)}\left[\sum_{t=0}^{T-1} Q^{\pi_\theta}(\boldsymbol{s}_t, a_t) \cdot \nabla_\theta \log \pi_\theta(a_t|\boldsymbol{s}_t)\right]$$

Beweis:

- Direkte Ableitung der Objective Function:

$$\nabla_\theta J(\theta) = \int R(\tau) \cdot \nabla_\theta p(\tau|\theta) \mathrm{d}\tau = \int R(\tau) \cdot p(\tau|\theta) \cdot \underbrace{\frac{\nabla_\theta p(\tau|\theta)}{p(\tau|\theta)}}_{\nabla_\theta \log p(\tau|\theta)} \mathrm{d}\tau = \mathsf{E}_{\tau\sim p(\tau|\theta)}\left[R(\tau) \cdot \nabla_\theta \log p(\tau|\theta)\right]$$

- Wir berechnen nun $\nabla_\theta \log p(\tau|\theta)$:

$$\begin{aligned}\nabla_\theta \log p(\tau|\theta) &= \nabla_\theta \log \left[p(\boldsymbol{s}_0) \cdot \prod_{t=0}^{T-1} \pi_\theta(a_t|\boldsymbol{s}_t) \cdot T_{a_t}(\boldsymbol{s}_t, \boldsymbol{s}_{t+1}) \right] \\ &= \nabla_\theta \left[\log p(\boldsymbol{s}_0) + \sum_{t=0}^{T-1} \log \pi_\theta(a_t|\boldsymbol{s}_t) + \log T_{a_t}(\boldsymbol{s}_t, \boldsymbol{s}_{t+1}) \right] \\ &= \sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|\boldsymbol{s}_t)\end{aligned}$$

- Anwendung der Tower Property

$$\begin{aligned}\mathsf{E}_{\tau\sim p(\tau|\theta)} \left[R(\tau) \cdot \underbrace{\nabla_\theta \log p(\tau|\theta)}_{=\sum_{t=0}^{T-1} \log \pi_\theta(a_t|\boldsymbol{s}_t)} \right] &= \sum_{t=0}^{T-1} \mathsf{E}_{\tau\sim p(\tau|\theta)} \left[R(\tau) \cdot \nabla_\theta \log \pi_\theta(a_t|\boldsymbol{s}_t) \right] \\ &= \sum_{t=0}^{T-1} \mathsf{E}_{\tau\sim p(\tau|\theta)} \left[\mathsf{E}_{\tau\sim p(\tau|\theta)} \left[R(\tau) \cdot \underbrace{\nabla_\theta \log \pi_\theta(a_t|\boldsymbol{s}_t)}_{\sigma(a_t,s_t)\text{- messbar}} \mid \boldsymbol{s}_t, a_t \right] \right] \\ &= \sum_{t=0}^{T-1} \mathsf{E}_{\tau\sim p(\tau|\theta)} \left[\nabla_\theta \log \pi_\theta(a_t|\boldsymbol{s}_t) \cdot \mathsf{E}_{\tau\sim p(\tau|\theta)} \left[R(\tau) \mid \boldsymbol{s}_t, a_t \right] \right]\end{aligned}$$

- Wir Zerlegen die Reward Funktion in vergangene und zukünftige Rewards:

$$R(\tau) := \sum_{k=0}^{T-1} \gamma^k r_{k+1}(a_k, \boldsymbol{s}_k) = \underbrace{\sum_{k=0}^{t-1} \gamma^k r_{k+1}(a_k, \boldsymbol{s}_k)}_{\text{Past Rewards}} + \underbrace{\sum_{k=t}^{T-1} \gamma^k r_{k+1}(a_k, \boldsymbol{s}_k)}_{\text{Future Rewards}} = R_{\text{past}}(t) + R_t$$

- Wir nutzen die Unabhängigkeit der vergangenen Rewards von a_t:

$$\mathsf{E}_{\tau\sim p(\tau|\theta)} \left[\nabla_\theta \log \pi_\theta(a_t|\boldsymbol{s}_t) \cdot \left(\underbrace{\mathsf{E}_{\tau\sim p(\tau|\theta)} \left[R_{\text{past}}(t) \mid \boldsymbol{s}_t, a_t \right]}_{\mathsf{E}_{\tau\sim p(\tau|\theta)}[R_{\text{past}}(t)|\boldsymbol{s}_t]} + \mathsf{E}_{\tau\sim p(\tau|\theta)} \left[R_t \mid \boldsymbol{s}_t, a_t \right] \right) \right] := \mathsf{E}_{\tau\sim p(\tau|\theta)}[Y]$$

- Die Zufallsvariable Y hängt ausschließlich von $\boldsymbol{s}_t$ und a_t ab. Die Vertelung von τ ist ein Produkt der Transitionen und der Policy also gilt

$$\mathsf{E}_{\tau\sim p(\tau|\theta)}[Y] = \mathsf{E}_{\boldsymbol{s}_t\sim p(\boldsymbol{s}_t|\theta)} \left[\mathsf{E}_{a_t\sim\pi_\theta(a_t|\boldsymbol{s}_t)}[Y] \right] \quad p(\boldsymbol{s}_t \mid \theta) = \sum_{\boldsymbol{s}_0,\cdots,\boldsymbol{s}_{t-1}} \sum_{a_0,\cdots,a_{t-1}} p(\tau|\theta)$$

- Wir erhalten nun für den Term mit den vergangenen Rewards $a_t \sim \pi_\theta(a_t|\boldsymbol{s}_t)$:

$$\mathsf{E}_{a_t} \left[\nabla_\theta \log \pi_\theta(a_t|\boldsymbol{s}_t) \right] = \int_{\mathcal{A}} \pi_\theta(a|\boldsymbol{s}_t) \cdot \underbrace{\nabla_\theta \log \pi_\theta(a|\boldsymbol{s}_t)}_{\frac{\nabla_\theta \pi_\theta(a|\boldsymbol{s}_t)}{\pi_\theta(a|\boldsymbol{s}_t)}} \mathrm{d}a = \nabla_\theta \underbrace{\int_{\mathcal{A}} \pi_\theta(a|\boldsymbol{s}_t) \mathrm{d}a}_{=1} = 0$$

$$\Rightarrow \mathsf{E}_{a_t} \left[\nabla_\theta \log \pi_\theta(a_t|\boldsymbol{s}_t) \cdot \mathsf{E}_{\tau\sim p(\tau|\theta)} \left[R_{\text{past}}(t) \mid \boldsymbol{s}_t \right] \right] = \mathsf{E}_{\tau\sim p(\tau|\theta)} \left[R_{\text{past}}(t) \mid \boldsymbol{s}_t \right] \cdot \underbrace{\mathsf{E}_{a_t} \left[\nabla_\theta \log \pi_\theta(a_t|\boldsymbol{s}_t) \right]}_{=0} = 0$$

- Somit verbleibt nur der Term mit den zukünftigen Rewards:

$$\nabla_\theta J(\theta) = \underbrace{\sum_{t=0}^{T-1} \mathsf{E}_{\boldsymbol{s}_t \sim p(\boldsymbol{s}_t|\theta)} \left[\mathsf{E}_{a_t \sim \pi_\theta(a_t|\boldsymbol{s}_t)} \left[\nabla_\theta \log \pi_\theta(a_t|\boldsymbol{s}_t) \cdot \underbrace{\mathsf{E}_{\tau \sim p(\tau|\theta)} \left[R_t \mid \boldsymbol{s}_t, a_t \right]}_{=Q^{\pi_\theta}(\boldsymbol{s}_t, a_t)} \right] \right]}_{\mathsf{E}_{\tau \sim p(\tau|\theta)} \left[\nabla_\theta \log \pi_\theta(a_t|\boldsymbol{s}_t) \cdot Q^{\pi_\theta}(\boldsymbol{s}_t, a_t) \right]}$$

□

Satz 5.3. Policy Gradient Theorem Für die Objective Function $J(\pi)$ gilt:

$$\nabla_\pi J(\pi) = \sum_{t=0}^{T-1} p(\boldsymbol{s}_t = \boldsymbol{s}) \cdot Q^\pi(\boldsymbol{s}, a_t)$$

Beweis:

- Wir nutzen den Variationsansatz $\pi_\epsilon = \pi + \epsilon \cdot h$

$$\frac{d}{d\epsilon} J(\pi_\epsilon) \mid_{\epsilon=0} = \int R(\tau) \cdot \underbrace{p(\tau|\pi_\epsilon) \cdot \frac{d}{d\epsilon} \log p(\tau|\pi_\epsilon)}_{\frac{d}{d\epsilon} p(\tau|\pi_\epsilon)} \mathrm{d}\tau \mid_{\epsilon=0} = \underbrace{\int R(\tau) \cdot p(\tau|\pi) \cdot \frac{d}{d\epsilon} \log p(\tau|\pi_\epsilon) \mid_{\epsilon=0} \mathrm{d}\tau}_{\mathsf{E}_{\tau \sim p(\tau|\pi)} \left[R(\tau) \cdot \frac{d}{d\epsilon} \log p(\tau|\pi_\epsilon)|_{\epsilon=0} \right]}$$

- Wir berechnen nun $\frac{d}{d\epsilon} \log p(\tau|\pi_\epsilon)$:

$$\begin{aligned}
\frac{d}{d\epsilon} \log p(\tau|\pi_\epsilon) &= \frac{d}{d\epsilon} \log \left[p(\boldsymbol{s}_0) \cdot \prod_{t=0}^{T-1} \pi_\epsilon(a_t|\boldsymbol{s}_t) \cdot T_{a_t}(\boldsymbol{s}_t, \boldsymbol{s}_{t+1}) \right] \\
&= \frac{d}{d\epsilon} \left[\log p(\boldsymbol{s}_0) + \sum_{t=0}^{T-1} \log \pi_\epsilon(a_t|\boldsymbol{s}_t) + \log T_{a_t}(\boldsymbol{s}_t, \boldsymbol{s}_{t+1}) \right] \\
&= \sum_{t=0}^{T-1} \frac{d}{d\epsilon} \log \left(\pi(a_t|\boldsymbol{s}_t) + \epsilon \cdot h(a_t|\boldsymbol{s}_t) \right) \\
&= \sum_{t=0}^{T-1} \frac{h(a_t|\boldsymbol{s}_t)}{\pi(a_t|\boldsymbol{s}_t) + \epsilon \cdot h(a_t|\boldsymbol{s}_t)} \overset{\epsilon=0}{=} \sum_{t=0}^{T-1} \frac{h(a_t|\boldsymbol{s}_t)}{\pi(a_t|\boldsymbol{s}_t)}
\end{aligned}$$

- Wir erhalten somit analog zum parametrisierten Fall:

$$\frac{d}{d\epsilon} J(\pi_\epsilon) \mid_{\epsilon=0} = \mathsf{E}_{\tau \sim p(\tau|\pi)} \left[R(\tau) \cdot \sum_{t=0}^{T-1} \frac{h(a_t|\boldsymbol{s}_t)}{\pi(a_t|\boldsymbol{s}_t)} \right] = \sum_{t=0}^{T-1} \mathsf{E}_{\tau \sim p(\tau|\pi)} \left[R(\tau) \cdot \frac{h(a_t|\boldsymbol{s}_t)}{\pi(a_t|\boldsymbol{s}_t)} \right]$$

- Für jeden festen Zeitschritt t verwenden wir die Turmeigenschaft:

$$\mathsf{E}_{\tau \sim p(\tau|\pi)} \left[R(\tau) \cdot \frac{h(a_t|\boldsymbol{s}_t)}{\pi(a_t|\boldsymbol{s}_t)} \right] = \mathsf{E}_{\tau \sim p(\tau|\pi)} \left[\mathsf{E}_{\tau \sim p(\tau|\boldsymbol{s}_t, a_t)} \left[\underbrace{(R_{\text{past}}(t) + R_{\text{future}}(t))}_{R(\tau)} \cdot \underbrace{\frac{h(a_t|\boldsymbol{s}_t)}{\pi(a_t|\boldsymbol{s}_t)}}_{\sigma(\boldsymbol{s}_t, a_t)\text{- messbar}} \mid \boldsymbol{s}_t, a_t \right] \right]$$

- Da π_ϵ eine Dichtefunktion ist, gilt ist das Integral für jedes feste $\boldsymbol{s}_t$ gleich 1:

$$1 = \int_{\mathcal{A}} \pi(a|\boldsymbol{s}_t) + \epsilon \cdot h(a|\boldsymbol{s}_t) \mathrm{d}a = \underbrace{\int_{\mathcal{A}} \pi(a|\boldsymbol{s}_t) \mathrm{d}a}_{=1} + \epsilon \cdot \underbrace{\int_{\mathcal{A}} h(a|\boldsymbol{s}_t) \mathrm{d}a}_{=0}$$

$$\Rightarrow \mathsf{E}_{a_t \sim \pi(a_t|\boldsymbol{s}_t)} \left[\frac{h(a_t|\boldsymbol{s}_t)}{\pi(a_t|\boldsymbol{s}_t)} \right] = \int_{\mathcal{A}} \underbrace{\pi(a|\boldsymbol{s}_t) \cdot \frac{h(a|\boldsymbol{s}_t)}{\pi(a|\boldsymbol{s}_t)}}_{=h(a|\boldsymbol{s}_t)} \mathrm{d}a = 0$$

$$\Rightarrow \mathsf{E}_{a_t \sim \pi(a_t|\boldsymbol{s}_t)} \left[\frac{h(a_t|\boldsymbol{s}_t)}{\pi(a_t|\boldsymbol{s}_t)} \cdot \underbrace{\mathsf{E}_{\tau \sim p(\tau|\pi)} \left[R_{\text{past}}(t) \mid \boldsymbol{s}_t \right]}_{\text{unabhängig von } a_t} \right] = \mathsf{E}_{\tau \sim p(\tau|\pi)} \left[R_{\text{past}}(t) \mid \boldsymbol{s}_t \right] \cdot \mathsf{E}_{a_t \sim \pi(a_t|\boldsymbol{s}_t)} \left[\frac{h(a_t|\boldsymbol{s}_t)}{\pi(a_t|\boldsymbol{s}_t)} \right] = 0$$

- Wie isolieren den Gradienten aus der Richtungsableitung:

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}\epsilon} J(\pi_\epsilon) \mid_{\epsilon=0} &= \sum_{t=0}^{T-1} \mathsf{E}_{\boldsymbol{s}_t \sim p(\boldsymbol{s}_t|\pi)} \left[\mathsf{E}_{a_t \sim \pi(a_t|\boldsymbol{s}_t)} \left[\frac{h(a_t|\boldsymbol{s}_t)}{\pi(a_t|\boldsymbol{s}_t)} \cdot \underbrace{\mathsf{E}_{\tau \sim p(\tau|\pi)} \left[R_t \mid \boldsymbol{s}_t, a_t \right]}_{=Q^\pi(\boldsymbol{s}_t, a_t)} \right] \right] \\
&= \sum_{t=0}^{T-1} \mathsf{E}_{\boldsymbol{s}_t \sim p(\boldsymbol{s}_t|\pi)} \left[\int_{\mathcal{A}} \pi(a|\boldsymbol{s}_t) \cdot \frac{h(a|\boldsymbol{s}_t)}{\pi(a|\boldsymbol{s}_t)} \cdot Q^\pi(\boldsymbol{s}_t, a) \mathrm{d}a \right] \\
&= \int_{\mathcal{S}} \int_{\mathcal{A}} h(a|\boldsymbol{s}) \cdot \underbrace{\sum_{t=0}^{T-1} p(\boldsymbol{s}_t = \boldsymbol{s}) \cdot Q^\pi(\boldsymbol{s}, a)}_{\nabla_\pi J(\pi(a|\boldsymbol{s}))} \mathrm{d}a \mathrm{d}\boldsymbol{s}
\end{aligned}$$

- Bilden der Ableitung nach θ ergibt:

$$\begin{aligned}
\nabla_\theta J(\pi_\theta) &= \int_{\mathcal{S}} \int_{\mathcal{A}} \underbrace{\sum_{t=0}^{T-1} p(\boldsymbol{s}_t = \boldsymbol{s}) \cdot Q^{\pi_\theta}(\boldsymbol{s}, a)}_{\nabla_\pi J(\pi_\theta(a|\boldsymbol{s}))} \cdot \underbrace{\pi_\theta(a|\boldsymbol{s}) \cdot \nabla_\theta \log \pi_\theta(a|\boldsymbol{s})}_{\nabla_\theta \pi_\theta(a|\boldsymbol{s})} \mathrm{d}a \, \mathrm{d}\boldsymbol{s} \\
&= \sum_{t=0}^{T-1} \int_{\mathcal{S}} \int_{\mathcal{A}} \underbrace{\mathsf{E}_{\tau \sim p(\tau|\theta)} \left[R_t \mid \boldsymbol{s}_t, a_t \right]}_{Q^{\pi_\theta}(\boldsymbol{s}, a)} \cdot \underbrace{p(\boldsymbol{s}_t = \boldsymbol{s}) \cdot \pi_\theta(a|\boldsymbol{s}) \cdot \nabla_\theta \log \pi_\theta(a|\boldsymbol{s})}_{\sigma(\boldsymbol{s}_t, a_t)\text{- messbar}} \mathrm{d}a \, \mathrm{d}\boldsymbol{s} \\
&= \sum_{t=0}^{T-1} \mathsf{E}_{\boldsymbol{s}_t \sim p(\boldsymbol{s}_t|\theta)} \left[\mathsf{E}_{a \sim \pi_\theta(a|\boldsymbol{s}_t)} \left[Q^{\pi_\theta}(\boldsymbol{s}_t, a_t) \cdot \nabla_\theta \log \pi_\theta(a_t|\boldsymbol{s}_t) \right] \right] \\
&= \mathsf{E}_{\tau \sim p(\tau|\theta)} \left[\sum_{t=0}^{T-1} Q^{\pi_\theta}(\boldsymbol{s}_t, a_t) \cdot \nabla_\theta \log \pi_\theta(a_t|\boldsymbol{s}_t) \right]
\end{aligned}$$

□

Algorithm 1 Policy Gradient Algorithm - Monte Carlo [6, p.110]

Differentiable Policy π_θ, Learning Rate $\alpha > 0$, threshold $\epsilon > 0$
Initialize parameters $\theta \in \mathbb{R}^d$ arbitrarily
while $\|\nabla_\theta J(\pi_\theta)\|_2 > \epsilon$ **do**
  Generate Trace τ following π_θ
  for $t \in 0, \cdots, T-1$ **do**
    $R_t \leftarrow \sum_{k=t}^{T-1} \gamma^{k-t} r_k$ ▷ $R_t \approx Q^{\pi_\theta}(\boldsymbol{s}_t, a_t)$
    $\theta \leftarrow \theta + \alpha \cdot R_t \cdot \nabla_\theta \log \pi_\theta(a_t|\boldsymbol{s}_t)$ ▷ $\nabla_\theta J(\theta) \approx \frac{1}{M} \sum_{i=1}^{M} \sum_{t=0}^{T-1} R_t^{(i)} \nabla_\theta \log \pi_\theta(a_t^{(i)}|\boldsymbol{s}_t^{(i)})$
  end for
end while

Beispiel 5.4: Optimal Control Problem. Wir betrachten das optimale Steuerungsproblem:

$$\min_u \int_0^1 x(t) + u(t)^2 \, \mathrm{d}t + x(1)^2 \quad \dot{x} = x(t) + u(t) + 1, \quad x(0) = 0$$

1. Diskretisierung mit dem expliziten Euler-Verfahren

$$\min_u \int_0^1 x(t) + u(t)^2 \, \mathrm{d}t + x(1)^2 \approx \max_u -\frac{1}{N} \sum_{i=0}^{N-1} x_i + u_i^2 - x_N^2$$

2. Formulierung als MDP

$$a_t \sim \pi_\theta(a_t|s_t) = \mathcal{N}\left(\mu(t,\theta), \sigma^2\right) \Rightarrow P\left[a_t \in A\right] = \int_A \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(x - \mu(t,\theta))^2}{2\sigma^2}\right] \mathrm{d}x$$

3. Formulierung des Policy Gradient Algorithmus:

$$\nabla J(\theta) \approx \frac{1}{M} \sum_{i=1}^{M} \sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t^{(i)}|\boldsymbol{s}_t^{(i)}) \cdot R_t^{(i)} = \frac{1}{M} \sum_{i=1}^{M} \sum_{t=0}^{T-1} \frac{a_t^{(i)} - \mu_\theta}{\sigma^2} \cdot \frac{d\mu_\theta}{d\theta} \cdot R_t^{(i)}$$

4. Wahl des Lösungsraums: $\mu(t,\theta) = \theta_0 \cdot e^{\theta_1 \cdot t} + \theta_2$ und $u(t) = a_t$ mit $M = 1$

$$\frac{\mathrm{d}\mu_\theta}{\mathrm{d}\theta} = \begin{bmatrix} e^{\theta_1 \cdot t} \\ \theta_0 \cdot t \cdot e^{\theta_1 \cdot t} \\ 1 \end{bmatrix} \Rightarrow \nabla J(\theta) \approx \sum_{t=0}^{T-1} \frac{a_t - \mu_\theta}{\sigma^2} \cdot \begin{bmatrix} e^{\theta_1 \cdot t} \\ \theta_0 \cdot t \cdot e^{\theta_1 \cdot t} \\ 1 \end{bmatrix} \cdot R_t$$

5. Simulation basierend auf einem Startwert $\theta = (0,0,0)^T$ und der Environment:

$$\mu(t_i,\theta) = \theta_0 \cdot e^{\theta_1 \cdot t_i} + \theta_2 \Rightarrow u(t_i) \sim \mathcal{N}(\mu(t_i,\theta), \sigma^2)$$
$$r_i = -\frac{1}{N}(x_i + u_i^2), r_N = -x_N^2 \Rightarrow R_i = \sum_{k=i}^{N} r_k$$
$$\begin{bmatrix} x_{i+1} \\ t_{i+1} \end{bmatrix} = \begin{bmatrix} x_i + \Delta t \cdot (x_i + u_i + 1) \\ t_i + \Delta t \end{bmatrix}, \quad \begin{bmatrix} x_0 \\ t_0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{Environment}$$

6. Formulierung des Gradientenverfahrens:

$$\theta_{k+1} = \theta_k + \alpha_k \cdot \nabla J(\theta_k)$$
$$\sigma \leftarrow \sigma \cdot 0.99998 \quad \text{Exploration zu Exploitation}$$

5.2. Neuronal Networks

In der einfachsten Form betrachten wir ein Hidden-Layer $\boldsymbol{W}_1$ und ein Outer-Layer $\boldsymbol{W}_N$ mit dem Bias $\boldsymbol{b}_1$ und $\boldsymbol{b}_N$. Wir betrachten zunächst die Approximation einer Value-Function $V : [0, T] \times \mathbb{R} \to \mathbb{R}$.

$$\boldsymbol{W}_1 = \begin{bmatrix} w_{11}^{(1)} & w_{12}^{(1)} \\ w_{21}^{(1)} & w_{22}^{(1)} \\ \vdots & \vdots \\ w_{n1}^{(1)} & w_{n2}^{(1)} \end{bmatrix} \in \mathbb{R}^{n \times 2} \quad \boldsymbol{b}_1 = \begin{bmatrix} b_1^{(1)} \\ b_2^{(1)} \\ \vdots \\ b_n^{(1)} \end{bmatrix} \in \mathbb{R}^n$$

Wir verwenden als Aktivierungsfunktion $\tanh : \mathbb{R} \to (-1, 1)$:

$$\boldsymbol{z}_1 = \boldsymbol{W}_1 \cdot \begin{bmatrix} t \\ x \end{bmatrix} + \boldsymbol{b}_1 = \begin{bmatrix} w_{11}^{(1)} t + w_{12}^{(1)} x + b_1^{(1)} \\ w_{21}^{(1)} t + w_{22}^{(1)} x + b_2^{(1)} \\ \vdots \\ w_{n1}^{(1)} t + w_{n2}^{(1)} x + b_n^{(1)} \end{bmatrix}$$

$$\boldsymbol{a}_1 = \tanh(\boldsymbol{z}_1) = \tanh\left(\boldsymbol{W}_1 \cdot \begin{bmatrix} t \\ x \end{bmatrix} + \boldsymbol{b}_1 \right)$$

Für das Outer-Layer gilt:

$$\boldsymbol{W}_N = \begin{bmatrix} w_1^{(N)} & w_2^{(N)} & \dots & w_n^{(N)} \end{bmatrix} \in \mathbb{R}^{1 \times n}, \quad b_N \in \mathbb{R}$$

$$\boldsymbol{z}_N = \boldsymbol{W}_N \cdot \boldsymbol{a}_1 + b_N = \sum_{j=1}^{n} w_j^{(N)} a_j^{(1)} + b_N$$

Für die Darstellung von V durch $\boldsymbol{z}_2$ benötigt man $4n + 1$ Parameter:

$$V_\phi(t, x) = \underbrace{\sum_{j=1}^{n} w_j^{(N)} \cdot \tanh\overbrace{\left(w_{j1}^{(1)} \cdot t + w_{j2}^{(1)} \cdot x + b_j^{(1)} \right)}^{\text{Neuron } j \text{ im Hidden-Layer}} + b_2}_{\text{Output-Layer}}$$

$$\text{Parameter } \phi = \{ w_{j1}^{(1)}, w_{j2}^{(1)}, b_j^{(1)}, w_j^{(N)}, b^{(N)} \}, \quad j = 1, \dots, n$$

Als nächstes Betrachten wir ein Neuronales Netz mit zwei Hidden-Layer $\boldsymbol{W}_1, \boldsymbol{W}_2$ und den zugehörigen Bias $\boldsymbol{b}_1, \boldsymbol{b}_2$.

$$\underbrace{\boldsymbol{W}_1 = \begin{bmatrix} w_{11}^{(1)} & w_{12}^{(1)} \\ w_{21}^{(1)} & w_{22}^{(1)} \\ \vdots & \vdots \\ w_{n_1 1}^{(1)} & w_{n_1 2}^{(1)} \end{bmatrix}}_{\in \mathbb{R}^{n_1 \times 2}} \quad \underbrace{\boldsymbol{b}_1 = \begin{bmatrix} b_1^{(1)} \\ b_2^{(1)} \\ \vdots \\ b_{n_1}^{(1)} \end{bmatrix}}_{\in \mathbb{R}^{n_1}} \quad \underbrace{\boldsymbol{W}_2 = \begin{bmatrix} w_{11}^{(2)} & \cdots & w_{1 n_1}^{(2)} \\ w_{21}^{(2)} & \cdots & w_{2 n_1}^{(2)} \\ \vdots & \ddots & \vdots \\ w_{n_2 1}^{(2)} & \cdots & w_{n_2 n_1}^{(2)} \end{bmatrix}}_{\in \mathbb{R}^{n_2 \times n_1}} \quad \underbrace{\boldsymbol{b}_2 = \begin{bmatrix} b_1^{(2)} \\ b_2^{(2)} \\ \vdots \\ b_{n_2}^{(2)} \end{bmatrix}}_{\in \mathbb{R}^{n_2}}$$

Für die Struktur basierend auf einem zweidimensionalen Input lautet somit:

$$\begin{aligned} \boldsymbol{z}_1 &= \boldsymbol{W}_1 \cdot \begin{bmatrix} t \\ x \end{bmatrix} + \boldsymbol{b}_1 \\ \boldsymbol{a}_1 &= \tanh(\boldsymbol{z}_1) \\ \boldsymbol{z}_2 &= \boldsymbol{W}_2 \cdot \boldsymbol{a}_1 + \boldsymbol{b}_2 = \boldsymbol{W}_2 \cdot \tanh\left(\boldsymbol{W}_1 \cdot \begin{bmatrix} t \\ x \end{bmatrix} + \boldsymbol{b}_1 \right) + \boldsymbol{b}_2 \\ \boldsymbol{a}_2 &= \tanh(\boldsymbol{z}_2) = \tanh\left[\boldsymbol{W}_2 \cdot \tanh\left(\boldsymbol{W}_1 \cdot \begin{bmatrix} t \\ x \end{bmatrix} + \boldsymbol{b}_1 \right) + \boldsymbol{b}_2 \right] \end{aligned}$$

Das Outer-Layer und der Output besitzt erneut folgende Struktur

$$\boldsymbol{W}_N = \begin{bmatrix} w_1^{(N)} & w_2^{(N)} & \dots & w_{n_2}^{(N)} \end{bmatrix} \in \mathbb{R}^{1\times n_2}, \quad b_N \in \mathbb{R}$$

$$V = \boldsymbol{z}_N = \boldsymbol{W}_N \cdot \boldsymbol{a}_2 + b_N = \sum_{j=1}^{n_2} w_j^{(N)} \boldsymbol{a}_j^{(2)} + b_N$$

Definition 5.5: Neuronales Netz. Bezeichne $V : \mathbb{R}^{\text{Input}} \to \mathbb{R}$ mit

- Eingabe $\boldsymbol{x} = \boldsymbol{a}^{(0)}$
- für $k = 1, \cdots, L$

$$\boldsymbol{z}^{(k)} = \boldsymbol{W}^{(k)} \cdot \boldsymbol{a}^{(k-1)} + \boldsymbol{b}^{(k)} \quad \boldsymbol{a}^{(k)} = \tanh\left(\boldsymbol{z}^{(k)}\right)$$

- Ausgabe $V = \boldsymbol{W}^N \cdot \boldsymbol{a}^{(L)} + \boldsymbol{b}^{(N)}$

als reellwertiges Neuronales Netz mit L Hidden-Layers.

Satz 5.6: Backpropagation. Sei $V : \mathbb{R}^{\text{Input}} \to \mathbb{R}$ ein Neuronales Netz mit L Hidden-Layers. Für das Output-Layer mit $N = L + 1$ gilt:

$$\frac{\partial V}{\partial \boldsymbol{W}^{(N)}} = \left(\boldsymbol{a}^{(L)}\right)^T \quad \frac{\partial V}{\partial \boldsymbol{b}^{(N)}} = 1 \quad \frac{\partial V}{\partial \boldsymbol{a}^{(L)}} = \boldsymbol{W}^{(N)}$$

Für die Fehlersignale gilt:

$$\boldsymbol{\delta}^{(L)} = \boldsymbol{W}^{(N)} \circ \left[1 - \left(\boldsymbol{a}^{(L)}\right)^2\right] \quad \boldsymbol{\delta}^{(k)} = \left(\boldsymbol{\delta}^{(k+1)} \cdot \boldsymbol{W}^{(k+1)}\right) \circ \left[1 - \left(\boldsymbol{a}^{(k)}\right)^2\right], \quad k = L-1, \cdots, 1$$

Für $k = L, L-1 \cdots, 1$ erhalten wir die Ableitungen nach den Parametern:

$$\frac{\partial V}{\partial \boldsymbol{W}^{(k)}} = \boldsymbol{\delta}^{(k)} \otimes \boldsymbol{a}^{(k-1)} = \left(\boldsymbol{\delta}^{(k)}\right)^T \cdot \left(\boldsymbol{a}^{(k-1)}\right)^T \quad \boldsymbol{a}^{(0)} = \boldsymbol{x}$$
$$\frac{\partial V}{\partial \boldsymbol{b}^{(k)}} = \boldsymbol{\delta}^{(k)}$$

Beweis: Es gilt mit der Kettenregel:

$$\frac{\partial V}{\partial \boldsymbol{W}^{(k)}} = \frac{\partial V}{\partial \boldsymbol{z}^{(k)}} \cdot \frac{\partial \boldsymbol{z}^{(k)}}{\partial \boldsymbol{W}^{(k)}}$$

Wir bezeichnen $\boldsymbol{\delta}^{(k)} := \frac{\partial V}{\partial \boldsymbol{z}^{(k)}}$ als das Fehlersignal in Schicht k und erhalten wieder mittels Kettenregel:

$$\boldsymbol{\delta}^{(k)} = \frac{\partial V}{\partial \boldsymbol{a}^{(k)}} \cdot \frac{\partial \boldsymbol{a}^{(k)}}{\partial \boldsymbol{z}^{(k)}}$$

Für unsere Aktivierungsfunktion $\boldsymbol{a}^{(k)} = \tanh(\boldsymbol{z}^{(k)})$ gilt komponentenweise

$$\frac{\partial \boldsymbol{a}_i^{(k)}}{\partial \boldsymbol{z}_j^{(k)}} = \frac{\partial \tanh\left(\boldsymbol{z}_i^{(k)}\right)}{\partial \boldsymbol{z}_j^{(k)}} = \begin{cases} 1 - \left(\boldsymbol{a}_i^{(k)}\right)^2, & i = j \\ 0 & i \neq j \end{cases}$$

Die Jacobi-Matrix ist somit eine Diagonalmatrix

$$\frac{\partial \boldsymbol{a}^{(k)}}{\partial \boldsymbol{z}^{(k)}} = \left[\frac{\partial \tanh\left(\boldsymbol{z}_i^{(k)}\right)}{\partial \boldsymbol{z}_j^{(k)}}\right]_{i,j=1,\cdots,n_k} = \underbrace{\begin{bmatrix} 1-\left(\boldsymbol{a}_1^{(k)}\right)^2 & 0 & \cdots & 0 \\ 0 & 1-\left(\boldsymbol{a}_2^{(k)}\right)^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1-\left(\boldsymbol{a}_{n_k}^{(k)}\right)^2 \end{bmatrix}}_{\mathsf{diag}\left[1-\left(\boldsymbol{a}_i^{(k)}\right)^2\right]_{i=1,\cdots,n_k}}$$

Für die Ausgabe $V = \boldsymbol{W}^{(N)} \cdot \boldsymbol{a}^{(L)} + b_N$ gilt:

$$\frac{\partial V}{\partial \boldsymbol{W}^{(N)}} = \left(\boldsymbol{a}^{(L)}\right)^T \quad \frac{\partial V}{\partial \boldsymbol{a}^{(L)}} = \boldsymbol{W}^{(N)}$$

Für Schicht L gilt entsprechend:

$$\frac{\partial V}{\partial \boldsymbol{W}^{(L)}} = \boldsymbol{\delta}^{(L)} \cdot \frac{\partial \boldsymbol{z}^{(L)}}{\partial \boldsymbol{W}^{(L)}} \quad \boldsymbol{\delta}^{(L)} = \underbrace{\frac{\partial V}{\partial \boldsymbol{a}^{(L)}}}_{\boldsymbol{W}^{(N)}} \cdot \underbrace{\frac{\partial \boldsymbol{a}^{(L)}}{\partial \boldsymbol{z}^{(L)}}}_{\mathsf{diag}\left[1-\left(\boldsymbol{a}_i^{(L)}\right)^2\right]} =: \boldsymbol{W}^{(N)} \circ \left[1-\left(\boldsymbol{a}^{(L)}\right)^2\right]$$

Um den Tensor zu vermeiden betrachten wir den Ausdruck komponentenweise:

$$\begin{aligned} \frac{\partial V}{\partial \boldsymbol{W}_{ij}^{(k)}} &= \sum_{p=1}^{n} \frac{\partial V}{\partial \boldsymbol{z}_p^{(k)}} \cdot \frac{\partial \boldsymbol{z}_p^{(k)}}{\partial \boldsymbol{W}_{ij}^{(k)}} \\ &= \sum_{p=1}^{n} \boldsymbol{\delta}_p^{(k)} \cdot \frac{\partial \boldsymbol{z}_p^{(k)}}{\partial \boldsymbol{W}_{ij}^{(k)}} \\ &= \boldsymbol{\delta}_i^{(k)} \cdot \boldsymbol{a}_j^{(k-1)} \\ \Rightarrow \frac{\partial V}{\partial \boldsymbol{W}^{(k)}} &= \left(\boldsymbol{\delta}^{(k)}\right)^T \cdot \left(\boldsymbol{a}^{(k-1)}\right)^T = \boldsymbol{\delta}^{(k)} \otimes \boldsymbol{a}^{(k-1)} \end{aligned}$$

Denn betrachten wir allgemein die Schicht k komponentenweise so gilt:

$$\boldsymbol{z}^{(k)} = \underbrace{\begin{bmatrix} \boldsymbol{W}_{11}^{(k)} & \cdots & \boldsymbol{W}_{1m}^{(k)} \\ \boldsymbol{W}_{21}^{(k)} & \cdots & \boldsymbol{W}_{2m}^{(k)} \\ \vdots & \ddots & \vdots \\ \boldsymbol{W}_{n1}^{(k)} & \cdots & \boldsymbol{W}_{nm}^{(k)} \end{bmatrix}}_{\boldsymbol{W}^{(k)}} \cdot \underbrace{\begin{bmatrix} \boldsymbol{a}_1^{(k-1)} \\ \boldsymbol{a}_2^{(k-1)} \\ \vdots \\ \boldsymbol{a}_m^{(k-1)} \end{bmatrix}}_{\boldsymbol{a}^{(k-1)}} + \underbrace{\begin{bmatrix} \boldsymbol{b}_1^{(k)} \\ \boldsymbol{b}_2^{(k)} \\ \vdots \\ \boldsymbol{b}_n^{(k)} \end{bmatrix}}_{\boldsymbol{b}^{(k)}} \Rightarrow \boldsymbol{z}_i^{(k)} = \sum_{j=1}^{m} \boldsymbol{W}_{ij}^{(k)} \cdot \boldsymbol{a}_j^{(k-1)} + \boldsymbol{b}_i^{(k)}$$

Für die Ableitung nach den Gewichten $\boldsymbol{W}_{ij}^{(k)}$ gilt demnach

$$\frac{\partial \boldsymbol{z}_p^{(k)}}{\partial \boldsymbol{W}_{ij}^{(k)}} = \begin{cases} \boldsymbol{a}_j^{(k-1)}, & p = i \\ 0, & p \neq i \end{cases}$$

Für den Zusammenhang zwischen $\boldsymbol{z}^{(k+1)}$ und $\boldsymbol{z}^{(k)}$ gilt:

$$\boldsymbol{z}^{(k+1)} = \boldsymbol{W}^{(k+1)} \cdot \boldsymbol{a}^{(k)} + \boldsymbol{b}^{(k+1)} \quad \boldsymbol{a}^{(k)} = \tanh(\boldsymbol{z}^{(k)})$$

und $\boldsymbol{\delta}^{(k+1)}$ ausgedrückt über $\boldsymbol{\delta}^{(k)} = \frac{\partial V}{\partial \boldsymbol{z}^{(k)}}$ ergibt

$$\begin{aligned}\boldsymbol{\delta}^{(k)} &= \frac{\partial V}{\partial \boldsymbol{z}^{(k+1)}} \cdot \frac{\partial \boldsymbol{z}^{(k+1)}}{\partial \boldsymbol{a}^{(k)}} \cdot \frac{\partial \boldsymbol{a}^{(k)}}{\partial \boldsymbol{z}^{(k)}} \\ &= \boldsymbol{\delta}^{(k+1)} \cdot \boldsymbol{W}^{(k+1)} \cdot \mathsf{diag}\left[1 - \left(\boldsymbol{a}^{(k)}\right)^2\right] \\ &= \left(\boldsymbol{\delta}^{(k+1)} \cdot \boldsymbol{W}^{(k+1)}\right) \circ \left[1 - \left(\boldsymbol{a}^{(k)}\right)^2\right]\end{aligned}$$

Für den Bias $\boldsymbol{b}^{(k)}$ gilt entsprechend:

$$\begin{aligned}\frac{\partial V}{\partial b^{(N)}} &= 1 \\ \frac{\partial V}{\partial \boldsymbol{b}^{(k)}} &= \frac{\partial V}{\partial \boldsymbol{a}^{(k)}} \cdot \frac{\partial \boldsymbol{a}^{(k)}}{\partial \boldsymbol{z}^{(k)}} \cdot \underbrace{\frac{\partial \boldsymbol{z}^{(k)}}{\partial \boldsymbol{b}^{(k)}}}_{I_n} = \boldsymbol{\delta}^{(k)} \quad k = 1, \cdots, L\end{aligned}$$

□

Satz 5.7. Sei $V : \mathbb{R}^{\text{Input}} \to \mathbb{R}$ ein neuronales Netz mit Input $\boldsymbol{x}$. Dann gilt:

$$\frac{\partial V}{\partial \boldsymbol{x}} = \boldsymbol{\delta}^{(1)} \cdot \boldsymbol{W}^{(1)} \quad \nabla_{\boldsymbol{x}} V(\boldsymbol{x}) = \left(\boldsymbol{\delta}^{(1)} \cdot \boldsymbol{W}^{(1)}\right)^T$$

Beweis: Es gilt nach der Kettenregel:

$$\frac{\partial V}{\partial \boldsymbol{x}} = \frac{\partial V}{\partial \boldsymbol{z}^{(1)}} \cdot \frac{\partial \boldsymbol{z}^{(1)}}{\partial \boldsymbol{x}} = \boldsymbol{\delta}^{(1)} \cdot \boldsymbol{W}^{(1)}$$

Für L Hidden-Layer erhalten wir mittels Backpropagation

$$\boldsymbol{\delta}^{(L)} = \boldsymbol{W}^{(N)} \circ \left[1 - \left(\boldsymbol{a}^{(L)}\right)^2\right] \quad \boldsymbol{\delta}^{(k)} = \left(\boldsymbol{\delta}^{(k+1)} \cdot \boldsymbol{W}^{(k+1)}\right) \circ \left[1 - \left(\boldsymbol{a}^{(k)}\right)^2\right], \quad k = L-1, \cdots, 1$$

□

5.3. Applications on Control Problems

Algorithm 2 REINFORCE-Monte Carlo [6]

1
for Number of episodes **do**
  for $i \in 0, \cdots, N-1$ **do**
    $a_{t_i} \sim \mathcal{N}(u(\boldsymbol{s}_{t_i}, \theta), b)$ $r_i = \left(X_{t_i} + a_{t_i}^2\right) \Delta t$
    $X_{t_{i+1}} = X_{t_i} + \Delta t \left(X_{t_i} + a_{t_i} + 1\right)$
  end for
  Cost $R_k = \sum_{i=k}^{N} r_i$ with $r_N = X_{t_N}^2$
  for $i \in 0, \cdots, N-1$ **do**

$$\nabla_\theta \log(\pi_\theta) = \frac{a_{t_i} - u(\boldsymbol{s}_{t_i}, \theta)}{b} \cdot \frac{\partial u(\boldsymbol{s}_{t_i}, \theta)}{\partial \theta}$$

  end for
end for

6. Neural Ordinary Differential Equations

Sei folgende unbekannten Dynamik gegeben:

$$\dot{\boldsymbol{x}}(t) = \boldsymbol{f}(\boldsymbol{x}(t)) \quad \boldsymbol{x}(t_0) = \boldsymbol{x}_0$$

Ziel ist es nun das Vektorfeld $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^n$ basierend auf Trainingsdaten zu lernen. Wir nutzen zur Modellierung des Vektorfeldes Multi-Layer Perceptron (MLP) nach [1]

$$\begin{aligned}
\dot{\boldsymbol{x}}(t) = \boldsymbol{f}(\boldsymbol{x}(t), \boldsymbol{u}) &= W^{(N)} \cdot \boldsymbol{\sigma}\left(W^{(1)}\boldsymbol{x} + b^{(1)}\right) + b^{(N)} \\
\boldsymbol{u} &= (W^{(1)}, b^{(1)}, W^{(N)}, b^{(N)}) \in \mathbb{R}^{2hn+h+n} \\
W^{(1)} \in \mathbb{R}^{h\times n}, W^{(N)} &\in \mathbb{R}^{n\times h}, b^{(1)} \in \mathbb{R}^h, b^{(N)} \in \mathbb{R}^n
\end{aligned}$$

Wir verwenden den Trajektorienloss J basierend auf den Trainingsdaten zu M Trajektorien zu N Zeitschritten durch:

$$J(\boldsymbol{u}) = \int_{t_0}^{t_f} \|\boldsymbol{x}(t) - \boldsymbol{x}_{\text{data}}(t)\|^2 \, \mathrm{d}t \approx \frac{1}{M \cdot N} \sum_{j=1}^{M} \sum_{i=1}^{N} \left(\boldsymbol{x}(t_i; \boldsymbol{u})^{(j)} - \boldsymbol{x}_{\text{data}}(t_i)^{(j)}\right)^2$$

Für die Ableitung von $\boldsymbol{f}$ nach Zustand $\boldsymbol{x}$ gilt:[3]

$$\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}} = W^{(N)} \cdot \mathsf{diag}\left(\boldsymbol{\alpha}\right) \cdot W^{(1)} \quad \boldsymbol{\alpha}_k = \left.\frac{\partial \sigma(z)}{\partial \boldsymbol{z}}\right|_{z=\boldsymbol{z}_k}$$

$$W^{(N)} \cdot \mathsf{diag}\left(\boldsymbol{\alpha}\right) \cdot W^{(1)} = \begin{bmatrix} W_{11}^{(N)}\boldsymbol{\alpha}_1 & \cdots & W_{1h}^{(N)}\boldsymbol{\alpha}_h \\ \vdots & \ddots & \vdots \\ W_{n1}^{(N)}\boldsymbol{\alpha}_1 & \cdots & W_{nh}^{(N)}\boldsymbol{\alpha}_h \end{bmatrix} \cdot \begin{bmatrix} W_{11}^{(1)} & \cdots & W_{1n}^{(1)} \\ \vdots & \ddots & \vdots \\ W_{h1}^{(1)} & \cdots & W_{hn}^{(1)} \end{bmatrix}$$

Für die Ableitung nach den Parametern gilt:

$$\begin{aligned}
\frac{\partial \boldsymbol{f}_i}{\partial W_{jk}^{(1)}} &= W_{ij}^{(N)} \cdot \left.\frac{\partial \sigma(z)}{\partial \boldsymbol{z}}\right|_{z=\boldsymbol{z}_j} \cdot \frac{\partial \boldsymbol{z}_j}{\partial W_{jk}^{(1)}} = W_{ij}^{(N)} \cdot \boldsymbol{\alpha}_j \cdot \boldsymbol{x}_k \\
\frac{\partial \boldsymbol{f}_i}{\partial b_j^{(1)}} &= W_{ij}^{(N)} \cdot \left.\frac{\partial \sigma(z)}{\partial \boldsymbol{z}}\right|_{z=\boldsymbol{z}_j} = W_{ij}^{(N)} \cdot \boldsymbol{\alpha}_j \\
\frac{\partial \boldsymbol{f}_i}{\partial W_{jk}^{(N)}} &= \delta_{ij} \cdot \sigma(\boldsymbol{z}_k) = \delta_{ij} \cdot \boldsymbol{a}_k \\
\frac{\partial \boldsymbol{f}_i}{\partial b_j^{(N)}} &= \delta_{ij}
\end{aligned}$$

[3] For the components $\boldsymbol{f}_i$ we get

$$\boldsymbol{f}_i = \sum_{k=1}^{h} W_{ik}^{(N)} \cdot \sigma\left(\boldsymbol{z}_k\right) + b_i^{(N)} \quad \boldsymbol{z}_k = \sum_{s=1}^{n} W_{ks}^{(1)} \boldsymbol{x}_s + b_k^{(1)} \quad \boldsymbol{a}_k := \sigma(\boldsymbol{z}_k)$$

applying the chain rule leads to

$$\frac{\partial \boldsymbol{f}_i}{\partial \boldsymbol{x}_j} = \sum_{l=1}^{h} W_{ik}^{(N)} \cdot \left.\frac{\partial \sigma(z)}{\partial z}\right|_{z=\boldsymbol{z}_k} \cdot \frac{\partial \boldsymbol{z}_k}{\partial \boldsymbol{x}_j} \qquad \boldsymbol{\alpha}_k := \left.\frac{\partial \sigma(z)}{\partial z}\right|_{z=\boldsymbol{z}_k} \quad \frac{\partial \boldsymbol{z}_k}{\partial \boldsymbol{x}_j} = W_{kj}^{(1)}$$

6.1. Sensitivity Approach

Für den Gradienten der Loss Funktion gilt:

$$\frac{\partial}{\partial \boldsymbol{u}} J(\boldsymbol{u}) = \int_{t_0}^{t_f} \frac{\partial}{\partial \boldsymbol{u}} \left\| \boldsymbol{x}(t) - \boldsymbol{x}_{\text{data}}(t) \right\|^2 \, \mathrm{d}t = \int_{t_0}^{t_f} 2 \left(\boldsymbol{x}(t) - \boldsymbol{x}_{\text{data}}(t) \right)^T \cdot \frac{\partial \boldsymbol{x}(t)}{\partial \boldsymbol{u}} \, \mathrm{d}t$$

Wir bezeichnen $\boldsymbol{S}(t) := \frac{\partial \boldsymbol{x}(t)}{\partial \boldsymbol{u}}$ als die Sensitivität und definieren blockweise:

$$[\boldsymbol{S}_{W^{(1)}}]_{ijk} := \frac{\partial \boldsymbol{x}_i(t)}{\partial W^{(1)}_{jk}} \quad [\boldsymbol{S}_{b^{(1)}}]_{ij} := \frac{\partial \boldsymbol{x}_i(t)}{\partial b^{(1)}_j} \quad [\boldsymbol{S}_{W^{(N)}}]_{ijk} := \frac{\partial \boldsymbol{x}_i(t)}{\partial W^{(N)}_{jk}} \quad [\boldsymbol{S}_{b^{(N)}}]_{ijk} := \frac{\partial \boldsymbol{x}_i(t)}{\partial b^{(N)}}$$

Für die Ableitung nach der Zeit gilt somit [4]

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t} [\boldsymbol{S}_{W^{(1)}}]_{ijk} &= \frac{\partial \boldsymbol{f}_i(\boldsymbol{x}(t), \boldsymbol{u})}{\partial W^{(1)}_{jk}} = \sum_{s=1}^{n} \frac{\partial \boldsymbol{f}_i}{\partial \boldsymbol{x}_s} \cdot [\boldsymbol{S}_{W^{(1)}}]_{sjk} + \frac{\partial \boldsymbol{f}_i}{\partial W^{(1)}_{jk}} \\
\frac{\mathrm{d}}{\mathrm{d}t} [\boldsymbol{S}_{b^{(1)}}]_{ij} &= \frac{\partial \boldsymbol{f}_i(\boldsymbol{x}(t), \boldsymbol{u})}{\partial b^{(1)}_j} = \sum_{s=1}^{n} \frac{\partial \boldsymbol{f}_i}{\partial \boldsymbol{x}_s} \cdot [\boldsymbol{S}_{b^{(1)}}]_{sj} + \frac{\partial \boldsymbol{f}_i}{\partial b^{(1)}_j} \\
\frac{\mathrm{d}}{\mathrm{d}t} [\boldsymbol{S}_{W^{(N)}}]_{ijk} &= \frac{\partial \boldsymbol{f}_i(\boldsymbol{x}(t), \boldsymbol{u})}{\partial W^{(N)}_{jk}} = \sum_{s=1}^{n} \frac{\partial \boldsymbol{f}_i}{\partial \boldsymbol{x}_s} \cdot [\boldsymbol{S}_{W^{(N)}}]_{sjk} + \frac{\partial \boldsymbol{f}_i}{\partial W^{(N)}_{jk}} \\
\frac{\mathrm{d}}{\mathrm{d}t} [\boldsymbol{S}_{b^{(N)}}]_{ij} &= \frac{\partial \boldsymbol{f}_i(\boldsymbol{x}(t), \boldsymbol{u})}{\partial b^{(N)}_j} = \sum_{s=1}^{n} \frac{\partial \boldsymbol{f}_i}{\partial \boldsymbol{x}_s} \cdot [\boldsymbol{S}_{b^{(N)}}]_{sj} + \frac{\partial \boldsymbol{f}_i}{\partial b^{(N)}_j}
\end{aligned}$$

Für jestes (j,k) lässt sich die Gleichung für $W^{(1)}_{jk}$ als Matrix Gleichung schreiben:

$$\frac{\mathrm{d}}{\mathrm{d}t} \boldsymbol{S}^{(j,k)}_{W^{(1)}} = \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}} \cdot \boldsymbol{S}^{(j,k)}_{W^{(1)}} + \boldsymbol{U}^{(j,k)}_{W^{(1)}} \qquad \boldsymbol{S}^{(j,k)}_{W^{(1)}} := \begin{pmatrix} [\boldsymbol{S}_{W^{(1)}}]_{1jk} \\ \vdots \\ [\boldsymbol{S}_{W^{(1)}}]_{njk} \end{pmatrix} \quad \boldsymbol{U}^{(j,k)}_{W^{(1)}} := \begin{pmatrix} \frac{\partial \boldsymbol{f}_1}{\partial W^{(1)}_{jk}} \\ \vdots \\ \frac{\partial \boldsymbol{f}_n}{\partial W^{(1)}_{jk}} \end{pmatrix}$$

Bezeichne $\left[W^{(N)}\right]^{(j)}$ die j-te Spalte von $W^{(N)}$, dann gilt mit der Komponentenweise Ableitung:

$$\frac{\partial \boldsymbol{f}_i}{\partial W^{(1)}_{jk}} = W^{(N)}_{ij} \cdot \boldsymbol{\alpha}_j \cdot \boldsymbol{x}_k \qquad \boldsymbol{U}^{(j,k)}_{W^{(1)}} = \begin{pmatrix} W^{(N)}_{1j} \boldsymbol{\alpha}_j \boldsymbol{x}_k \\ \vdots \\ W^{(N)}_{nj} \boldsymbol{\alpha}_j \boldsymbol{x}_k \end{pmatrix} = \boldsymbol{\alpha}_j \cdot \left[W^{(N)}\right]^{(j)} \cdot \boldsymbol{x}_k$$

Anordnen jeder Spalte (j,k) in den Matrizen $\boldsymbol{S}_{W^{(1)}}, \boldsymbol{U}_{W^{(1)}} \in \mathbb{R}^{n \times (hn)}$ liefert:[5]

---

[4] For a given dynamics $\dot{\boldsymbol{x}}(t) = \boldsymbol{f}(\boldsymbol{x}(t), \boldsymbol{u})$ the sensitivity is defined as $\boldsymbol{S}(t) := \frac{\partial \boldsymbol{x}(t)}{\partial \boldsymbol{u}}$. Using the Chain rule leads to:

$$\frac{\mathrm{d}}{\mathrm{d}t} \boldsymbol{S}(t) = \frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial \boldsymbol{x}(t)}{\partial \boldsymbol{u}} = \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}} \cdot \frac{\partial \boldsymbol{x}(t)}{\partial \boldsymbol{u}} + \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{u}} = \boldsymbol{f}_{\boldsymbol{x}} \cdot \boldsymbol{S}(t) + \boldsymbol{f}_{\boldsymbol{u}} \quad \boldsymbol{S}(0) = \boldsymbol{0}$$

[5] Let $\boldsymbol{A} \in \mathbb{R}^{n \times m}$ and $\boldsymbol{B} \in \mathbb{R}^{r \times p}$, then the Kronecker product $\otimes$ is defined as

$$\boldsymbol{A} \otimes \boldsymbol{B} = \begin{bmatrix} \boldsymbol{A} \cdot (\boldsymbol{B}_{11}) & \cdots & \boldsymbol{A} \cdot (\boldsymbol{B}_{1p}) \\ \vdots & \ddots & \vdots \\ \boldsymbol{A} \cdot (\boldsymbol{B}_{r1}) & \cdots & \boldsymbol{A} \cdot (\boldsymbol{B}_{rp}) \end{bmatrix} \in \mathbb{R}^{nr \times mp}$$

$$
\begin{aligned}
\boldsymbol{U}_{W^{(1)}} &:= \begin{bmatrix} | & | & & | \\ \boldsymbol{U}_{W^{(1)}}^{(1,1)} & \boldsymbol{U}_{W^{(1)}}^{(1,2)} & \cdots & \boldsymbol{U}_{W^{(1)}}^{(h,n)} \\ | & | & & | \end{bmatrix} \\
&= \begin{bmatrix} \boldsymbol{\alpha}_1 \left[W^{(N)}\right]^{(1)} \boldsymbol{x}_1 & \boldsymbol{\alpha}_1 \left[W^{(N)}\right]^{(1)} \boldsymbol{x}_2 & \cdots & \boldsymbol{\alpha}_h \left[W^{(N)}\right]^{(h)} \boldsymbol{x}_n \end{bmatrix} \\
&= \left[W^{(N)} \cdot \mathsf{diag}(\boldsymbol{\alpha})\right] \otimes \boldsymbol{x}^T \\
\boldsymbol{S}_{W^{(1)}} &= \begin{bmatrix} | & | & & | \\ \boldsymbol{S}_{W^{(1)}}^{(1,1)} & \boldsymbol{S}_{W^{(1)}}^{(1,2)} & \cdots & \boldsymbol{S}_{W^{(1)}}^{(h,n)} \\ | & | & & | \end{bmatrix}
\end{aligned}
$$

Die Gleichung lässt sich somit schreiben durch:

$$
\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{S}_{W^{(1)}} = W^{(N)} \cdot \mathsf{diag}\left(\boldsymbol{\alpha}\right) \cdot W^{(1)} \cdot \boldsymbol{S}_{W^{(1)}} + \left[W^{(N)} \cdot \mathsf{diag}(\boldsymbol{\alpha})\right] \otimes \boldsymbol{x}^T
$$

Für $\boldsymbol{U}_{W^{(N)}}$ erhalten wir:

$$
\begin{aligned}
\boldsymbol{U}_{W^{(N)}}^{(j,k)} &= \begin{pmatrix} \delta_{1j}\sigma(\boldsymbol{z}_k) \\ \vdots \\ \delta_{nj}\sigma(\boldsymbol{z}_k) \end{pmatrix} = \sigma(\boldsymbol{z}_k)\boldsymbol{e}_j = \boldsymbol{a}_k\boldsymbol{e}_j \quad \boldsymbol{e}_j \in \mathbb{R}^n \\
\boldsymbol{U}_{W^{(N)}} &= \left[\begin{array}{cccc|ccc} | & | & & | & | & & | \\ \boldsymbol{a}_1\boldsymbol{e}_1 & \boldsymbol{a}_2\boldsymbol{e}_1 & \cdots & \boldsymbol{a}_h\boldsymbol{e}_1 & \boldsymbol{a}_1\boldsymbol{e}_2 & \cdots & \boldsymbol{a}_h\boldsymbol{e}_n \\ | & | & & | & | & & | \end{array}\right] \\
&= \begin{bmatrix} | & & | \\ \boldsymbol{e}_1 \otimes \boldsymbol{a}^T & \cdots & \boldsymbol{e}_n \otimes \boldsymbol{a}^T \\ | & & | \end{bmatrix} \\
&= \boldsymbol{I}_n \otimes \boldsymbol{a}^T \in \mathbb{R}^{n \times nh}
\end{aligned}
$$

Wir erhalten für die anderen Matrizen $\boldsymbol{U}$:

Block	Ableitung Komponentenform	Ableitung Matrizenform
$W^{(1)}$	$W_{ij}^{(N)} \cdot \boldsymbol{\alpha}_j \cdot \boldsymbol{x}_k$	$\left(W^{(N)} \cdot \mathsf{diag}(\boldsymbol{\alpha})\right) \otimes \boldsymbol{x}^T$
$b^{(1)}$	$W_{ij}^{(N)} \cdot \boldsymbol{\alpha}_j$	$W^{(N)} \cdot \mathsf{diag}\left(\boldsymbol{\alpha}\right)$
$W^{(N)}$	$\delta_{ij} \cdot \boldsymbol{a}_k$	$\boldsymbol{I}_n \otimes \boldsymbol{a}^T$
$b^{(N)}$	δ_{ij}	$\boldsymbol{I}_n$

Analog erhalten wir für die anderen Sensitivitäten:

$$
\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{S}_{b^{(1)}} &= W^{(N)} \cdot \mathsf{diag}\left(\boldsymbol{\alpha}\right) \cdot W^{(1)} \cdot \boldsymbol{S}_{b^{(1)}} + W^{(N)} \cdot \mathsf{diag}\left(\boldsymbol{\alpha}\right) \\
\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{S}_{W^{(N)}} &= W^{(N)} \cdot \mathsf{diag}\left(\boldsymbol{\alpha}\right) \cdot W^{(1)} \cdot \boldsymbol{S}_{W^{(N)}} + \boldsymbol{I}_n \otimes \boldsymbol{a}^T \\
\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{S}_{b^{(N)}} &= W^{(N)} \cdot \mathsf{diag}\left(\boldsymbol{\alpha}\right) \cdot W^{(1)} \cdot \boldsymbol{S}_{b^{(N)}} + \boldsymbol{I}_n
\end{aligned}
$$

Algorithm 3 Training of a MLP Neural ODE with Sensitivity Approach

Dataset: M Trajectories with N time steps. Where $\boldsymbol{X}_0 \in \mathbb{R}^{n \times M}$ are the initial values

Set $\boldsymbol{X}_0$ as $\left[\boldsymbol{x}_{0,\text{data}}^{(j)}\right]_{j=0,\cdots,M-1}$

for epoch in Episodes **do**

  Forward Pass: Save the Trajectories $\boldsymbol{X}_i, \boldsymbol{A}_i, \boldsymbol{\alpha}_i$

  for $i = 0$ to $N-1$ **do** ▷ Forward Pass

$$\begin{aligned}
\boldsymbol{Z}_i &= W^{(1)}\boldsymbol{X}_i + b^{(1)} \cdot \mathbf{1}_M^T \\
\boldsymbol{A}_i &= \tanh\left(\boldsymbol{Z}_i\right) \\
\boldsymbol{\alpha}_i &= \mathbf{1}_h \cdot \mathbf{1}_M^T - \boldsymbol{A}_i \odot \boldsymbol{A}_i \\
\boldsymbol{X}_{i+1} &= \boldsymbol{X}_i + \Delta t_i \cdot \left(W^{(N)}\boldsymbol{A}_i + b^{(N)} \cdot \mathbf{1}_M^T\right)
\end{aligned}$$

  end for

  Sensitivity ODE: Save the Compututed Sensitivities $\boldsymbol{S}_\theta$ ▷ Sensitivity

  for $j = 0$ to $M-1$ **do**

    for $i = 0$ to $N-1$ **do**

$$\begin{aligned}
\boldsymbol{f}_{\boldsymbol{x},i}^{(j)} &= W^{(N)}\mathsf{diag}\left(\boldsymbol{\alpha}_i^{(j)}\right)W^{(1)} \\
\left[\boldsymbol{S}_{W^{(1)}}^{(j)}\right]_{i+1} &= \left[\boldsymbol{S}_{W^{(1)}}^{(j)}\right]_i + \Delta t_i \cdot \boldsymbol{f}_{\boldsymbol{x},i}^{(j)}\left[\boldsymbol{S}_{W^{(1)}}^{(j)}\right]_i + \Delta t_i \cdot \left[W^{(N)}\mathsf{diag}(\boldsymbol{\alpha}_i^{(j)})\right] \otimes \left(\boldsymbol{X}_i^{(j)}\right)^T \\
\left[\boldsymbol{S}_{b^{(1)}}^{(j)}\right]_{i+1} &= \left[\boldsymbol{S}_{b^{(1)}}^{(j)}\right]_i + \Delta t_i \cdot \boldsymbol{f}_{\boldsymbol{x},i}^{(j)}\left[\boldsymbol{S}_{b^{(1)}}^{(j)}\right]_i + \Delta t_i \cdot W^{(N)}\mathsf{diag}\left(\boldsymbol{\alpha}_i^{(j)}\right) \\
\left[\boldsymbol{S}_{W^{(N)}}^{(j)}\right]_{i+1} &= \left[\boldsymbol{S}_{W^{(N)}}^{(j)}\right]_i + \Delta t_i \cdot \boldsymbol{f}_{\boldsymbol{x},i}^{(j)}\left[\boldsymbol{S}_{W^{(N)}}^{(j)}\right]_i + \Delta t_i \cdot \boldsymbol{I}_n \otimes \left(\boldsymbol{A}_i^{(j)}\right)^T \\
\left[\boldsymbol{S}_{b^{(N)}}^{(j)}\right]_{i+1} &= \left[\boldsymbol{S}_{b^{(N)}}^{(j)}\right]_i + \Delta t_i \cdot \boldsymbol{f}_{\boldsymbol{x},i}^{(j)}\left[\boldsymbol{S}_{b^{(N)}}^{(j)}\right]_i + \Delta t_i \cdot \boldsymbol{I}_n
\end{aligned}$$

    end for

  end for

  Gradient Computation: For each θ a unique Gradient will be computed

$$\nabla_\theta J = \frac{1}{NM}\sum_{i=0}^{N-1}\Delta t_i \sum_{j=0}^{M-1} 2\left(\boldsymbol{X}_i^{(j)} - \boldsymbol{X}_{i,\text{data}}^{(j)}\right)^T \left[\boldsymbol{S}_\theta^{(j)}\right]_i$$

  Update the Parameter θ using gradient descent

end for

Output: Trained Parameters $\theta = (W^{(1)}, b^{(1)}, W^{(N)}, b^{(N)})$

6.2. Adjoint Approach

Das Maximumprinzip liefert den Gradienten 2.3 und den adjungierten Zustand $\boldsymbol{p} \in \mathbb{R}^n$:

$$\nabla_{\boldsymbol{u}} J(\boldsymbol{u}) = \int_{t_0}^{t_f} \left[\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{u}}\right]^T \boldsymbol{p}(t)\,\mathrm{d}t$$

$$\dot{\boldsymbol{p}}(t) = -2 \cdot (\boldsymbol{x}(t) - \boldsymbol{x}_{\text{data}}(t)) - \left(\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right)^T \cdot \boldsymbol{p}(t) \quad \boldsymbol{p}(t_f) = 0$$

Für den Ausdruck $\left(\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right)^T \cdot \boldsymbol{p}(t)$ in einem Zeitpunkt t_i gilt somit:

$$\begin{aligned}
\boldsymbol{f}_{\boldsymbol{x}}(\boldsymbol{x}(t_i))^T \cdot \boldsymbol{p}(t_i) &= \left[W^{(N)} \cdot \mathsf{diag}\left(\boldsymbol{\alpha}(t_i)\right) \cdot W^{(1)}\right]^T \cdot \boldsymbol{p}(t_i) \\
&= W^{(1)T} \mathsf{diag}\left(\boldsymbol{\alpha}(t_i)\right) W^{(N)T} \cdot \boldsymbol{p}(t_i) \\
&= W^{(1)T} \cdot \left[\boldsymbol{\alpha}(t_i) \odot \left(W^{(N)T} \cdot \boldsymbol{p}(t_i)\right)\right]
\end{aligned}$$

Für den Ausdruck $\boldsymbol{f}_{\boldsymbol{u}}^T \cdot \boldsymbol{p}$ berechnen wir blockweise[6]

$$\sum_{i=1}^{n} \frac{\partial \boldsymbol{f}_i}{\partial W_{jk}^{(1)}} \cdot \boldsymbol{p}_i = \boldsymbol{\alpha}_j \cdot \boldsymbol{\delta}_j \cdot \boldsymbol{x}_k = (\boldsymbol{\alpha} \odot \boldsymbol{\delta})_j \cdot \boldsymbol{x}_k \qquad \boldsymbol{\delta} := \left(W^{(N)}\right)^T \cdot \boldsymbol{p}$$

$$\sum_{i=1}^{n} \frac{\partial \boldsymbol{f}_i}{\partial b_j^{(1)}} \cdot \boldsymbol{p}_i = \boldsymbol{\alpha}_j \cdot \sum_{i=1}^{n} W_{ij}^{(N)} \boldsymbol{p}_i = \boldsymbol{\alpha}_j \boldsymbol{\delta}_j$$

$$\sum_{i=1}^{n} \frac{\partial \boldsymbol{f}_i}{\partial W_{jk}^{(N)}} \cdot \boldsymbol{p}_i = \boldsymbol{a}_k \cdot \sum_{i=1}^{n} \delta_{ij} \cdot \boldsymbol{p}_i = \boldsymbol{a}_k \cdot \boldsymbol{p}_j$$

$$\sum_{i=1}^{n} \frac{\partial \boldsymbol{f}_i}{\partial b_j^{(N)}} \cdot \boldsymbol{p}_i = \sum_{i=1}^{n} \delta_{ij} \cdot \boldsymbol{p}_i = \boldsymbol{p}_j$$

Wir erhalten für die Matrizen:

Block	Ableitung Komponentenform	Ableitung Matrizenform
$W^{(1)}$	$\boldsymbol{\alpha}_j \cdot \boldsymbol{\delta}_j \cdot \boldsymbol{x}_k$	$(\boldsymbol{\alpha} \odot \boldsymbol{\delta}) \cdot \boldsymbol{x}^T$
$b^{(1)}$	$\boldsymbol{\alpha}_j \boldsymbol{\delta}_j$	$\boldsymbol{\alpha} \odot \boldsymbol{\delta}$
$W^{(N)}$	$\boldsymbol{a}_k \cdot \boldsymbol{p}_j$	$\boldsymbol{p} \odot \boldsymbol{a}^T$
$b^{(N)}$	$\boldsymbol{p}_j$	$\boldsymbol{p}$

[6] Let $A, B \in \mathbb{R}^{n \times m}$, then the direct product (or Hadamard product) $\odot$ is defined as

$$(A \odot B)_{ij} = A_{ij} \cdot B_{ij}$$

Algorithm 4 Training of a MLP Neural ODE with Adjoint Approach

Dataset: M Trajectories with N time steps. Where $\boldsymbol{X}_0 \in \mathbb{R}^{n\times M}$ are the initial values

Set $\boldsymbol{X}_0$ as $\left[\boldsymbol{x}_{0,\text{data}}^{(j)}\right]_{j=0,\cdots,M-1}$

for epoch in Episodes **do**

  Forward Pass: Save the Trajectories $\boldsymbol{X}_i, \boldsymbol{A}_i, \boldsymbol{\alpha}_i$

  for $i = 0$ to $N-1$ **do** ▷ Forward Pass

$$\begin{aligned}\boldsymbol{Z}_i &= W^{(1)}\boldsymbol{X}_i + b^{(1)}\cdot \mathbf{1}_M^T\\ \boldsymbol{A}_i &= \tanh\left(\boldsymbol{Z}_i\right)\\ \boldsymbol{\alpha}_i &= \mathbf{1}_h\cdot\mathbf{1}_M^T - \boldsymbol{A}_i\odot\boldsymbol{A}_i\\ \boldsymbol{X}_{i+1} &= \boldsymbol{X}_i + \Delta t_i\cdot\left(W^{(N)}\boldsymbol{A}_i + b^{(N)}\cdot\mathbf{1}_M^T\right)\end{aligned}$$

  end for

  Backward Pass: Using the terminal condition $\boldsymbol{P}_{N-1} = \mathbf{0}_{n\times M}$ ▷ Adjoint

  for $i = N-2$ to 0 **do**

$$\begin{aligned}\boldsymbol{\delta}_{i+1} &= \left(W^{(N)}\right)^T\cdot\boldsymbol{P}_{i+1}\\ \boldsymbol{P}_i &= \boldsymbol{P}_{i+1} + \Delta t_i\cdot\left\{2\cdot\left(\boldsymbol{X}_{i+1} - \boldsymbol{X}_{i+1}^{(data)}\right) + \left(W^{(1)}\right)^T\cdot\left[\boldsymbol{\alpha}_{i+1}\odot\boldsymbol{\delta}_{i+1}\right]\right\}\end{aligned}$$

  end for

  Set $\boldsymbol{delta}_0 = \left(W^{(N)}\right)^T\cdot\boldsymbol{P}_0$

  Gradient Computation:

$$\nabla_{W^{(1)}}J(\boldsymbol{u}) = \sum_{i=0}^{N-2}\Delta t_i\cdot\sum_{j=0}^{M-1}\left[\underbrace{\boldsymbol{\alpha}_i^{(j)}\odot\boldsymbol{\delta}_i^{(j)}}_{h\times 1}\right]\cdot\underbrace{\left(\boldsymbol{X}_i^{(j)}\right)^T}_{1\times n} = \sum_{i=0}^{N-2}\Delta t_i\cdot\left[\underbrace{\boldsymbol{\alpha}_i\odot\boldsymbol{\delta}_i}_{h\times M}\right]\cdot\underbrace{\left(\boldsymbol{X}_i\right)^T}_{M\times n}$$

$$\nabla_{b^{(1)}}J(\boldsymbol{u}) = \sum_{i=0}^{N-2}\Delta t_i\cdot\sum_{j=0}^{M-1}\left[\underbrace{\boldsymbol{\alpha}_i^{(j)}\odot\boldsymbol{\delta}_i^{(j)}}_{h\times 1}\right] = \sum_{i=0}^{N-2}\Delta t_i\cdot\left[\underbrace{\boldsymbol{\alpha}_i\odot\boldsymbol{\delta}_i}_{h\times M}\right]\cdot\mathbf{1}_M$$

$$\nabla_{W^{(N)}}J(\boldsymbol{u}) = \sum_{i=0}^{N-2}\Delta t_i\cdot\sum_{j=0}^{M-1}\left[\underbrace{\boldsymbol{P}_i^{(j)}\cdot\boldsymbol{A}_i^{(j)T}}_{n\times h}\right] = \sum_{i=0}^{N-2}\Delta t_i\cdot\underbrace{\boldsymbol{P}_i\cdot\left(\boldsymbol{A}_i\right)^T}_{n\times h}$$

$$\nabla_{b^{(N)}}J(\boldsymbol{u}) = \sum_{i=0}^{N-2}\Delta t_i\cdot\sum_{j=0}^{M-1}\underbrace{\boldsymbol{P}_i^{(j)}}_{n\times 1} = \sum_{i=0}^{N-2}\Delta t_i\cdot\underbrace{\boldsymbol{P}_i}_{n\times M}\cdot\underbrace{\mathbf{1}_M}_{M\times 1}$$

  Update the Parameter θ using gradient descent

end for

Output: Trained Parameters $\theta = (W^{(1)}, b^{(1)}, W^{(N)}, b^{(N)})$

6.2.1. Numerisches Beispiel

Wir betrachten konkret den Fluss Φ der logistischen Gleichung: [7]

$$\dot{\boldsymbol{x}}(t) = f(\boldsymbol{x}(t)) = \frac{1}{2} \cdot \boldsymbol{x}(t) \cdot (2 - \boldsymbol{x}(t))$$

besitzt die eindeutige Abbildung

$$\boldsymbol{x}(t) = \Phi_t(\boldsymbol{x}_0) = \frac{2\boldsymbol{x}_0}{(2 - \boldsymbol{x}_0)e^{-(t-t_0)} + \boldsymbol{x}_0}$$

Wir nutzen 20 hidden Layer mit einer linearen Ausgabeschicht und erhalten beim Training folgende Entwicklung von Loss und Lernrate:

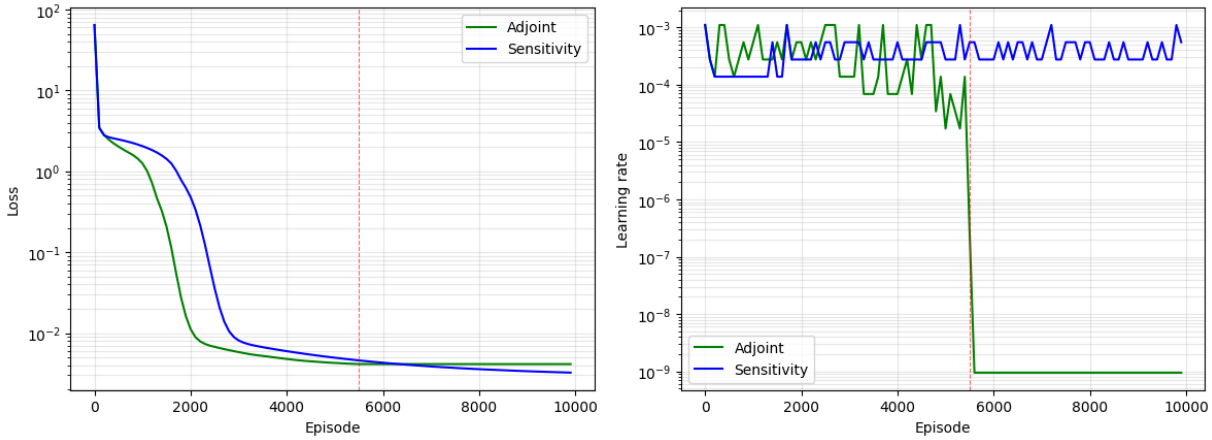


Abbildung 1: Lernrate und Loss Entwicklung im Training

Die Simulation des Startzustandes $x_0 = \frac{1}{2}$ gegen die analytische Lösung sieht wie folgt aus:

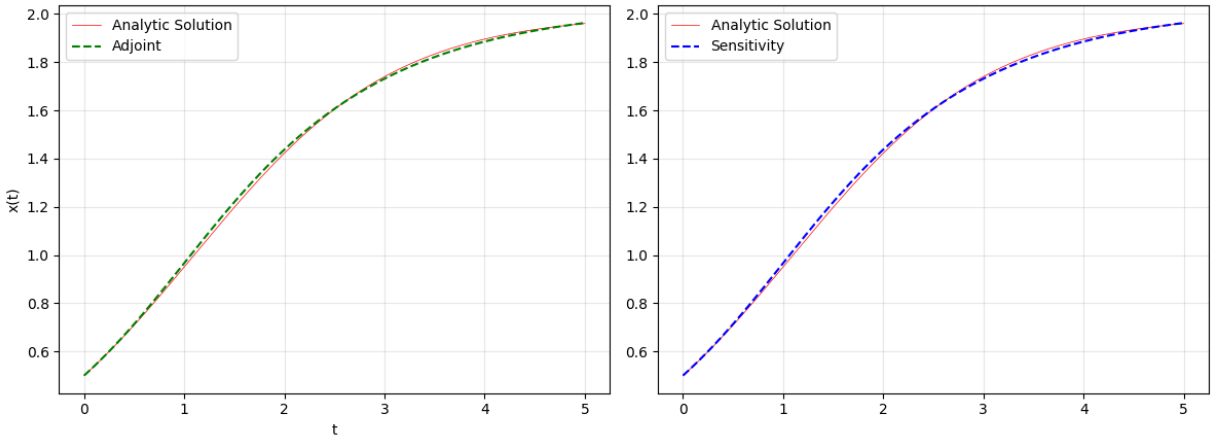


Abbildung 2: Simulation der logistischen ODE mittels gelernter Dynamik

[7] Let $\boldsymbol{x}(t_0) = \boldsymbol{x}_0$, multiplication the equation with $\boldsymbol{x}^{-2}$ and substitute $\boldsymbol{y}(t) := \boldsymbol{x}(t)^{-1}$ $\dot{\boldsymbol{y}}(t) = -\dot{\boldsymbol{x}}(t)\boldsymbol{x}(t)^{-2}$ leads to

$$\underbrace{\dot{\boldsymbol{x}}(t)\boldsymbol{x}(t)^{-2}}_{-\dot{\boldsymbol{y}}(t)} = \underbrace{\Big[a \cdot \boldsymbol{x}(t) \cdot (K - \boldsymbol{x}(t))\Big] \cdot \boldsymbol{x}(t)^{-2}}_{aK\boldsymbol{y}(t) - a} \Rightarrow \underbrace{\boldsymbol{y}(t)}_{\frac{1}{\boldsymbol{x}(t)}} = \frac{1}{\boldsymbol{x}_0} \underbrace{e^{\int_{t_0}^{t} -aK\,\mathrm{d}s}}_{e^{-aK(t-t_0)}} + \underbrace{e^{-aK(t-t_0)} \cdot \int_{t_0}^{t} a e^{aK(s-t_0)}\,\mathrm{d}s}_{\frac{1-e^{-aK(t-t_0)}}{K}}$$

6.3. Diskretisierung der kontinierlichen Gleichung

Wir betrachten auf $[0,T]$ die Dynamik:

$$\dot{\boldsymbol{a}}(t) = \boldsymbol{f}(\boldsymbol{a}(t), \boldsymbol{u}(t)) := \tanh\left[\boldsymbol{W}\boldsymbol{a}(t) + \boldsymbol{b}\right] - \boldsymbol{a}(t)$$

Die Steuerungen sind gegeben durch $\boldsymbol{u} = (\boldsymbol{W}, \boldsymbol{b})$ und wir formulieren das System:

$$\boldsymbol{a}(T) = \boldsymbol{a}(0) + \int_0^T \boldsymbol{f}(\boldsymbol{a}(t), \boldsymbol{u})\,\mathrm{d}t \quad \boldsymbol{a}(0) = \boldsymbol{x}_0$$
$$\Phi(\boldsymbol{a}(T)) = \boldsymbol{W}^{(N)} \cdot \boldsymbol{a}(T) + b^{(N)}$$

Unser adjungiertes System lautet:

$$\dot{\boldsymbol{p}}(t) = -\boldsymbol{p}(t) \cdot \left[\mathsf{diag}\left(1 - \tanh\left(\sum_{j=1}^{n} \boldsymbol{W}_{ij}\boldsymbol{a}_j(t) + \boldsymbol{b}_i\right)^2\right)_{i=1,\cdots,n} \cdot \boldsymbol{W}(t) - \boldsymbol{I}_n\right] \quad \boldsymbol{p}(T) = \boldsymbol{W}^N$$
$$\dot{\boldsymbol{a}}(t) = \tanh\left(\sum_{j=1}^{n} \boldsymbol{W}_{ij}\boldsymbol{a}_j(t) + \boldsymbol{b}_i\right)_{i=1,\cdots,n} - \boldsymbol{a}(t) \quad \boldsymbol{a}(0) = \boldsymbol{x}_0$$

Für die Ableitung der Ausgabe $\Phi = \boldsymbol{W}^{(N)} \cdot \boldsymbol{a}(T) + b^N$ gilt:

$$\frac{\partial \Phi}{\partial \boldsymbol{u}} = \int_0^T \boldsymbol{p}(t) \cdot \boldsymbol{f}_{\boldsymbol{u}}(\boldsymbol{a}(t), \boldsymbol{u})\,\mathrm{d}t \quad \frac{\partial \Phi}{\partial \boldsymbol{W}^{(N)}} = \boldsymbol{a}(T) \quad \frac{\partial \Phi}{\partial b^{(N)}} = 1$$

Es gilt für die Komponente i

$$\boldsymbol{f}_i(\boldsymbol{a}(t), \boldsymbol{u}) = \tanh\left[\sum_{k=1}^{n} \boldsymbol{W}_{ik}\boldsymbol{a}_k(t) + \boldsymbol{b}_i\right] - \boldsymbol{a}_i(t)$$

Für die partielle Ableitung nach $\boldsymbol{a}_j$ und die Jacobi-Matrix $\boldsymbol{f}_{\boldsymbol{a}}$ erhalten wir

$$\frac{\partial \boldsymbol{f}_i}{\partial \boldsymbol{a}_j} = \left[1 - \tanh\left(\sum_{k=1}^{n} \boldsymbol{W}_{ik}\boldsymbol{a}_k(t) + \boldsymbol{b}_i\right)^2\right] \cdot \boldsymbol{W}_{ij} - \frac{\partial \boldsymbol{a}_i(t)}{\partial \boldsymbol{a}_j(t)}$$
$$\left(\boldsymbol{f}_{\boldsymbol{a}}(\boldsymbol{a}(t), \boldsymbol{u})\right)_{ij} = \left[1 - \tanh\left(\sum_{k=1}^{n} \boldsymbol{W}_{ik}\boldsymbol{a}_k(t) + \boldsymbol{b}_i\right)^2\right] \cdot \boldsymbol{W}_{ij} - \delta_{ij}$$

Für die partiellen Ableitungen nach den Parametern $\boldsymbol{W}$ und $\boldsymbol{b}$ gilt:

$$\frac{\partial \boldsymbol{f}_i}{\partial \boldsymbol{W}_{lj}} = \begin{cases} \left[1 - \tanh\left(\sum_{k=1}^{n} \boldsymbol{W}_{ik}\boldsymbol{a}_k(t) + \boldsymbol{b}_i\right)^2\right] \cdot \boldsymbol{a}_j(t), & i = l \\ 0, & i \neq l \end{cases}$$
$$\frac{\partial \boldsymbol{f}_i}{\partial \boldsymbol{b}_j} = \begin{cases} 1 - \tanh\left(\sum_{k=1}^{n} \boldsymbol{W}_{ik}\boldsymbol{a}_k(t) + \boldsymbol{b}_i\right)^2, & i = j \\ 0, & i \neq j \end{cases}$$

Diskretisierung mittels Schrittweite $h = 1$ und $\boldsymbol{u}(t) = (\boldsymbol{W}(t), \boldsymbol{b}(t))$:

$$\boldsymbol{a}(t_k) = \boldsymbol{a}(t_{k-1}) + \boldsymbol{f}(\boldsymbol{a}(t_{k-1}), \boldsymbol{u}(t_{k-1})) = \tanh\left[\boldsymbol{W}(t_{k-1}) \cdot \boldsymbol{a}(t_{k-1}) + \boldsymbol{b}(t_{k-1})\right]$$
$$\boldsymbol{a}(t_k) := \boldsymbol{a}^{(k)} \quad \boldsymbol{a}^{(0)} = \boldsymbol{x}_0 \quad \boldsymbol{W}(t_{k-1}) := \boldsymbol{W}^k \quad \boldsymbol{b}(t_{k-1}) := \boldsymbol{b}^{(k)}$$
$$\boldsymbol{a}^{(k)} = \tanh\left[\boldsymbol{W}^{(k)} \cdot \boldsymbol{a}^{(k-1)} + \boldsymbol{b}^{(k)}\right] := \boldsymbol{F}_k(\boldsymbol{a}^{(k-1)}, \boldsymbol{u}^{(k)})$$

Für eine Diskretisierung $\{t_0, \cdots, t_L = T\}$ erhalten wir somit das Neuronale Netz

$$\begin{aligned}
\boldsymbol{a}^{(0)} &= \boldsymbol{x}_0 \\
\boldsymbol{a}^{(k)} &= \tanh\left[\boldsymbol{W}^{(k)}\boldsymbol{a}^{(k-1)} + \boldsymbol{b}^{(k)}\right] =: \boldsymbol{F}_k(\boldsymbol{a}^{(k-1)}, \boldsymbol{u}^{(k)}) \quad k = 1, \ldots, L \\
a^{(N)} &= \Phi\left(\boldsymbol{a}^{(L)}\right) := \boldsymbol{W}^{(N)}\boldsymbol{a}^{(L)} + b^{(N)}
\end{aligned}$$

Für die adjungierte Gleichung erhalten wir:

$$\begin{aligned}
\boldsymbol{p}(t_k) &= \boldsymbol{p}(t_{k-1}) - \boldsymbol{p}(t_k) \cdot \boldsymbol{f}_{\boldsymbol{a}}(\boldsymbol{a}(t_{k-1}), \boldsymbol{u}(t_{k-1})) \\
\boldsymbol{p}^{(k-1)} &= \boldsymbol{p}^{(k)} + \boldsymbol{p}^{(k)} \cdot \operatorname{diag}\left[1 - \tanh\left(\boldsymbol{W}^{(k)} \cdot \boldsymbol{a}^{(k-1)} + \boldsymbol{b}^{(k)}\right)_i^2\right]_{i=1,\cdots,n} \cdot \boldsymbol{W}^{(k)} - \boldsymbol{p}^{(k)} \cdot \boldsymbol{I}_n \\
\boldsymbol{p}^{(k-1)} &= \boldsymbol{p}^{(k)} \cdot \frac{\partial \boldsymbol{F}_k}{\partial \boldsymbol{a}^{(k-1)}} \quad \boldsymbol{p}^{(L)} = \frac{\partial \Phi}{\partial \boldsymbol{a}^{(L)}} = \boldsymbol{W}^{(N)}
\end{aligned}$$

Das Maximumprinzip des Steuerungproblems lautet dann:

$$\begin{aligned}
H_k\left(\boldsymbol{a}^{(k-1)}, \boldsymbol{u}^{(k)}, \boldsymbol{p}^{(k)}\right) &= \boldsymbol{p}^{(k)} \cdot \boldsymbol{F}_k(\boldsymbol{a}^{(k-1)}, \boldsymbol{u}^{(k)}) \\
\frac{\partial J}{\partial \boldsymbol{u}^{(k)}} &= \frac{\partial H_k}{\partial \boldsymbol{u}^{(k)}} = \boldsymbol{p}^{(k)} \cdot \frac{\partial \boldsymbol{F}_k}{\partial \boldsymbol{u}^{(k)}}, \quad k = 1, \ldots, L. \\
\frac{\partial J}{\partial \boldsymbol{W}^{(N)}} &= (\boldsymbol{a}^{(L)})^T \quad \frac{\partial J}{\partial b^{(N)}} = 1
\end{aligned}$$

Mit $\boldsymbol{a}^{(k)} = \tanh(\boldsymbol{W}^{(k)}\boldsymbol{a}^{(k-1)} + \boldsymbol{b}^{(k)})$ erhalten wir:

$$\begin{aligned}
\frac{\partial \boldsymbol{F}_k}{\partial \boldsymbol{a}^{(k-1)}} &= \operatorname{diag}\left[1 - \left(\boldsymbol{a}^{(k)}\right)^2\right] \cdot \boldsymbol{W}^{(k)} \\
\boldsymbol{p}^{(k-1)} &= \boldsymbol{p}^{(k)} \cdot \frac{\partial \boldsymbol{F}_k}{\partial \boldsymbol{a}^{(k-1)}} \\
\boldsymbol{p}^{(k-1)} &= \underbrace{\boldsymbol{p}^{(k)} \circ \operatorname{diag}\left[1 - \left(\boldsymbol{a}^{(k)}\right)^2\right]}_{:=\boldsymbol{\delta}^{(k)}} \cdot \boldsymbol{W}^{(k)} \\
\underbrace{\boldsymbol{p}^{(k-1)} \circ \left[1 - \left(\boldsymbol{a}^{(k-1)}\right)^2\right]}_{\boldsymbol{\delta}^{(k-1)}} &= \boldsymbol{\delta}^{(k)} \cdot \boldsymbol{W}^{(k)} \circ \left[1 - \left(\boldsymbol{a}^{(k-1)}\right)^2\right]
\end{aligned}$$

Wobei $\left[1 - \left(\boldsymbol{a}^{(k)}\right)^2\right]$ der Vektor mit den Elementen $(1 - \boldsymbol{a}_i^{(k)})_{i=1,\cdots,n}$ bezeichnet. Für die Hidden-Layers gilt entsprechend

$$\begin{aligned}
\frac{\partial J}{\partial \boldsymbol{W}_{ij}^{(k)}} &= \sum_m \boldsymbol{p}_m^{(k)} \cdot \frac{\partial (\boldsymbol{F}_k)_m}{\partial \boldsymbol{W}_{ij}^{(k)}} \quad (\boldsymbol{F}_k)_m = \tanh\left(\sum_j \boldsymbol{W}_{mj}^{(k)} \cdot \boldsymbol{a}_j^{(k-1)} + \boldsymbol{b}_m^{(k)}\right) \\
&= \sum_{m=i} \boldsymbol{p}_m^{(k)} \cdot \left[1 - \tanh\left(\sum_j \boldsymbol{W}_{ij}^{(k)} \cdot \boldsymbol{a}_j^{(k-1)} + \boldsymbol{b}_m^{(k)}\right)^2\right] \cdot \boldsymbol{a}_j^{(k-1)} \\
&= \underbrace{\boldsymbol{p}_i^{(k)} \cdot \left[1 - \left(\boldsymbol{a}_i^{(k)}\right)^2\right]}_{\boldsymbol{\delta}_i^{(k)}} \cdot \boldsymbol{a}_j^{(k-1)}
\end{aligned}$$

Als Matrix erhalten wir somit erneut

$$\frac{\partial J}{\partial \boldsymbol{W}^{(k)}} = (\boldsymbol{\delta}^{(k)})^T (\boldsymbol{a}^{(k-1)})^T := \boldsymbol{\delta}^{(k)} \otimes \boldsymbol{a}^{(k-1)}$$

Für den Bias gilt analog

$$\frac{\partial J}{\partial \boldsymbol{b}_i^{(k)}} = \sum_m \boldsymbol{p}_m^{(k)} \cdot \frac{\partial (\boldsymbol{F}_k)_m}{\partial \boldsymbol{b}_i^{(k)}} = \boldsymbol{p}_i^{(k)} \cdot \left[1 - \left(\boldsymbol{a}_i^{(k)}\right)^2\right] \Rightarrow \frac{\partial J}{\partial \boldsymbol{b}^{(k)}} = \boldsymbol{\delta}^{(k)}$$

Algorithm 5 Backpropagation in Neural Networks 1 Hidden-Layers

---

Given $\boldsymbol{W}^{(1)}, \boldsymbol{W}^{(N)}, \boldsymbol{b}^{(1)}, b^{(N)}, \begin{bmatrix} t_i & X_{t_i} \end{bmatrix}^T$
Foward Computation

$$\begin{aligned} \boldsymbol{z}^{(1)} &= \boldsymbol{W}^{(1)} \cdot \begin{bmatrix} t_i \\ X_{t_i} \end{bmatrix} + \boldsymbol{b}^{(1)} \\ \boldsymbol{a}^{(1)} &= \tanh\left(\boldsymbol{z}^{(1)}\right) \\ V &= \boldsymbol{W}^{(N)} \cdot \boldsymbol{a}^{(1)} + b^{(N)} \end{aligned}$$

Backpropagation

$$\begin{aligned} \boldsymbol{\delta}^{(1)} &= \boldsymbol{W}^{(N)} \circ \left[1 - \left(\boldsymbol{a}^{(1)}\right)^2\right] \\ \frac{\partial V}{\partial \boldsymbol{W}^{(N)}} &= \boldsymbol{a}^{(1)} \\ \frac{\partial V}{\partial b_N} &= 1 \\ \frac{\partial V}{\partial \boldsymbol{W}^{(1)}} &= \left(\boldsymbol{\delta}^{(1)}\right)^T \cdot \begin{bmatrix} t_i & X_{t_i} \end{bmatrix} \\ \frac{\partial V}{\partial \boldsymbol{b}^{(1)}} &= \left(\boldsymbol{\delta}^{(1)}\right)^T \end{aligned}$$

return

$$\nabla_\phi V = \begin{bmatrix} \left(\boldsymbol{\delta}^{(1)}\right)^T \cdot t_i \\ \left(\boldsymbol{\delta}^{(1)}\right)^T \cdot X_{t_i} \\ \left(\boldsymbol{\delta}^{(1)}\right)^T \\ \boldsymbol{a}^{(1)} \\ 1 \end{bmatrix} \quad \phi = \begin{bmatrix} \boldsymbol{W}^{(1)} \cdot \begin{bmatrix} 1 & 0 \end{bmatrix}^T \\ \boldsymbol{W}^{(1)} \cdot \begin{bmatrix} 0 & 1 \end{bmatrix}^T \\ \boldsymbol{b}^{(1)} \\ \left(\boldsymbol{W}^{(N)}\right)^T \\ b^{(N)} \end{bmatrix}$$

---

7. Lagrangian Neural Networks

Gegeben sei $\boldsymbol{q} : [t_0, t_f] \to \mathbb{R}^n$, dann bezeichnen wir mit J das Wirkfunktional:

$$J\left[\boldsymbol{q}(t)\right] = \int_{t_0}^{t_f} L(\boldsymbol{q}(t), \dot{\boldsymbol{q}}(t))\,\mathrm{d}t = \int_{t_0}^{t_f} T(\dot{\boldsymbol{q}}(t)) - V(\boldsymbol{q}(t))\,\mathrm{d}t$$

wobei $T : \mathbb{R}^n \to \mathbb{R}$ die kinetische Energie und $V : \mathbb{R}^n \to \mathbb{R}$ die potenzielle Energie beschreibt. Weiter sei externe Steuerung $\boldsymbol{u} : \mathbb{R}^n \to \mathbb{R}$ gegeben. Das Prinzip der kleinsten Wirkung verlangt, dass die Euler-Lagrange-Gleichungen erfüllt werden müssen:

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L(\boldsymbol{q}, \dot{\boldsymbol{q}})}{\partial \dot{\boldsymbol{q}}_i} - \frac{\partial L(\boldsymbol{q}, \dot{\boldsymbol{q}})}{\partial \boldsymbol{q}_i} = \boldsymbol{u}_i(t) \quad i = 1, \cdots, n$$

Anwendung der Kettenregel auf $\boldsymbol{F}_i(\boldsymbol{q}, \dot{\boldsymbol{q}}) := \frac{\partial L(\boldsymbol{q}, \dot{\boldsymbol{q}})}{\partial \dot{\boldsymbol{q}}_i}$ liefert:

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{F}_i(\boldsymbol{q}, \dot{\boldsymbol{q}}) = \sum_{j=1}^{n} \frac{\partial \boldsymbol{F}_i(\boldsymbol{q}, \dot{\boldsymbol{q}})}{\partial \boldsymbol{q}_j} \cdot \dot{\boldsymbol{q}}_j + \frac{\partial \boldsymbol{F}_i(\boldsymbol{q}, \dot{\boldsymbol{q}})}{\partial \dot{\boldsymbol{q}}_j} \cdot \ddot{\boldsymbol{q}}_j$$

und somit gilt:

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{\boldsymbol{q}}_i} = \sum_{j=1}^{n} \frac{\partial^2 L}{\partial \boldsymbol{q}_j \partial \dot{\boldsymbol{q}}_i} \cdot \dot{\boldsymbol{q}}_j + \frac{\partial^2 L}{\partial \dot{\boldsymbol{q}}_j \partial \dot{\boldsymbol{q}}_i} \cdot \ddot{\boldsymbol{q}}_j \quad i = 1, \cdots, n$$

Die vektorisierte Gleichung lautet:

$$\frac{\mathrm{d}}{\mathrm{d}t}\nabla_{\dot{\boldsymbol{q}}} L = \nabla_{\boldsymbol{q}}\nabla_{\dot{\boldsymbol{q}}}^T L \cdot \dot{\boldsymbol{q}} + \nabla_{\dot{\boldsymbol{q}}}\nabla_{\dot{\boldsymbol{q}}}^T L \cdot \ddot{\boldsymbol{q}}$$

Eingesetzt in die Euler-Lagrange Gleichung in vektorisierter Form:

$$\frac{\mathrm{d}}{\mathrm{d}t}\nabla_{\dot{\boldsymbol{q}}} L = \underbrace{\nabla_{\boldsymbol{q}}\nabla_{\dot{\boldsymbol{q}}}^T L}_{:=C} \cdot \dot{\boldsymbol{q}} + \underbrace{\nabla_{\dot{\boldsymbol{q}}}\nabla_{\dot{\boldsymbol{q}}}^T L}_{:=H} \cdot \ddot{\boldsymbol{q}} = \nabla_{\boldsymbol{q}} L + \boldsymbol{u}$$

$$\ddot{\boldsymbol{q}} = H^{-1} \cdot \left[\nabla_{\boldsymbol{q}} L - C\dot{\boldsymbol{q}} + \boldsymbol{u}\right]$$

Wir nutzen zur Parametrisierung von L erneut ein MLP wie in [2] etabliert.

$$L_{\boldsymbol{\theta}}(\boldsymbol{q}, \dot{\boldsymbol{q}}) = W^{(N)}\boldsymbol{\sigma}\left(W^{(1)}\begin{bmatrix}\boldsymbol{q}\\ \dot{\boldsymbol{q}}\end{bmatrix} + b^{(1)}\right) + b^{(N)}$$

$$W^{(1)} \in \mathbb{R}^{h \times 2n}, W^{(N)} \in \mathbb{R}^{1 \times h}, b^{(1)} \in \mathbb{R}^h, b^{(N)} \in \mathbb{R}^1$$

$$\boldsymbol{\theta} = \left(W^{(1)}, b^{(1)}, W^{(N)}, b^{(N)}\right) \in \mathbb{R}^{2hn+2h+1}$$

Wir erhalten also das Problem der Neural ODE:

$$\underbrace{\begin{bmatrix}\dot{\boldsymbol{q}}\\ \ddot{\boldsymbol{q}}\end{bmatrix}}_{\dot{\boldsymbol{x}}(t)} = \underbrace{\begin{bmatrix}\dot{\boldsymbol{q}}\\ H_{\boldsymbol{\theta}}^{-1} \cdot \left[\nabla_{\boldsymbol{q}} L_{\boldsymbol{\theta}} - C_{\boldsymbol{\theta}}\dot{\boldsymbol{q}} + \boldsymbol{u}\right]\end{bmatrix}}_{\boldsymbol{f}_{\boldsymbol{\theta}}(\boldsymbol{x}, \boldsymbol{u})}$$

7.1. Problemformulierung

Gegeben seien M Trajektorien mit N Datenpunkten, dann definieren wir Loss Funktion über den Mean-Squared-Error:

	Regression	Trajektorien
Loss	$\sum_{j=0}^{M-1}\sum_{i=0}^{N-1}\left\|\ddot{\boldsymbol{q}}_{i,\boldsymbol{\theta}}^{(j)}-\ddot{\boldsymbol{q}}_{i,\text{data}}^{(j)}\right\|^2$	$\sum_{j=0}^{M-1}\int_{t_0}^{t_f}\left\|\boldsymbol{q}_{\boldsymbol{\theta}}^{(j)}(t)-\boldsymbol{q}_{data}^{(j)}(t)\right\|^2\,\mathrm{d}t$
Gradient	$\frac{1}{M}\sum_{j=0}^{M-1}\sum_{i=0}^{N-1}\left(\boldsymbol{p}_i^{(j)}\right)^T\cdot\left[-\frac{\partial H_{\boldsymbol{\theta}}}{\partial\boldsymbol{\theta}}\ddot{\boldsymbol{q}}_{\boldsymbol{\theta}}+\frac{\partial N_{\boldsymbol{\theta}}}{\partial\boldsymbol{\theta}}\right]$	$\frac{1}{M}\sum_{j=0}^{M-1}\int_{t_0}^{t_f}\left[\frac{\partial\boldsymbol{f}}{\partial\boldsymbol{\theta}}(\boldsymbol{x}^{(j)}(t))\right]^T\boldsymbol{p}^{(j)}(t)\,\mathrm{d}t$
Adjoint	$\boldsymbol{p}:=2H_{\boldsymbol{\theta}}^{-1}\left(\ddot{\boldsymbol{q}}-\ddot{\boldsymbol{q}}_{(\text{data})}\right)$	$\dot{\boldsymbol{p}}(t)=-2\begin{bmatrix}\boldsymbol{q}_{\boldsymbol{\theta}}^{(j)}(t)-\boldsymbol{q}_{data}^{(j)}(t)\\ \mathbf{0}_{n\times1}\end{bmatrix}-\left(\frac{\partial\boldsymbol{f}}{\partial\boldsymbol{x}}\right)^T\boldsymbol{p}(t)$ $\boldsymbol{p}(t_f)=\mathbf{0}_{2n\times1}$

Für die Berechnung des Gradienten:

$$\nabla_{\boldsymbol{\theta}}J(\boldsymbol{\theta})=\sum_{j=0}^{M-1}\sum_{i=0}^{N-1}2\cdot\left[\ddot{\boldsymbol{q}}_{\boldsymbol{\theta}}(\boldsymbol{q}_i^{(j)},\dot{\boldsymbol{q}}_i^{(j)},\boldsymbol{u}_i^{(j)})-\ddot{\boldsymbol{q}}_i^{(j)}\right]^T\cdot\frac{\partial}{\partial\boldsymbol{\theta}}\ddot{\boldsymbol{q}}_{\boldsymbol{\theta}}(\boldsymbol{q}_i^{(j)},\dot{\boldsymbol{q}}_i^{(j)},\boldsymbol{u}_i^{(j)})$$

Für die Ableitung nach den Parametern gilt nach der Produktregel:

$$\begin{aligned}\frac{\partial\ddot{\boldsymbol{q}}_{\boldsymbol{\theta}}}{\partial\boldsymbol{\theta}}&=\frac{\partial}{\partial\boldsymbol{\theta}}\left(H_{\boldsymbol{\theta}}^{-1}\cdot\left[\underbrace{\nabla_{\boldsymbol{q}}L_{\boldsymbol{\theta}}-C_{\boldsymbol{\theta}}\dot{\boldsymbol{q}}+\boldsymbol{u}}_{:=N_{\boldsymbol{\theta}}}\right]\right)\\&=\frac{\partial}{\partial\boldsymbol{\theta}}\left(H_{\boldsymbol{\theta}}^{-1}\right)\cdot N_{\boldsymbol{\theta}}+H_{\boldsymbol{\theta}}^{-1}\cdot\frac{\partial}{\partial\boldsymbol{\theta}}N_{\boldsymbol{\theta}}\\&=-H_{\boldsymbol{\theta}}^{-1}\frac{\partial H_{\boldsymbol{\theta}}}{\partial\boldsymbol{\theta}}\underbrace{H_{\boldsymbol{\theta}}^{-1}N_{\boldsymbol{\theta}}}_{\ddot{\boldsymbol{q}}_{\boldsymbol{\theta}}}+H_{\boldsymbol{\theta}}^{-1}\frac{\partial N_{\boldsymbol{\theta}}}{\partial\boldsymbol{\theta}}\\\frac{\partial\ddot{\boldsymbol{q}}_{\boldsymbol{\theta}}}{\partial\boldsymbol{\theta}}&=H_{\boldsymbol{\theta}}^{-1}\cdot\left[-\frac{\partial H_{\boldsymbol{\theta}}}{\partial\boldsymbol{\theta}}\ddot{\boldsymbol{q}}_{\boldsymbol{\theta}}+\frac{\partial N_{\boldsymbol{\theta}}}{\partial\boldsymbol{\theta}}\right]\end{aligned}$$

Da H als Hesse-Matrix symmetrisch ist, gilt $H^{-1}=H^{-T}$. Damit:

$$\nabla_{\boldsymbol{\theta}}J(\boldsymbol{\theta})=\sum_{j=0}^{M-1}\sum_{i=0}^{N-1}\underbrace{2\cdot\left[\ddot{\boldsymbol{q}}_{\boldsymbol{\theta}}(\boldsymbol{q}_i^{(j)},\dot{\boldsymbol{q}}_i^{(j)},\boldsymbol{u}_i^{(j)})-\ddot{\boldsymbol{q}}_i^{(j)}\right]^T H_{\boldsymbol{\theta}}^{-1}}_{\left(\boldsymbol{p}_i^{(j)}\right)^T}\cdot\left[-\frac{\partial H_{\boldsymbol{\theta}}}{\partial\boldsymbol{\theta}}\ddot{\boldsymbol{q}}_{\boldsymbol{\theta}}+\frac{\partial N_{\boldsymbol{\theta}}}{\partial\boldsymbol{\theta}}\right]$$

$$\nabla_{\boldsymbol{\theta}}J(\boldsymbol{\theta})=\sum_{j=0}^{M-1}\sum_{i=0}^{N-1}\left(\boldsymbol{p}_i^{(j)}\right)^T\cdot\left[-\frac{\partial H_{\boldsymbol{\theta}}}{\partial\boldsymbol{\theta}}\ddot{\boldsymbol{q}}_{\boldsymbol{\theta}}+\frac{\partial N_{\boldsymbol{\theta}}}{\partial\boldsymbol{\theta}}\right]\quad\boldsymbol{p}:=2H_{\boldsymbol{\theta}}^{-1}\left(\ddot{\boldsymbol{q}}-\ddot{\boldsymbol{q}}_{(\text{data})}\right)$$

7.2. Herleitung im eindimensionalen Fall

Sei $\boldsymbol{x}(t)=\begin{bmatrix}q(t) & \dot{q}(t)\end{bmatrix}^T\in\mathbb{R}^2$ gegeben. Für die Lagrangefunktion ergibt sich die folgende Struktur:

$$\begin{aligned}L(\boldsymbol{x})=L_{\boldsymbol{\theta}}(q(t),\dot{q})&=W^{(N)}\boldsymbol{\sigma}\left(W^{(1)}\begin{bmatrix}q(t)\\\dot{q}(t)\end{bmatrix}+b^{(1)}\right)+b^{(N)}\\W^{(1)}\in\mathbb{R}^{h\times2},b^{(1)}\in\mathbb{R}^h,W^{(N)}&\in\mathbb{R}^{1\times h},b^{(N)}\in\mathbb{R}\end{aligned}$$

Wir verwenden für die Aktivierungsfunktion tanh und softplus:[8]

	tanh	softplus
$\boldsymbol{A}_k := \sigma(\boldsymbol{z}_k)$	$\tanh(\boldsymbol{z}_k)$	$\log(1+\exp(\boldsymbol{z}_k))$
$\boldsymbol{\alpha}_k := \left.\frac{\partial \sigma(z)}{\partial z}\right\|_{z=\boldsymbol{z}_k}$	$1-\boldsymbol{A}_k^2$	$\frac{\exp(\boldsymbol{z}_k)}{1+\exp(\boldsymbol{z}_k)}$
$\boldsymbol{\beta}_k := \left.\frac{\partial \boldsymbol{\alpha}_k(z)}{\partial z}\right\|_{z=\boldsymbol{z}_k}$	$(-2)\cdot \boldsymbol{A}_k \cdot \boldsymbol{\alpha}_k$	$\boldsymbol{\alpha}_k \cdot (1-\boldsymbol{\alpha}_k)$
$\boldsymbol{\gamma}_k := \left.\frac{\partial \boldsymbol{\beta}_k(z)}{\partial z}\right\|_{z=\boldsymbol{z}_k}$	$2\cdot \boldsymbol{\alpha}_k \cdot \left(3\cdot \boldsymbol{A}_k^2 - 1\right)$	$\boldsymbol{\beta}_k \cdot (1-2\cdot \boldsymbol{\alpha}_k)$

Bezeichne $\left[W^{(1)}\right]^{(s)}$ Spalte s der Matrix $W^{(1)}$. Für die ersten Ableitungen der Lagrange-Funktion nach dem Zustand gilt:

$$\frac{\partial L(\boldsymbol{x})}{\partial q} = \sum_{k=1}^{h} W_{1k}^{(N)} \cdot \frac{\partial \sigma}{\partial z}(\boldsymbol{z}_k) \cdot W_{k1}^{(1)} = W^{(N)} \cdot \left(\boldsymbol{\alpha} \odot \left[W^{(1)}\right]^{(1)}\right)$$

$$\frac{\partial L(\boldsymbol{x})}{\partial \dot{q}} = \sum_{k=1}^{h} W_{1k}^{(N)} \cdot \frac{\partial \sigma}{\partial z}(\boldsymbol{z}_k) \cdot W_{k2}^{(1)} = W^{(N)} \cdot \left(\boldsymbol{\alpha} \odot \left[W^{(1)}\right]^{(2)}\right)$$

Für die Ableitung nach $\left[W^{(1)}\right]^{(1)}$ gilt komponentenweise und in Vektorform:

$$\frac{\partial}{\partial W_{k1}^{(1)}} \frac{\partial L(\boldsymbol{x})}{\partial q} = W_{1k}^{(N)} \cdot \left[\left.\frac{\partial \boldsymbol{\alpha}_k(z)}{\partial z}\right|_{z=\boldsymbol{z}_k} \cdot \frac{\partial \boldsymbol{z}_k}{\partial W_{k1}^{(1)}} \cdot W_{k1}^{(1)} + \boldsymbol{\alpha}_k \cdot \frac{\partial W_{k1}^{(1)}}{\partial W_{k1}^{(1)}}\right] = W_{1k}^{(N)} \cdot \left[\boldsymbol{\beta}_k \cdot \boldsymbol{x}_1 \cdot W_{k1}^{(1)} + \boldsymbol{\alpha}_k\right]$$

$$\frac{\partial}{\partial \left[W^{(1)}\right]^{(1)}} \frac{\partial L(\boldsymbol{x})}{\partial q} = \begin{bmatrix} W_{11}^{(N)} \\ \vdots \\ W_{1h}^{(N)} \end{bmatrix} \odot \left[\boldsymbol{\beta} \odot \left[W^{(1)}\right]^{(1)} \cdot \boldsymbol{x}_1 + \boldsymbol{\alpha}\right] = \left(W^{(N)}\right)^T \odot \left[\boldsymbol{\beta} \odot \left[W^{(1)}\right]^{(1)} \cdot \boldsymbol{x}_1 + \boldsymbol{\alpha}\right]$$

Zusammenfassen der Ableitungen nach den Parametern $\boldsymbol{\theta}$:

$\boldsymbol{\theta}$	$\frac{\partial}{\partial \theta}\frac{\partial L}{\partial q}$
$\left[W^{(1)}\right]^{(1)}$	$\left(W^{(N)}\right)^T \odot \left[\boldsymbol{\beta} \odot \left[W^{(1)}\right]^{(1)} \cdot \boldsymbol{x}_1 + \boldsymbol{\alpha}\right]$
$\left[W^{(1)}\right]^{(2)}$	$\left(W^{(N)}\right)^T \odot \left[\boldsymbol{\beta} \odot \left[W^{(1)}\right]^{(1)} \cdot \boldsymbol{x}_2\right]$
$b^{(1)}$	$\left(W^{(N)}\right)^T \odot \left[\boldsymbol{\beta} \odot \left[W^{(1)}\right]^{(1)}\right]$
$W^{(N)}$	$\boldsymbol{\alpha} \odot \left[W^{(1)}\right]^{(1)}$

[8] Let f, g be defined as $f(z) := \tanh(z)$ and $g(z) = \log(1+e^z)$, the derivatives are given by

$$f'(z) = 1-\tanh(z)^2 \quad f''(z) = (-2)\cdot f(z)\cdot f'(z) \quad f^{(3)}(z) = (-2)\cdot f'(z)\left(1-f(z)^2\right) + 4\cdot f(z)^2 f'(z) = 6f'(z)f(z)^2 - 2f'(z)$$

$$g'(z) = \frac{e^z}{1+e^z} \quad g''(z) = \frac{e^z}{(1+e^z)^2} = g'(z)\cdot(1-g'(z)) \quad g^{(3)}(z) = g''(z)\cdot[1-2\cdot g'(z)]$$

Für die zweiten Ableitungen der Lagrange-Funktion nach dem Zustand gilt:

$$C = \frac{\partial^2 L(\boldsymbol{x})}{\partial \dot{q} \partial q} = \sum_{k=1}^{h} W_{1k}^{(N)} \cdot \frac{\partial \boldsymbol{\alpha}_k}{\partial z}(\boldsymbol{z}_k) \cdot \frac{\partial \boldsymbol{z}_k}{\partial \boldsymbol{x}_2} \cdot W_{k1}^{(1)} = W^{(N)} \cdot \left(\boldsymbol{\beta} \odot \left[W^{(1)} \right]^{(1)} \odot \left[W^{(1)} \right]^{(2)} \right)$$

$$H = \frac{\partial^2 L(\boldsymbol{x})}{\partial \dot{q}^2} = \sum_{k=1}^{h} W_{1k}^{(N)} \cdot \frac{\partial \boldsymbol{\alpha}_k}{\partial z}(\boldsymbol{z}_k) \cdot \frac{\partial \boldsymbol{z}_k}{\partial \boldsymbol{x}_2} \cdot W_{k2}^{(1)} = W^{(N)} \cdot \left(\boldsymbol{\beta} \odot \left[W^{(1)} \right]^{(2)} \odot \left[W^{(1)} \right]^{(2)} \right)$$

Es gilt für die Ableitung nach $W^{(1)}$:

$$\frac{\partial C}{\partial W_{k1}^{(1)}} = W_{1k}^{(N)} \cdot W_{k2}^{(1)} \cdot \left[\frac{\partial \boldsymbol{\beta}_k}{\partial z}(\boldsymbol{z}_k) \cdot \frac{\partial \boldsymbol{z}_k}{\partial W_{k1}^{(1)}} \cdot W_{k1}^{(1)} + \boldsymbol{\beta}_k \right] = W_{1k}^{(N)} \cdot W_{k2}^{(1)} \cdot \left[\boldsymbol{\gamma}_k \cdot \boldsymbol{x}_1 \cdot W_{k1}^{(1)} + \boldsymbol{\beta}_k \right]$$

$$\frac{\partial H}{\partial W_{k1}^{(1)}} = W_{1k}^{(N)} \cdot \left(W_{k2}^{(1)} \right)^2 \cdot \left[\frac{\partial \boldsymbol{\beta}_k}{\partial z}(\boldsymbol{z}_k) \cdot \frac{\partial \boldsymbol{z}_k}{\partial W_{k1}^{(1)}} \right] = W_{1k}^{(N)} \cdot \left(W_{k2}^{(1)} \right)^2 \cdot \left[\boldsymbol{\gamma}_k \cdot \boldsymbol{x}_1 \right]$$

Wir fassen die Ableitungen zusammen:

$\boldsymbol{\theta}$	$\frac{\partial C}{\partial \theta}$	$\frac{\partial H}{\partial \theta}$
$\left[W^{(1)} \right]^{(1)}$	$\left(W^{(N)} \right)^T \odot \left[W^{(1)} \right]^{(2)} \odot \left(\boldsymbol{\gamma} \odot \left[W^{(1)} \right]^{(1)} \boldsymbol{x}_1 + \boldsymbol{\beta} \right)$	$\left(W^{(N)} \right)^T \odot \left[W^{(1)} \right]^{(2)} \odot \left[W^{(1)} \right]^{(2)} \odot \boldsymbol{\gamma} \boldsymbol{x}_1$
$\left[W^{(1)} \right]^{(2)}$	$\left(W^{(N)} \right)^T \odot \left[W^{(1)} \right]^{(1)} \odot \left(\boldsymbol{\gamma} \odot \left[W^{(1)} \right]^{(2)} \boldsymbol{x}_2 + \boldsymbol{\beta} \right)$	$\left(W^{(N)} \right)^T \odot \left[W^{(1)} \right]^{(2)} \odot \left(\boldsymbol{\gamma} \odot \left[W^{(1)} \right]^{(2)} \boldsymbol{x}_2 + 2\boldsymbol{\beta} \right)$
$b^{(1)}$	$\left(W^{(N)} \right)^T \odot \boldsymbol{\gamma} \odot \left[W^{(1)} \right]^{(1)} \odot \left[W^{(1)} \right]^{(2)}$	$\left(W^{(N)} \right)^T \odot \boldsymbol{\gamma} \odot \left[W^{(1)} \right]^{(2)} \odot \left[W^{(1)} \right]^{(2)}$
$W^{(N)}$	$\boldsymbol{\beta} \odot \left[W^{(1)} \right]^{(1)} \odot \left[W^{(1)} \right]^{(2)}$	$\boldsymbol{\beta} \odot \left[W^{(1)} \right]^{(2)} \odot \left[W^{(1)} \right]^{(2)}$

Wir verwenden als Aktivierungsfunktion softplus und erhalten folgenen Algorithmus:

Algorithm 6 LNN with pointwise regression loss

for epoch in Episodes **do**
  Forward Pass: Given M trajectories with N time points, $\boldsymbol{X} = \begin{bmatrix} \boldsymbol{q} & \dot{\boldsymbol{q}} \end{bmatrix}^T \in \mathbb{R}^{2 \times K}$
    $\boldsymbol{Z} = W^{(1)} \cdot \boldsymbol{X} + b^{(1)} \cdot \mathbf{1}_K^T$
    $\boldsymbol{A} = \log\left(\mathbf{1}_{h \times K} + \exp\left(\boldsymbol{Z} \right) \right)$
    $\boldsymbol{\alpha} = \dfrac{\exp\left(\boldsymbol{Z} \right)}{\mathbf{1}_{h \times K} + \exp\left(\boldsymbol{Z} \right)}$
    $\boldsymbol{\beta} = \boldsymbol{\alpha} \odot \left(\mathbf{1}_{h \times K} - \boldsymbol{\alpha} \right)$
    $\boldsymbol{\gamma} = \boldsymbol{\beta} \odot \left(\mathbf{1}_{h \times K} - \mathbf{2}_{h \times K} \odot \boldsymbol{\alpha} \right)$
  Compute derivatives of L:
    $\partial_q \boldsymbol{L} = W^{(N)} \left[\boldsymbol{\alpha} \odot \left(\left[W^{(1)} \right]^{(1)} \cdot \mathbf{1}_K^T \right) \right]$
    $\boldsymbol{C} = W^{(N)} \cdot \left[\boldsymbol{\beta} \odot \left(\left[W^{(1)} \right]^{(1)} \cdot \mathbf{1}_K^T \right) \odot \left(\left[W^{(1)} \right]^{(2)} \cdot \mathbf{1}_K^T \right) \right]$
    $\boldsymbol{H} = W^{(N)} \cdot \left[\boldsymbol{\beta} \odot \left(\left[W^{(1)} \right]^{(2)} \cdot \mathbf{1}_K^T \right) \odot \left(\left[W^{(1)} \right]^{(2)} \cdot \mathbf{1}_K^T \right) \right]$
  Compute the prediction $\ddot{\boldsymbol{q}} = \dfrac{\partial_q \boldsymbol{L} - \boldsymbol{C} \dot{\boldsymbol{q}} + \boldsymbol{u}}{\boldsymbol{H}}$ ▷ Possible because H is a scalar component wise
  Gradient Computation: $\dfrac{1}{K} \sum_{k=1}^{K} 2 \cdot \dfrac{\ddot{\boldsymbol{q}} - \ddot{\boldsymbol{q}}_{\text{data}}}{\boldsymbol{H}} \cdot \left[-\dfrac{\partial \boldsymbol{H}}{\partial \boldsymbol{\theta}} \cdot \ddot{\boldsymbol{q}} + \dfrac{\partial}{\partial \boldsymbol{\theta}} \partial_q \boldsymbol{L} - \dfrac{\partial \boldsymbol{C}}{\partial \boldsymbol{\theta}} \cdot \dot{\boldsymbol{q}} \right]$
  Update the Parameter $\boldsymbol{\theta}$ using gradient descent
end for
Output: Trained Parameters $\boldsymbol{\theta} = (W^{(1)}, b^{(1)}, W^{(N)}, b^{(N)})$

Für die Ableitung der Funktion $\boldsymbol{f}$ gilt:

$$\begin{aligned}\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}} &= \begin{bmatrix} \partial_q \dot{q} & \partial_{\dot{q}} \dot{q} \\ \partial_q \ddot{q} & \partial_{\dot{q}} \ddot{q} \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1 \\ \frac{\partial}{\partial q} H^{-1} \left(\frac{\partial L}{\partial q} - C\dot{q} + u \right) & \frac{\partial}{\partial \dot{q}} H^{-1} \left(\frac{\partial L}{\partial q} - C\dot{q} + u \right) \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1 \\ \frac{-1}{H^2} \frac{\partial H}{\partial q} \left(\frac{\partial L}{\partial q} - C\dot{q} + u \right) + \frac{1}{H} \left(\frac{\partial^2 L}{\partial q^2} - \frac{\partial C}{\partial q} \dot{q} \right) & \frac{-1}{H^2} \frac{\partial H}{\partial \dot{q}} \left(\frac{\partial L}{\partial q} - C\dot{q} + u \right) + \frac{1}{H} \left(C - \frac{\partial C}{\partial \dot{q}} \dot{q} - C \right) \end{bmatrix}\end{aligned}$$

Wir verwenden erneut die Kettenregel mit $\frac{\partial \boldsymbol{z}}{\partial q} = \left[W^{(1)}\right]^{(1)}$ und $\frac{\partial \boldsymbol{z}}{\partial \dot{q}} = \left[W^{(1)}\right]^{(2)}$

Ausdruck	Vektorform
$\frac{\partial H}{\partial q}$	$W^{(N)} \left(\boldsymbol{\gamma} \odot \left[W^{(1)}\right]^{(1)} \odot \left[W^{(1)}\right]^{(2)} \odot \left[W^{(1)}\right]^{(2)} \right)$
$\frac{\partial^2 L}{\partial q^2}$	$W^{(N)} \left(\boldsymbol{\beta} \odot \left[W^{(1)}\right]^{(1)} \odot \left[W^{(1)}\right]^{(1)} \right)$
$\frac{\partial C}{\partial q}$	$W^{(N)} \left(\boldsymbol{\gamma} \odot \left[W^{(1)}\right]^{(1)} \odot \left[W^{(1)}\right]^{(1)} \odot \left[W^{(1)}\right]^{(1)} \right)$
$\frac{\partial H}{\partial \dot{q}}$	$W^{(N)} \left(\boldsymbol{\gamma} \odot \left[W^{(1)}\right]^{(2)} \odot \left[W^{(1)}\right]^{(2)} \odot \left[W^{(1)}\right]^{(2)} \right)$
$\frac{\partial C}{\partial \dot{q}}$	$W^{(N)} \left(\boldsymbol{\gamma} \odot \left[W^{(1)}\right]^{(2)} \odot \left[W^{(1)}\right]^{(1)} \odot \left[W^{(1)}\right]^{(2)} \right)$

Es gilt für die Kontraktion $\boldsymbol{f}_{\boldsymbol{x}}^T \boldsymbol{p}$:

$$\boldsymbol{f}_{\boldsymbol{x}}^T \cdot \begin{bmatrix} \boldsymbol{p}_1 \\ \boldsymbol{p}_2 \end{bmatrix} = \begin{bmatrix} \boldsymbol{p}_2 \cdot \left(-\frac{1}{H^2} \frac{\partial H}{\partial q} \left(\frac{\partial L}{\partial q} - C\dot{q} + u \right) + \frac{1}{H} \left(\frac{\partial^2 L}{\partial q^2} - \frac{\partial C}{\partial q} \dot{q} \right) \right) \\ \boldsymbol{p}_1 + \boldsymbol{p}_2 \cdot \left(-\frac{1}{H^2} \frac{\partial H}{\partial \dot{q}} \left(\frac{\partial L}{\partial q} - C\dot{q} + u \right) + \frac{1}{H} \left(C - \frac{\partial C}{\partial \dot{q}} \dot{q} - C \right) \right) \end{bmatrix}$$

Für die adjungierte Gleichung gilt:

$$\begin{bmatrix} \dot{\boldsymbol{p}}_1 \\ \dot{\boldsymbol{p}}_2 \end{bmatrix} = -2 \cdot \begin{bmatrix} q_{\boldsymbol{\theta}}(t) - q_{\text{data}}(t) \\ 0 \end{bmatrix} - \begin{bmatrix} \boldsymbol{p}_2 \cdot \left(-\frac{1}{H^2} \frac{\partial H}{\partial q} \left(\frac{\partial L}{\partial q} - C\dot{q} + u \right) + \frac{1}{H} \left(\frac{\partial^2 L}{\partial q^2} - \frac{\partial C}{\partial q} \dot{q} \right) \right) \\ \boldsymbol{p}_1 + \boldsymbol{p}_2 \cdot \left(-\frac{1}{H^2} \frac{\partial H}{\partial \dot{q}} \left(\frac{\partial L}{\partial q} - C\dot{q} + u \right) + \frac{1}{H} \left(C - \frac{\partial C}{\partial \dot{q}} \dot{q} - C \right) \right) \end{bmatrix}$$

Es gilt für die Kontraktion $\boldsymbol{f}_{\boldsymbol{\theta}}^T \boldsymbol{p}$:

$$\begin{bmatrix} \boldsymbol{0}_{1 \times (4h+1)} \\ (\partial_{\boldsymbol{\theta}} \ddot{q}) \end{bmatrix}^T \cdot \begin{bmatrix} \boldsymbol{p}_1 \\ \boldsymbol{p}_2 \end{bmatrix} = \partial_{\boldsymbol{\theta}} \ddot{q} \cdot \boldsymbol{p}_2 = \frac{1}{H} \cdot \left[-\frac{\partial H}{\partial \boldsymbol{\theta}} \ddot{q} + \frac{\partial}{\partial \boldsymbol{\theta}} \left(\partial_q L - C\dot{q} + u \right) \right] \boldsymbol{p}_2$$

Wir verwenden als Aktivierungsfunktion erneut softplus.

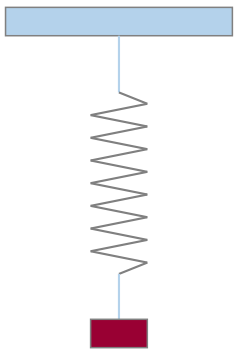
Algorithm 7 LNN with trajectory based adjoint approach

Dataset: M Trajectories with N time steps.
Set $\boldsymbol{X}_0 \in \mathbb{R}^{2\times M}$ initial values of the trajectories
for epoch in Episodes **do**
  Forward Pass: Save $\boldsymbol{X} \in \mathbb{R}^{N\times 2\times M}$
  for i from 0 to $N-2$ **do**
    $\boldsymbol{Z}_i = W^{(1)}\boldsymbol{X}_i + b^{(1)} \cdot \mathbf{1}_M^T$
    $\boldsymbol{A}_i = \log\left(\mathbf{1}_{h\times M} + \exp\left(\boldsymbol{Z}_i\right)\right)$
    $\boldsymbol{\alpha}_i = \dfrac{\exp\left(\boldsymbol{Z}_i\right)}{\mathbf{1}_{h\times M} + \exp\left(\boldsymbol{Z}_i\right)}$
    $\boldsymbol{\beta}_i = \boldsymbol{\alpha}_i \cdot \left(\mathbf{1}_{h\times M} - \boldsymbol{\alpha}_i\right)$
    $\boldsymbol{\gamma}_i = \boldsymbol{\beta}_i \cdot \left(\mathbf{1}_{h\times M} - 2\cdot\boldsymbol{\alpha}_i\right)$
    Compute $\boldsymbol{H}_i, \boldsymbol{C}_i, \partial_q \boldsymbol{L}_i$ based on $\boldsymbol{\alpha}_i, \boldsymbol{\beta}_i, \boldsymbol{\gamma}_i$
    $\boldsymbol{X}_{i+1} = \begin{bmatrix}\boldsymbol{Q}_{i+1}\\ \dot{\boldsymbol{Q}}_{i+1}\end{bmatrix} = \begin{bmatrix}\boldsymbol{Q}_i\\ \dot{\boldsymbol{Q}}_i\end{bmatrix} + \Delta t \begin{bmatrix}\dot{\boldsymbol{Q}}_i\\ \frac{1}{\boldsymbol{H}_i}\left(\partial_q \boldsymbol{L}_i - \boldsymbol{C}_i\dot{\boldsymbol{Q}}_i + \boldsymbol{U}_i\right)\end{bmatrix}$
    Compute $\ddot{\boldsymbol{Q}}_i = \frac{\partial_q \boldsymbol{L}_i - \boldsymbol{C}_i\dot{\boldsymbol{Q}}_i + \boldsymbol{U}_i}{\boldsymbol{H}_i}$
  end for
  Backward Pass: Save $\boldsymbol{P} \in \mathbb{R}^{N\times 2\times M}$
  for i from $N-1$ to 1 **do**
    $\begin{bmatrix}\boldsymbol{P}_{1,i-1}\\ \boldsymbol{P}_{2,i-1}\end{bmatrix} = \begin{bmatrix}\boldsymbol{P}_{1,i}\\ \boldsymbol{P}_{2,i}\end{bmatrix} + \Delta t \cdot \left(2\cdot\begin{bmatrix}\boldsymbol{Q}_{i,\boldsymbol{\theta}} - \boldsymbol{Q}_{i,\text{data}}\\ \mathbf{0}_{1\times M}\end{bmatrix} + \boldsymbol{f}_{\boldsymbol{x}}\left(\boldsymbol{X}_i\right)^T \boldsymbol{P}_i\right) \quad \boldsymbol{P}_{N-1} = \mathbf{0}_{2\times M}$
  end for
  Perform gradient descent with $\nabla_{\boldsymbol{\theta}} J = \sum_{i=0}^{N-1} \Delta t_i \cdot \frac{1}{\boldsymbol{H}_i} \cdot \left[-\frac{\partial \boldsymbol{H}_i}{\partial \boldsymbol{\theta}}\ddot{\boldsymbol{Q}}_i + \frac{\partial}{\partial \boldsymbol{\theta}}\left(\partial_q \boldsymbol{L}_i - \boldsymbol{C}_i\dot{\boldsymbol{Q}}_i + \boldsymbol{U}_i\right)\right]\boldsymbol{P}_{2,i}^T$
end for
Output: Trained Parameters $\boldsymbol{\theta} = (W^{(1)}, b^{(1)}, W^{(N)}, b^{(N)})$

7.3. Experiment: Federpendel

Wir betrachten das Federpendel mit Steuerung u,[9] dann lautet die Lagrange-Formulierung des Problems:

$$\min_q J[q(t)] = \min_q \int_{t_0}^{t_f} \frac{1}{2} m \cdot \dot{q}(t)^2 - \frac{1}{2} k q(t)^2 \, \mathrm{d}t$$



Euler-Lagrange Gleichung des Systems lautet:

$$m\ddot{q}(t) + kq(t) = u(t) \Leftrightarrow \ddot{q}(t) = -\frac{k}{m}\cdot q(t) + \frac{1}{m}\cdot u(t)$$

Für den erweitereten Zustand $\boldsymbol{x} \in \mathbb{R}^2$ gilt:

$$\boldsymbol{x}(t) := \begin{bmatrix} q(t)\\ \dot{q}(t)\end{bmatrix} \quad \dot{\boldsymbol{x}}(t) = \begin{bmatrix} 0 & 1\\ -\frac{k}{m} & 0\end{bmatrix}\cdot \boldsymbol{x}(t) + \begin{bmatrix} 0\\ \frac{1}{m}\end{bmatrix}\cdot u(t)$$

Abbildung 3: Federpendel mit Steuerung $u(t)$.

Für die Wahl von $m = 2$ und $k = 0.5$ ohne externe Steuerung erhalten wir die Gleichung $\ddot{q}(t) + \frac{1}{4}q(t) = 0$.

[9] Hooke's Law states that the force of a spring is given by $F(q) = -k \cdot q$, where q denotes the displacement of the spring and k is the spring constant. The potential energy V is defined by:

$$-\frac{\partial V}{\partial q} = F(q) = -k\cdot q \Rightarrow V(q) = \frac{1}{2}kq^2$$

The kinetic energy T ist given by $T(\dot{q}) = \frac{1}{2}m\dot{q}^2$

Die analytische Lösung ist gegeben durch:

$$q(t) = 2\dot{q}(0) \cdot \sin\left(\frac{t}{2}\right) + q(0) \cdot \cos\left(\frac{t}{2}\right)$$

- Wir erhalten somit mit der tanh Aktivierung

$$\begin{aligned}
\boldsymbol{\alpha}_i &= \mathbf{1}_h \cdot \mathbf{1}_M^T - \boldsymbol{A}_i \odot \boldsymbol{A}_i \\
\boldsymbol{\beta}_i &= (-2) \cdot \boldsymbol{A}_i \odot \boldsymbol{\alpha}_i \\
\boldsymbol{\gamma}_i &= 2 \cdot \boldsymbol{\alpha}_i \odot \left(3 \cdot \boldsymbol{A}_i \odot \boldsymbol{A}_i - \mathbf{1}_h \cdot \mathbf{1}_M^T\right)
\end{aligned}$$

- Für den Ausdruck für $\boldsymbol{H}$ gilt:

$$\boldsymbol{H}_i^{(j)} = \sum_{k=1}^{h} W_{1k}^{(N)} \left[\boldsymbol{\beta}_i^{(j)}\right]_k \underbrace{\left(W_{k2}^{(1)}\right)^2}_{\geq 0} \quad \left[\boldsymbol{\beta}_i^{(j)}\right]_k = (-2) \cdot \tanh\left(\left[\boldsymbol{z}_i^{(j)}\right]_k\right) \cdot \left(1 - \tanh\left(\left[\boldsymbol{z}_i^{(j)}\right]_k\right)\right)$$

- Mit den Initialisierungen

$$W_{ij}^{(1)} \sim \mathsf{N}\left(0, 10^{-2}\right) \quad q(0), \dot{q}(0) \sim \mathsf{U}\left[-\frac{3}{2}, \frac{3}{2}\right] \quad b^{(1)} = -1 \Rightarrow \left[\boldsymbol{z}_0^{(j)}\right]_k \approx (-4, 2)$$

- Für β gilt allerdings

$$\beta(z < 0) < 0 \quad \beta(z = 0) = 0 \quad \beta(z > 0) > 0 \tag{2}$$

- Mit der softplus Aktivierung:
- Konvergenz unter günstigen Bedingungen:
- Divergenz bei negativer Massenmatrix: Bei 50 Seeds
- Keine Identifikation der Lagrange-Funktion:

8. Stochastische Steuerungsprobleme

Seien $\boldsymbol{f} : \mathbb{R}^{n_x \times n_u} \to \mathbb{R}^{n_x}$, $L : \mathbb{R}^{n_x \times n_u} \to \mathbb{R}$, $\Phi : \mathbb{R}^{n_x} \to \mathbb{R}$, $\boldsymbol{\sigma} : \mathbb{R}^{n_x \times n_u} \to \mathbb{R}^{n_x}$ sowie die Zustandsfunktion $\boldsymbol{X} : [t_0, t_f] \times \Omega \to \mathbb{R}^{n_x}$ und die Steuerung $\boldsymbol{u} : [t_0, t_f] \to \mathbb{R}^{n_u}$ gegeben, dann definieren war nachfolgend das stochastische Steuerungsproblem: [10]

$$\begin{aligned} \min_{\boldsymbol{u} \in \mathcal{U}} J(\boldsymbol{u}) &= \mathsf{E}\left[\int_{t_0}^{t_f} L(\boldsymbol{X}_t, \boldsymbol{u}_t)\,\mathrm{d}t + \Phi(\boldsymbol{X}_{t_f})\right] \\ \mathrm{d}\boldsymbol{X}_t &= \boldsymbol{f}(\boldsymbol{X}_t, \boldsymbol{u}_t)\,\mathrm{d}t + \boldsymbol{\sigma}(\boldsymbol{X}_t, \boldsymbol{u}_t)\mathrm{d}W_t, \end{aligned}$$

Für $(t, \boldsymbol{x}) \in [t_0, t_f] \times \mathbb{R}^n$ definieren wir die Value-Function:

$$V(t, \boldsymbol{x}) := \min_{\boldsymbol{u} \in \mathcal{U}} \mathsf{E}\left[\int_t^{t_f} L(\boldsymbol{X}_s, \boldsymbol{u}_s)\,\mathrm{d}s + \Phi(\boldsymbol{X}_{t_f}) \mid \boldsymbol{X}_t = \boldsymbol{x}\right]$$

Für $X_t \in \mathbb{R}^1$ erhalten wir die Hamilton-Jacobi-Bellman Gleichung:

Satz 8.1. Die Value-Function $V(t, x)$ erfüllt:

$$-V_t = \min_{u \in \mathcal{U}} \left[L + f \cdot V_x(t, x) + \frac{1}{2}\sigma^2 \cdot V_{xx}(t, x)\right], \quad V(T, x) = \Phi(x)$$

Beweis: Anwendung der Itô-Formel [1] auf $\mathrm{d}V(s, X_s)$:

$$\begin{aligned} \mathrm{d}V(s, X_s) &= \left[V_t + f(s, X_s, u_s)V_x + \frac{1}{2}\sigma^2(s, X_s, u_s)V_{xx}\right]\mathrm{d}s + \sigma(s, X_s, u_s)V_x\,\mathrm{d}W_s \\ \Rightarrow V(t + \Delta t, X_{t+\Delta t}) &= V(t, x) + \int_t^{t+\Delta t}\left[V_t + f \cdot V_x + \frac{1}{2}\sigma^2 \cdot V_{xx}\right]\mathrm{d}s + \int_t^{t+\Delta t} \sigma \cdot V_x\,\mathrm{d}W_s \end{aligned}$$

Wir wechseln zum Zeitpunkt $t + \Delta t$ zur optimalen Steuerung:

$$\begin{aligned} V(t, x) &\le \mathsf{E}\left[\int_t^{t+\Delta t} L(s, X_s, u_s)\,\mathrm{d}s + V(t + \Delta t, X_{t+\Delta t}) \mid X_t = x\right] \\ 0 &\le \mathsf{E}\left[\int_t^{t+\Delta t} L + \underbrace{V_t + f \cdot V_x + \frac{1}{2}\sigma^2 \cdot V_{xx}\,\mathrm{d}s + \int_t^{t+\Delta t} \sigma \cdot V_x\,\mathrm{d}W_s}_{=V(t+\Delta t, X_{t+\Delta t}) - V(t,x)} \mid X_t = x\right] \end{aligned}$$

Wir verwenden die Martingaleigenschaft des Itô-Integrals [a]:

$$\mathsf{E}\left[\int_t^{t+\Delta t} \underbrace{\sigma \cdot V_x}_{H_s}\,\mathrm{d}W_s \mid \underbrace{X_t = x}_{\mathcal{F}_t}\right] = \underbrace{\mathsf{E}\left[\int_0^{t+\Delta t} H_s\,\mathrm{d}W_s \mid \mathcal{F}_t\right]}_{=\int_0^t H_s\,\mathrm{d}W_s} - \underbrace{\mathsf{E}\left[\underbrace{\int_0^t H_s\,\mathrm{d}W_s}_{\mathcal{F}_t\text{-messbar}} \mid \mathcal{F}_t\right]}_{=\int_0^t H_s\,\mathrm{d}W_s} = 0$$

[10] Let $W : [0, \infty) \times \Omega \to \mathbb{R}$ be a stochastic process on a probability space $(\Omega, \{\mathcal{F}_t\}_{t \ge 0}, \mathsf{P})$. With $\mathsf{P}(W_0 = 0) = 1$ and

$$\forall t > s \ge 0: \quad W_t - W_s \perp \mathcal{F}_s = \sigma(W_r : r \le s) \quad W_t - W_s \sim \mathcal{N}(0, t - s)$$

The map $t \mapsto W_t(\omega)$ is continuous and almost sure nowhere differentiable. This Process is called a Brownian Motion.

Unter der Regularitätsbedingung $\int_0^T \mathsf{E}[H_s^2]\,\mathrm{d}s = \int_0^T \mathsf{E}[\sigma^2 V_x^2]\,\mathrm{d}s < \infty$ für jedes T. Somit gilt

$$0 \leq \mathsf{E}\left[\int_t^{t+\Delta t} L + V_t + f \cdot V_x + \frac{1}{2}\sigma^2 \cdot V_{xx}\,\mathrm{d}s \mid X_t = x\right] + \underbrace{\mathsf{E}\left[\int_t^{t+\Delta t} \sigma \cdot V_x\,\mathrm{d}W_s \mid X_t = x\right]}_{=0}$$

Division durch Δt und Grenzübergang $\Delta t \to 0$ liefert (mittels dominierter Konvergenz):

$$0 \leq \lim_{\Delta t \to 0} \frac{1}{\Delta t} \mathsf{E}\left[\underbrace{\int_t^{t+\Delta t} L + V_t + f \cdot V_x + \frac{1}{2}\sigma^2 \cdot V_{xx}\,\mathrm{d}s}_{=(L[\tau]+V_t[\tau]+f[\tau]\cdot V_x[\tau]+\frac{1}{2}\sigma^2\cdot V_{xx}[\tau])\Delta t} \mid X_t = x\right] \quad \tau \in [t, t+\Delta t]$$

$$0 \leq L(t,x,u_t) + V_t(t,x) + f(t,x,u_t)V_x(t,x) + \frac{1}{2}\sigma^2(t,x,u_t)V_{xx}(t,x)$$

da $\lim_{\Delta t \to 0} X_{t+\Delta t} = X_t = x$

unter der Stetigkeitsbedingung der Funktionen L, f, σ und des Prozesses X_t. Für die optimale Steuerung u^* gilt:

$$0 = \min_{u \in \mathcal{U}} \left[L(t,x,u) + V_t(t,x) + f(t,x,u)V_x(t,x) + \frac{1}{2}\sigma^2(t,x,u)V_{xx}(t,x)\right]$$

$$-V_t(t,x) = L(t,x,u^*) + f(t,x,u^*)V_x(t,x) + \frac{1}{2}\sigma^2(t,x,u^*)V_{xx}(t,x)$$

□

Beispiel 8.2. Wir betrachten das Steuerungsproblem:

$$\min_{u \in \mathcal{U}} J(u) = \mathsf{E}\left[\int_0^1 u_t^2\,\mathrm{d}t + X_1^2\right]$$
$$\mathrm{d}X_t = u_t\,\mathrm{d}t + \sigma X_t\,\mathrm{d}W_t, \quad X_0 = x_0$$

Die Value-Function ist definiert als:

$$V(t,x) = \min_{u \in \mathcal{U}} \mathsf{E}\left[\int_t^1 u_s^2\,\mathrm{d}s + X_1^2 \mid X_t = x\right]$$

[a] Sei H_t mit $\mathsf{E}\left[\int_0^T H_s^2\,\mathrm{d}s\right] < \infty$, dann ist $M_t := \int_0^t H_s\,\mathrm{d}W_s$ ein Martingal bezüglich $(\mathcal{F}_t)_{t\geq 0}$ und es gilt

$$\mathsf{E}\left[\int_0^t H_s\,\mathrm{d}W_s \mid \mathcal{F}_s\right] = \int_0^s H_u\,\mathrm{d}W_u, s \leq t \Rightarrow \mathsf{E}\left[\int_0^t H_s\,\mathrm{d}W_s\right] = 0$$

[1] Sei X_t gegeben durch

$$\mathrm{d}X_t = \mu(t, X_t)\,\mathrm{d}t + \sigma(t, X_t)\,\mathrm{d}W_t \quad X_0 = x_0,$$

dann gilt für jede differenzierbare Funktion $f : [0,\infty) \times \mathbb{R} \to \mathbb{R}$:

$$\mathrm{d}f(t, X_t) = \left[f_t + \mu f_x + \frac{1}{2}\sigma^2 f_{xx}\right]\mathrm{d}t + \sigma f_x \mathrm{d}W_t$$

Die HJB-Gleichung lautet:

$$-V_t(t,x) = \min_{u\in\mathcal{U}} \left[u^2 + uV_x(t,x) + \frac{1}{2}\sigma^2 x^2 V_{xx}(t,x)\right] \quad V(1,x) = x^2$$

Wir bestimmen das Minimum durch Ableiten nach u:

$$\frac{\partial}{\partial u}\left[u^2 + uV_x + \frac{1}{2}\sigma^2 x^2 V_{xx}\right] = 2u + V_x = 0 \Rightarrow u^* = -\frac{1}{2}V_x$$

Wir erhalten also die Gleichung:

$$0 = V_t - \frac{1}{4}V_x^2 + \frac{1}{2}\sigma^2 x^2 V_{xx}, \quad V(1,x) = x^2$$

Als Ansatzfunktion wählen wir: $V(t,x) = P(t)x^2$

$$0 = \underbrace{P'(t)x^2}_{V_t} - \frac{1}{4}\left[\underbrace{2P(t)x}_{V_x}\right]^2 + \frac{1}{2}\sigma^2 x^2 \cdot \underbrace{2P(t)}_{V_{xx}} \quad V(1,x) = P(1)x^2 = x^2 \Rightarrow P(1) = 1$$
$$0 = P'(t) - P(t)^2 + \sigma^2 P(t) \Rightarrow P'(t) = P(t)\cdot(P(t) - \sigma^2)$$

1. Lösung im deterministischen Fall $\sigma = 0$:

$$P'(t) = P(t)^2 \Rightarrow \int_t^1 \frac{P'(s)}{P(s)^2}\,\mathrm{d}s = \int_t^1 1\,\mathrm{d}s \Rightarrow \underbrace{\left[-\frac{1}{P(s)}\right]_t^1}_{-1+\frac{1}{P(t)}} = 1 - t \Rightarrow P(t) = \frac{1}{2-t}$$

   Für die Steuerung im deterministischen Fall ergibt sich somit:

$$u^*(t,x) = -\frac{1}{2}V_x(t,x) = -\frac{1}{2}\cdot 2P(t)x = -\frac{x}{2-t}$$

2. Lösung im stochastischen Fall $\sigma = \frac{1}{2}$:

$$P'(t) = P(t)\cdot\left(P(t) - \frac{1}{4}\right) \Rightarrow \int_t^1 \underbrace{\frac{P'(s)}{P(s)(P(s)-\frac{1}{4})}}_{=-\frac{4}{P(s)}+\frac{4}{P(s)-\frac{1}{4}}}\,\mathrm{d}s = \int_t^1 1\,\mathrm{d}s$$
$$\Rightarrow \underbrace{\left[-4\log(P(s)) + 4\log\left(P(s) - \frac{1}{4}\right)\right]_t^1}_{4\log\left(\frac{3}{4}\right)+4\log(P(t))-4\log\left(P(t)-\frac{1}{4}\right)} = 1 - t$$
$$\Rightarrow \log\left(\frac{\frac{3}{4}\cdot P(t)}{P(t)-\frac{1}{4}}\right) = \frac{1-t}{4} \Rightarrow P(t) = \frac{\frac{1}{4}}{1 - \frac{3}{4}\cdot\exp\left(-\frac{1-t}{4}\right)}$$

   Für die Steuerung im stochastischen Fall ergibt sich somit:

$$u^*(t,x) = -\frac{1}{2}V_x(t,x) = -\frac{1}{2}\cdot 2P(t)x = -\frac{\frac{1}{4}}{1 - \frac{3}{4}\cdot\exp\left(-\frac{1-t}{4}\right)}\cdot x$$

Satz 8.3: Infinite Horizon HJB Gleichung. Für das Steuerungsproblem:

$$\min_{u\in\mathcal{U}} J(u) = \mathsf{E}\left[\int_0^\infty e^{-\rho t} L(X_t, u_t)\,\mathrm{d}t\right] \qquad \mathrm{d}X_t = f(X_t, u_t)\,\mathrm{d}t + \sigma(X_t, u_t)\,\mathrm{d}W_t,$$

gilt für die Value-Function $V(x)$

$$V(x) = \min_{u\in\mathcal{U}} \mathsf{E}\left[\int_0^\infty e^{-\beta t} L(X_t, u_t)\,\mathrm{d}t \mid X_0 = x\right]$$

die HJB-Gleichung:

$$\beta V(x) = \min_{u\in\mathcal{U}}\left[L(x,u) + f(x,u)\cdot V_x(x) + \frac{1}{2}\sigma^2(x,u)\cdot V_{xx}(x)\right]$$

Beweis: Analog zum Beweis der HJB-Gleichung 8.1 mit dem Unterschied, dass wir den Zeitparameter nicht mehr berücksichtigen müssen.

1. Diskontierung des Kostenfunktionals um den Zeitpunkt Δt mit $s' = s - \Delta t$

$$\mathsf{E}\left[\int_{\Delta t}^\infty e^{-\beta s} L(X_s, u_s)\,\mathrm{d}s\right] = \mathsf{E}\left[e^{-\beta\Delta t}\int_0^\infty e^{-\beta s'} L(X_{s'+\Delta t}, u_{s'+\Delta t})\,\mathrm{d}s'\right]$$

2. Wechsel zur optimalen Steuerung zum Zeitpunkt Δt, wobei $X_{\Delta t} = x'$

$$V(x) \le \mathsf{E}\left[\int_0^{\Delta t} e^{-\beta s} L(X_s, u_s)\,\mathrm{d}s + e^{-\beta\Delta t}\underbrace{\int_0^\infty e^{-\beta s'} L(X_{s'+\Delta t}, u_{s'+\Delta t})\,\mathrm{d}s'}_{V(X_{\Delta t}=x')} \mid X_0 = x\right]$$

$$0 \le \mathsf{E}\left[\int_0^{\Delta t} e^{-\beta s} L(X_s, u_s)\,\mathrm{d}s + e^{-\beta\Delta t} V(x') - V(x) \mid X_0 = x\right]$$

3. Taylorentwicklung von $V(X_{\Delta t} = x')$

$$V(x') = V(x) + \int_0^{\Delta t}\left[f\cdot V_x + \frac{1}{2}\sigma^2\cdot V_{xx}\right]\,\mathrm{d}s + \int_0^{\Delta t}\sigma\cdot V_x\,\mathrm{d}W_s$$

$$e^{-\beta\Delta t}V(x') - V(x) = \left(e^{-\beta\Delta t} - 1\right)V(x) + e^{-\beta\Delta t}\left(\int_0^{\Delta t}\left[f\cdot V_x + \frac{\sigma^2}{2}V_{xx}\right]\,\mathrm{d}s + \int_0^{\Delta t}\sigma V_x\,\mathrm{d}W_s\right)$$

4. Martingaleigenschaft des Itô-Integrals **??** anwenden liefert: $\mathsf{E}\left[\int_0^{\Delta t}\sigma V_x\,\mathrm{d}W_s\right] = 0$

$$0 \le \lim_{\Delta t\to 0}\frac{1}{\Delta t}\mathsf{E}\left[\int_0^{\Delta t}\underbrace{e^{-\beta s}L + e^{-\beta\Delta t}\left[f\cdot V_x + \frac{\sigma^2}{2}\cdot V_{xx}\right]}_{\left(e^{-\beta\tau}L[\tau] + e^{-\beta\Delta t}\left[f[\tau]\cdot V_x[\tau] + \frac{\sigma^2}{2}\cdot V_{xx}[\tau]\right]\right)\cdot\Delta t}\,\mathrm{d}s + \left(e^{-\beta\Delta t} - 1\right)V(x) \mid X_0 = x\right]$$

$$0 \le L(x,u) + f(x,u)\cdot V_x(x) + \frac{1}{2}\sigma^2(x,u)\cdot V_{xx}(x) + \underbrace{\lim_{\Delta t\to 0}\frac{e^{-\beta\Delta t} - 1}{\Delta t}}_{-\beta} V(x)$$

da $\lim_{\Delta t\to 0} X_{\Delta t} = X_0 = x$

5. Für die optimale Steuerung u^* gilt:

$$0 = \min_{u \in \mathcal{U}} \left[L(x,u) + f(x,u) \cdot V_x(x) + \frac{1}{2}\sigma^2(x,u) \cdot V_{xx}(x) - \beta V(x) \right]$$
$$\beta V(x) = L(x,u^*) + f(x,u^*) \cdot V_x(x) + \frac{1}{2}\sigma^2(x,u^*) \cdot V_{xx}(x)$$

□

Beispiel 8.4: Optimality Equation für V. Wir diskretisieren die SDE mittels Euler-Maruyama:

$$X_{t+\Delta t} \approx X_t + f(X_t,u_t)\Delta t + \sigma(X_t,u_t)\Delta W_t, \quad \Delta W_t \sim \mathcal{N}(0,\Delta t)$$

Starten wir in Zustand $X_t = x$ und wählen eine Steuerung $u \in \mathcal{U}$

$$V(x) = \min_{u \in \mathcal{U}} \mathsf{E}\left[\underbrace{\frac{L(x',u) - L(x,u)}{\Delta t}}_{\approx \int_0^{\Delta t} e^{-\beta s} L(X_s,u_s)\,\mathrm{d}s} + \underbrace{e^{-\beta \Delta t}}_{\gamma} \cdot V\left(\underbrace{x + f(x,u)\Delta t + \sigma(x,u)\Delta W_t}_{\approx x'} \right) \mid X_0 = x \right]$$
$$= \min_{\pi} \mathsf{E}_{s' \sim T_a(s)} \left[R_a(s,s') + \gamma V(s') \mid s_t = s \right] \quad X_t = s_t = s, \quad X_{t+\Delta t} = s' \sim T_a(s), \quad u_t = a_t \sim \pi(s_t)$$

Wir erhalten in dieser Notation die Bellman-Gleichung aus dem Reinforcement Learning.

- Die unbekannte Environment

$$s' \sim T_a(s) = \mathcal{N}\left(s + f(s,a)\Delta t, \sigma^2(s,a)\Delta t\right)$$

- Für Policy Gradient Methoden wählt man häufig ebenfalls eine Normalverteilung:

$$a_t \sim \pi(\theta \mid s_t) = \mathcal{N}\left(\mu(s_t,\theta), \sigma(s_t,\theta)\right)$$

8.1. Policy-Gradient Methoden

Beispiel 8.5: Policy Gradient Methode für SDEs. Gegeben sei das Problem:

$$\min_u \int_0^1 X_t + u_t^2\,\mathrm{d}t + X_1^2 \quad \mathrm{d}X_t = X_t + u_t + 1, \quad X_0 = 0$$

dann lautet das stochastische Hilfsproblem:

$$\min_\theta \mathsf{E}_{u \sim \pi_\theta} \left[\int_0^1 X_t + u_t^2\,\mathrm{d}t + X_1^2 \right] \quad \mathrm{d}X_t = (X_t + u_t + 1)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \quad X_0 = 0$$

1. Die Verteilung π_θ ist durch θ parametrisiert

$$\nabla_\theta \mathsf{E}_{u \sim \pi_\theta} \left[\int_0^1 X_t + u_t^2\,\mathrm{d}t + X_1^2 \right] = \int_U \underbrace{\pi_\theta(u) \cdot \nabla_\theta \log(\pi_\theta(u))}_{\nabla_\theta \pi_\theta(u)} \left[\int_0^1 X_t + u_t^2\,\mathrm{d}t + X_1^2 \right] \mathrm{d}u$$
$$= \mathsf{E}_{u \sim \pi_\theta} \left[\nabla_\theta \log(\pi_\theta(u)) \left(\int_0^1 X_t + u_t^2\,\mathrm{d}t + X_1^2 \right) \right]$$

2. Sei $u_t \sim \mathcal{N}\left(\mu(\theta,t), b\right)$ dann gilt mit der Diskretisierung: $\boldsymbol{u} = (u_i)_{i=0}^{N-1}$

$$\begin{aligned}
\pi_\theta(\boldsymbol{u}) &= \prod_{i=0}^{N-1} \pi_\theta(u_i) = \prod_{i=0}^{N-1} \frac{1}{\sqrt{2\pi b}} \exp\left(-\frac{(u_i - \mu(\theta,t_i))^2}{2b}\right) \\
\log(\pi_\theta(\boldsymbol{u})) &= \sum_{i=0}^{N-1} \log(\pi_\theta(u_i)) = \sum_{i=0}^{N-1} \left[-\frac{1}{2}\log(2\pi b) - \frac{(u_i - \mu(\theta,t_i))^2}{2b}\right] \\
\nabla_\theta \log(\pi_\theta(\boldsymbol{u})) &= \sum_{i=0}^{N-1} \nabla_\theta \log(\pi_\theta(u_i)) = \sum_{i=0}^{N-1} \frac{u_i - \mu(\theta,t_i)}{b} \cdot \nabla_\theta \mu(\theta,t_i)
\end{aligned}$$

3. Diskretisierung des Problems mit Zeitschritt Δt

$$\int_0^1 X_t + u_t^2 \,\mathrm{d}t + X_1^2 \approx \sum_{i=0}^{N-1} \left(X_{t_i} + u_i^2\right) \Delta t + X_{t_N}^2 \tag{3}$$

4. Simulation des Zustandes $X_{t_{i+1}} \sim \mathcal{N}\left(X_{t_i} + (X_{t_i} + u_i + 1)\Delta t, \sigma \Delta t\right)$

5. Abschätzung des Gradienten durch Monte-Carlo:

$$\begin{aligned}
\nabla_\theta J(\theta) &= \mathsf{E}_{u\sim\pi_\theta}\left[\sum_{i=0}^{N-1} \nabla_\theta \log(\pi_\theta(u_i)) \cdot \underbrace{\left(\sum_{i=0}^{N-1} \left(X_{t_i} + (u_i)^2\right)\Delta t + X_{t_N}^2\right)}_{C(\boldsymbol{u}) = C(u_0,\ldots,u_{N-1})}\right] \\
&= \mathsf{E}_{u\sim\pi_\theta} \sum_{i=0}^{N-1} \left[\nabla_\theta \log(\pi_\theta(u_i)) \cdot \sum_{j=i}^{N} C_j\right]
\end{aligned}$$

Zerlegung von $C(\boldsymbol{u})$

$$C(\boldsymbol{u}) = \sum_{i=0}^{N-1} \underbrace{\left(X_{t_i} + (u_i)^2\right)\Delta t}_{C_i} + \underbrace{X_{t_N}^2}_{C_N} \Rightarrow C(\boldsymbol{u}) = \sum_{i=0}^{N} C_i$$

Für ein festes j gilt: u_j ist unabhängig von C_k für $k < j$, da die u_i paarweise unabhängig sind.

$$\mathsf{E}\left[\underbrace{\nabla_\theta \log(\pi_\theta(u_j))}_{\frac{\nabla_\theta \pi_\theta(u_j)}{\pi_\theta(u_j)}} \cdot C_k\right] = \underbrace{\int_{\mathbb{R}} \nabla_\theta \pi_\theta(u)\,\mathrm{d}u}_{\mathsf{E}\left[\frac{\nabla_\theta \pi_\theta(u_j)}{\pi_\theta(u_j)}\right]} \cdot \mathsf{E}\left[C_k\right] = \nabla_\theta \underbrace{\int_{\mathbb{R}} \pi_\theta(u)\,\mathrm{d}u}_{=1} \cdot \mathsf{E}\left[C_k\right] = 0$$

6. Monte Carlo Schätzung des Gradienten für die Optimierung

$$\begin{aligned}\nabla_\theta J(\theta) &= \mathsf{E}_{u\sim\pi_\theta}\left[\sum_{i=0}^{N-1}\nabla_\theta \log(\pi_\theta(u_i))\cdot \underbrace{\sum_{j=i}^{N} C_j}_{C(u_i,\dots,u_{N-1})}\right]\\ &\approx \frac{1}{M}\sum_{m=1}^{M}\left[\sum_{i=0}^{N-1}\frac{u_i^{(m)}-\mu(\theta,t_i)}{b}\cdot\nabla_\theta\mu(\theta,t_i)\cdot\left(\sum_{j=i}^{N-1}\left(X_{t_j}^{(m)}+(u_j^{(m)})^2\right)\Delta t+\left(X_{t_N}^{(m)}\right)^2\right)\right]\\ &\to \left[\sum_{i=0}^{N-1}\frac{u_i-\mu(\theta,t_i)}{b}\cdot\nabla_\theta\mu(\theta,t_i)\cdot\left(\sum_{j=i}^{N-1}\left(X_{t_j}+(u_j)^2\right)\Delta t+(X_{t_N})^2\right)\right]\quad (M=1)\end{aligned}$$

7. Gradientenverfahren:

$$\theta_{k+1}=\theta_k-\alpha_k\nabla_\theta J(\theta_k)$$

Beweis: Sei die Diskretisierung $\boldsymbol{u}=\{u_i\}_{i=0}^{N-1}$ und die Parametrisierung der Policy π_θ gegeben $u_i\sim\pi_\theta(\cdot\mid t_i)$:

1. Für den Gradienten unseres Problems gilt:

$$\nabla_\theta\mathsf{E}\left[C(\boldsymbol{u})\right]=\sum_{i=0}^{N-1}\mathsf{E}\left[\mathsf{E}\left[\underbrace{\nabla_\theta\log(\pi_\theta(u_i))}_{\sigma(X_{t_i},u_i)\text{-messbar}}\cdot C(u_i,\dots,u_{N-1})\mid\underbrace{X_{t_i}=x,u_i=u}_{\sigma(X_{t_i},u_i)}\right]\right]$$

$$\nabla_\theta J(\theta)=\sum_{i=0}^{N-1}\mathsf{E}\left[\nabla_\theta\log(\pi_\theta(u_i))\cdot\underbrace{\mathsf{E}\left[C(u_i,\dots,u_{N-1})\mid X_{t_i}=x,u_i=u\right]}_{Q^{\pi_\theta}(t_i,X_{t_i},u_i)}\right]$$

2. Für jeden festen Index i und die Value-Function gilt:

$$\mathsf{E}_{u\sim\pi_\theta}\left[\underbrace{\frac{\nabla_\theta\pi_\theta(u_i)}{\pi_\theta(u_i)}}_{\nabla_\theta\log(\pi_\theta(u_i))}\cdot V^{\pi_\theta}(t_i,X_{t_i})\right]=\int_{\mathbb{R}}\nabla_\theta\pi_\theta(u)\cdot\underbrace{V^{\pi_\theta}(t_i,X_{t_i})}_{\text{unabhängig von }u}\,\mathrm{d}u=V^{\pi_\theta}(t_i,X_{t_i})\cdot\nabla_\theta\underbrace{\int_{\mathbb{R}}\pi_\theta(u)\,\mathrm{d}u}_{=1}=0$$

Mittels der Advantage-Function $A:=Q-V$ können wir den Gradienten umschreiben:

$$\nabla_\theta J(\theta)=\mathsf{E}_{u\sim\pi_\theta}\left[\sum_{i=0}^{N-1}\nabla_\theta\log(\pi_\theta(u_i))\cdot A^{\pi_\theta}(t_i,X_{t_i},u_i)\right]$$

3. Wir parametrisieren nun die Value-Function $V_\phi^{\pi_\theta}(t_i,X_{t_i})$ und definieren den n-Schritt Q-Schätzer:

$$Q_n(t_i,X_{t_i},u_i):=\int_{t_i}^{t_i+n\cdot\Delta t}L(X_t,u_t)\,\mathrm{d}t+V_\phi(t_i+n\cdot\Delta t,X_{t_i+n\cdot\Delta t})\approx\sum_{k=0}^{n-1}C_{k+i}+V_\phi(t_{i+n},X_{t_{i+n}})$$

4. Die Value-Function V^{π_θ} erfüllt die Bellman-Gleichung, d.h. es gilt:

$$V^{\pi_\theta}(t_i, X_{t_i}) = \mathsf{E}_{u\sim\pi_\theta}\left[\int_{t_i}^{t_i+n\cdot\Delta t} L(X_t, u_t)\,\mathrm{d}t + V^{\pi_\theta}(t_i + n\cdot\Delta t, X_{t_i+n\cdot\Delta t}) \mid X_{t_i} = x\right]$$

für die parametrisierte Value-Function $V_\phi^{\pi_\theta}$ fordern wir somit näherungsweise:

$$V_\phi(t_i, X_{t_i}) \approx \mathsf{E}_{u\sim\pi_\theta}\left[\underbrace{\sum_{k=0}^{N-1} C_{i+k} + V_\phi(t_{i+N}, X_{t_{i+N}})}_{Q_n(t_i, X_{t_i}, u_i)} \mid X_{t_i} = x\right]$$

Für $V_\phi = V^{\pi_\theta}$ gilt $\mathsf{E}\left[Q_n(t_i, X_{t_i}, u_i) \mid X_{t_i} = x\right] = V_\phi(t_i, x)$. Der bedingte Erwartungswert ist aber genau der Minimierer des mittleren quadratischen Fehlers:

$$\min_\phi \mathsf{E}\left[\left(V_\phi(t_i, X_{t_i}) - Q_n(t_i, X_{t_i}, u_i)\right)^2\right] \Leftrightarrow V_\phi^*(t_i, X_{t_i}) = \mathsf{E}\left[Q_n \mid X_{t_i} = x\right]$$

□

8.2. Actor-Critic Methoden

Beispiel 8.6: Actor-Critic Methode für SDEs. Gegeben sei das Problem:

$$\min_\pi \mathsf{E}\left[\int_0^1 X_t + u_t^2\,\mathrm{d}t + X_1^2\right] \quad \mathrm{d}X_t = (X_t + u_t + 1)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \quad u_t \sim \pi_\theta(\cdot \mid t)$$

Für die Value-Function parametrisiert mit ϕ führen wir folgendes neuronales Netz ein:
Schätzung der Q-Function $Q_n^{\pi_\theta}$

$$Q_n(t_i, X_{t_i}, u_i) := \sum_{k=i}^{N-1} C_{k+i} + C_N = \underbrace{\sum_{k=i}^{N-1}(X_{t_k} + u_k^2)\cdot\Delta t + X_{t_N}^2}_{=R_i}$$

$$\min_\phi \mathsf{E}\left[(V_\phi - Q_n)^2\right] \Rightarrow \min_\phi F(\phi) = \frac{1}{2M}\sum_{m=1}^{M}\sum_{i=0}^{N-1}(V_\phi(t_i, X_{t_i}^{(m)}) - R_i^{(m)})^2$$

$$\Rightarrow \nabla_\phi F(\phi) = \frac{1}{M}\sum_{m=1}^{M}\sum_{i=0}^{N-1}\left(V_\phi(t_i, X_{t_i}^{(m)}) - R_i^{(m)}\right)\cdot\nabla_\phi V_\phi(t_i, X_{t_i}^{(m)})$$

Algorithm 8 Actor-Critic Algorithm for a stochastic Control Problem

Given $\theta_0, \phi_0, \alpha_0, \beta_0, M$

while $\|\theta_{k+1} - \theta_k\| > \epsilon$ **do**

for each episode $m = 1$ to M **do** $X_0^{(m)} = x_0 = 0$

for $i = 0$ to $N - 1$ **do**

Sample action

$$u_i^{(m)} \sim \pi_\theta(\cdot \mid t_i) = \mathcal{N}(\mu(\theta_k, t_i, X_{t_i}^{(m)}), b_k)$$

Simulate state:

$$X_{t_{i+1}}^{(m)} = X_{t_i}^{(m)} + (X_{t_i}^{(m)} + u_i^{(m)} + 1)\Delta t + \sigma \Delta W_{t_i}$$

Compute running costs:

$$C_i^{(m)} = (X_{t_i}^{(m)} + (u_i^{(m)})^2) \cdot \Delta t$$

end for

Compute returns:

$$R_i^{(m)} = \sum_{k=i}^{N-1} C_k^{(m)} + C_N^{(m)} = \sum_{k=i}^{N-1} \left[X_{t_k}^{(m)} + \left(u_k^{(m)} \right)^2 \right] \cdot \Delta t + (X_{t_N}^{(m)})^2$$

▷ MC

end for

Update Actor:

$$\theta_{k+1} \leftarrow \theta_k - \alpha_k \cdot \frac{1}{M} \sum_{m=1}^{M} \sum_{i=0}^{N-1} \underbrace{\frac{u_i^{(m)} - \mu(\theta_k, t_i)}{b} \cdot \nabla_\theta \mu(\theta_k, t_i)}_{\nabla_\theta \log(\pi_{\theta_k}(u_i^{(m)}))} \cdot \left[\underbrace{R_i^{(m)} - V_{\phi_k}(t_i, X_{t_i}^{(m)})}_{A^{\pi_{\theta_k}}(t_i, X_{t_i}^{(m)}, u_i^{(m)})} \right]$$

Update Critic:

$$\phi_{k+1} \leftarrow \phi_k - \beta_k \cdot \frac{1}{M} \sum_{m=1}^{M} \sum_{i=0}^{N-1} (V_{\phi_k}(t_i, X_{t_i}^{(m)}) - R_i^{(m)}) \cdot \nabla_\phi V_{\phi_k}(t_i, X_{t_i}^{(m)})$$

$k \leftarrow k + 1$

end while

8.3. Deterministic Policy Gradient

Wir betrachten das stochastische Steuerungsproblem mit *colored noise* (Ornstein-Uhlenbeck Prozess):

$$\begin{aligned}\min_u J(u) = \min_u \mathsf{E}\left[\int_0^T L(X_t, u_t)\,\mathrm{d}t + \Phi(X_T)\right]\\ \dot{X}_t = f(X_t, u_t + Y_t)\,\mathrm{d}t \quad X_0 = x_0\\ \mathrm{d}Y_t = -\alpha Y_t\,\mathrm{d}t + \sigma\,\mathrm{d}W_t \quad Y_0 = 0 \quad \alpha > 0, \sigma > 0\end{aligned}$$

Wir definieren die Zustandsgröße $\boldsymbol{s} = \begin{bmatrix} t & X_t \end{bmatrix}^T$ und die deterministische Policy $a = u(t, X_t) =: u(\boldsymbol{s})$, parametrisiert durch den Vektor $\theta \in \mathbb{R}^n$, d.h. $u(\boldsymbol{s}) = u_\theta(\boldsymbol{s})$. Die Diskretisierung mittels $\{t_i\}_{i=0}^{N-1}$ liefert:

- Zielfunktional

$$\int_0^T L(X_t, u_t)\,\mathrm{d}t + \Phi(X_N) \approx \Delta t \cdot \sum_{i=0}^{N-1} L(\boldsymbol{s}_i, u_\theta(\boldsymbol{s}_i)) + \Phi(\boldsymbol{s}_N)$$

- Dynamik mittels Euler-Verfahren und der Normalverteilung [11] von Y_t, wobei $\Delta t = t_i - t_{i-1}$:

$$\begin{aligned}Y_{t_{i+1}} &= Y_{t_i} \cdot e^{-\alpha\Delta t} + \sqrt{\frac{\sigma^2}{2\alpha}\left(1 - e^{-2\alpha\Delta t}\right)} \cdot Z_i, \quad Z_i \sim \mathcal{N}(0,1) \quad Y_0 = 0\\ X_{t_{i+1}} &= X_{t_i} + f(X_{t_i}, u_\theta(\boldsymbol{s}_i) + Y_{t_{i+1}}) \cdot \Delta t \quad X_0 = x_0\end{aligned}$$

Für unser Zielfunktional gilt:

$$J(\theta) = \mathsf{E}\left[\sum_{i=0}^{N-1} \underbrace{\Delta t \cdot L(\boldsymbol{s}_i, u_\theta(\boldsymbol{s}_i))}_{-r(\boldsymbol{s}_i, u_\theta(\boldsymbol{s}_i))} + \underbrace{\Phi(\boldsymbol{s}_N)}_{-r_N(\boldsymbol{s}_N)}\right] = \sum_{i=0}^{N-1} \mathsf{E}_{\boldsymbol{s}\sim P_i^{u_\theta}}\left[r(\boldsymbol{s}, u_\theta(\boldsymbol{s}))\right] + \mathsf{E}_{\boldsymbol{s}\sim P_N^{u_\theta}}\left[r_N(\boldsymbol{s})\right]$$

Wir definieren die Q-Function unter der Policy $a_t = u_\theta(\boldsymbol{s}_t)$ als:

$$Q^{u_\theta}(\boldsymbol{s}, a) = \mathsf{E}\left[r(\boldsymbol{s}, a) + \sum_{k=i+1}^{N-1} r(\boldsymbol{s}_k, a_k) + r_N \mid \boldsymbol{s}_i = \boldsymbol{s}, a_i = a\right]$$

[11] Für $\mathrm{d}Y_t = -aY_t\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$, und $\alpha, \sigma \in \mathbb{R}_{>0}$ ist $H_t = e^{-\alpha t}$ die Lösung der homogenen Gleichung $\mathrm{d}H_t = -\alpha H_t\,\mathrm{d}t$. Der Ansatz Ansatz $Y_t = C_t \cdot e^{-\alpha t}$ liefert:

$$\mathrm{d}Y_t = \mathrm{d}\left(C_t \cdot e^{-\alpha t}\right) = \underbrace{\left(\mu_C\,\mathrm{d}t + \sigma_C\,\mathrm{d}W_t\right) \cdot e^{-\alpha t}}_{\mathrm{d}C_t \cdot e^{-\alpha t}} + \underbrace{C_t \cdot \left(-\alpha e^{-\alpha t}\,\mathrm{d}t\right)}_{\mathrm{d}(e^{-\alpha t}) \cdot C_t} + \underbrace{\sigma_C \sigma_H\,\mathrm{d}t}_{\mathrm{d}C_t \cdot \mathrm{d}H_t = 0} \quad \sigma_H = 0$$

Einsetzen des Ansatzes $Y_t = C_t \cdot e^{-\alpha t}$ in die SDE $\mathrm{d}Y_t = -\alpha C_t e^{-\alpha t}\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$ und $Y_0 = C_0$

$$\mu_C = 0 \quad \sigma_C = \sigma e^{\alpha t} \Rightarrow Y_t = Y_0 \cdot e^{-\alpha t} + \int_0^t \sigma e^{\alpha(s-t)}\,\mathrm{d}W_s$$

Y_t ist normalverteilt mit $\mathsf{E}[Y_t] = Y_0 \cdot e^{-\alpha t}$. Die Ito-Isometrie liefert:

$$\mathsf{E}\left[Y_t^2\right] = \int_0^t \mathsf{E}\left[\sigma^2 e^{2\alpha(s-t)}\right]\,\mathrm{d}s = \frac{\sigma^2}{2\alpha}(1 - e^{-2\alpha t}) \Rightarrow Y_t \sim \mathcal{N}\left(Y_0 \cdot e^{-\alpha t}, \frac{\sigma^2}{2\alpha}(1 - e^{-2\alpha t})\right)$$

Satz 8.7: Deterministic Policy Gradient. Sei $J : \mathbb{R}^n \to \mathbb{R}$ gegeben durch:

$$J(\theta) = \sum_{i=0}^{N-1} \mathsf{E}_{\boldsymbol{s}\sim P_i^{u_\theta}} \left[r(\boldsymbol{s}, u_\theta(\boldsymbol{s})) \right] + \mathsf{E}_{\boldsymbol{s}\sim P_N^{u_\theta}} \left[r_N(\boldsymbol{s}) \right]$$

dann ist der Gradient gegeben durch

$$\nabla_\theta J(\theta) = \sum_{i=0}^{N-1} \mathsf{E}_{\boldsymbol{s}\sim P_i^{u_\theta}} \left[\nabla_\theta u_\theta(\boldsymbol{s}) \cdot \nabla_a Q^{u_\theta}(\boldsymbol{s}, a)\Big|_{a=u_\theta(\boldsymbol{s})} \right]$$

Beweis: Da die Policy determistisch ist, gilt für die Value-Function

$$V^{u_\theta}(\boldsymbol{s}) = \mathsf{E}_{a\sim u_\theta(\boldsymbol{s})} \left[Q^{u_\theta}(a, \boldsymbol{s}) \right] = Q^{u_\theta}(a, \boldsymbol{s})\Big|_{a=u_\theta(\boldsymbol{s})}$$

Wir erhalten für unser Zielfunktional

$$J(\theta) = V^{u_\theta}(\boldsymbol{s}_0) = Q^{u_\theta}(a, \boldsymbol{s}_0)\Big|_{a=u_\theta(\boldsymbol{s}_0)} \quad \boldsymbol{s}_0 = \begin{bmatrix} 0 \\ X_0 \end{bmatrix}$$

Diese erfüllt die Bellmanngleichung

$$\begin{aligned} Q^{u_\theta}(\boldsymbol{s}, a)\Big|_{a=u_\theta(\boldsymbol{s})} &= r(\boldsymbol{s}, a) + \mathsf{E}_{\boldsymbol{s}'\sim P_a(\boldsymbol{s},\boldsymbol{s}')} \left[Q^{u_\theta}(\boldsymbol{s}', u_\theta(\boldsymbol{s}')) \right] \Big|_{a=u_\theta(\boldsymbol{s})} \\ \Leftrightarrow V^{u_\theta}(\boldsymbol{s}) &= r(\boldsymbol{s}, u_\theta(\boldsymbol{s})) + \int_{\mathcal{S}} P(\boldsymbol{s}', \boldsymbol{s}, u_\theta(\boldsymbol{s})) \cdot \underbrace{Q^{u_\theta}(\boldsymbol{s}', u_\theta(\boldsymbol{s}'))}_{V^{u_\theta}(\boldsymbol{s}')} \mathrm{d}\boldsymbol{s}' \end{aligned}$$

Differenzieren nach dem Parameter θ liefert:

$$\begin{aligned} \nabla_\theta V^{u_\theta}(\boldsymbol{s}) &= \nabla_a r(\boldsymbol{s}, a)\Big|_{a=u_\theta(\boldsymbol{s})} \cdot \frac{\partial u_\theta(\boldsymbol{s})}{\partial \theta} \\ &\quad + \int_{\mathcal{S}} \nabla_a P(\boldsymbol{s}', \boldsymbol{s}, a)\Big|_{a=u_\theta(\boldsymbol{s})} \cdot \frac{\partial u_\theta(\boldsymbol{s})}{\partial \theta} \cdot V^{u_\theta}(\boldsymbol{s}') + P(\boldsymbol{s}', \boldsymbol{s}, u_\theta(\boldsymbol{s})) \cdot \nabla_\theta V^{u_\theta}(\boldsymbol{s}') \mathrm{d}\boldsymbol{s}' \\ &= \frac{\partial u_\theta(\boldsymbol{s})}{\partial \theta} \cdot \nabla_a \left[r(\boldsymbol{s}, a) + \int_{\mathcal{S}} P(\boldsymbol{s}', \boldsymbol{s}, a) \cdot V^{u_\theta}(\boldsymbol{s}') \mathrm{d}\boldsymbol{s}' \right]_{a=u_\theta(\boldsymbol{s})} + \int_{\mathcal{S}} P(\boldsymbol{s}', \boldsymbol{s}, u_\theta(\boldsymbol{s})) \cdot \nabla_\theta V^{u_\theta}(\boldsymbol{s}') \mathrm{d}\boldsymbol{s}' \end{aligned}$$

Einführen der Q-Function liefert nun:

$$\begin{aligned} \nabla_\theta V^{u_\theta}(\boldsymbol{s}) &= \frac{\partial u_\theta(\boldsymbol{s})}{\partial \theta} \cdot \nabla_a Q^{u_\theta}(\boldsymbol{s}, a)\Big|_{a=u_\theta(\boldsymbol{s})} \\ &\quad + \int_{\mathcal{S}} P(\boldsymbol{s}', \boldsymbol{s}, u_\theta(\boldsymbol{s})) \cdot \nabla_\theta V^{u_\theta}(\boldsymbol{s}') \mathrm{d}\boldsymbol{s}' \end{aligned}$$

Für den Ausdruck $\nabla_\theta V^{u_\theta}(\boldsymbol{s}')$ gilt entsprechend

$$\begin{aligned} \nabla_\theta V^{u_\theta}(\boldsymbol{s}') &= \frac{\partial u_\theta(\boldsymbol{s}')}{\partial \theta} \cdot Q^{u_\theta}(\boldsymbol{s}', a)\Big|_{a=u_\theta(\boldsymbol{s}')} \\ &\quad + \int_{\mathcal{S}} P(\boldsymbol{s}'', \boldsymbol{s}', u_\theta(\boldsymbol{s}')) \cdot \nabla_\theta V^{u_\theta}(\boldsymbol{s}'') \mathrm{d}\boldsymbol{s}'' \end{aligned}$$

Einsetzen in den Ausdruck für $\nabla_\theta V^{u_\theta}(\boldsymbol{s})$ liefert

$$\begin{aligned}
\nabla_\theta V^{u_\theta}(\boldsymbol{s}) &= \frac{\partial u_\theta(\boldsymbol{s})}{\partial \theta} \cdot \nabla_a Q^{u_\theta}(\boldsymbol{s}, a)\Big|_{a=u_\theta(\boldsymbol{s})} \\
&\quad + \int_{\mathcal{S}} P(\boldsymbol{s}', \boldsymbol{s}, u_\theta(\boldsymbol{s})) \cdot \frac{\partial u_\theta(\boldsymbol{s}')}{\partial \theta} \cdot \nabla_a Q^{u_\theta}(\boldsymbol{s}', a)\Big|_{a=u_\theta(\boldsymbol{s}')} \mathrm{d}\boldsymbol{s}' \\
&\quad + \int_{\mathcal{S}} P(\boldsymbol{s}', \boldsymbol{s}, u_\theta(\boldsymbol{s})) \int_{\mathcal{S}} P(\boldsymbol{s}'', \boldsymbol{s}', u_\theta(\boldsymbol{s}')) \cdot \nabla_\theta V^{u_\theta}(\boldsymbol{s}'') \mathrm{d}\boldsymbol{s}'' \mathrm{d}\boldsymbol{s}' \\
&= \int_{\mathcal{S}} \underbrace{P^{(0)}(\boldsymbol{s}', \boldsymbol{s}, u_\theta(\boldsymbol{s}))}_{=1 \ \boldsymbol{s}'=\boldsymbol{s}} \cdot \frac{\partial u_\theta(\boldsymbol{s}')}{\partial \theta} \cdot \nabla_a Q^{u_\theta}(\boldsymbol{s}', a)\Big|_{a=u_\theta(\boldsymbol{s}')} \mathrm{d}\boldsymbol{s}' \\
&\quad + \int_{\mathcal{S}} P^{(1)}(\boldsymbol{s}', \boldsymbol{s}) \cdot \frac{\partial u_\theta(\boldsymbol{s}')}{\partial \theta} \cdot \nabla_a Q^{u_\theta}(\boldsymbol{s}', a)\Big|_{a=u_\theta(\boldsymbol{s}')} \mathrm{d}\boldsymbol{s}' \\
&\quad + \int_{\mathcal{S}} P^{(2)}(\boldsymbol{s}', \boldsymbol{s}) \cdot \nabla_\theta V^{u_\theta}(\boldsymbol{s}') \mathrm{d}\boldsymbol{s}'
\end{aligned}$$

Hierbei bezeichnet $P^{(i)}(\boldsymbol{s}', \boldsymbol{s}, u_\theta)$ die Übergangsdichte von $\boldsymbol{s}$ nach $\boldsymbol{s}'$ in i Zeitschritten unter u_θ. Nach $N-1$ Zeitschritten erreichen wir den Zustand $\boldsymbol{s}_N$ und der Prozess endet.

$$\nabla_\theta V^{u_\theta}(\boldsymbol{s}) = \sum_{i=0}^{N-1} \int_{\mathcal{S}} P^{(i)}(\boldsymbol{s}', \boldsymbol{s}, u_\theta) \cdot \frac{\partial u_\theta(\boldsymbol{s}')}{\partial \theta} \cdot \nabla_a Q^{u_\theta}(\boldsymbol{s}', a)\Big|_{a=u_\theta(\boldsymbol{s}')} \mathrm{d}\boldsymbol{s}'$$

Für einen bekannten Startzustand $P\left(\boldsymbol{s}_0 = \begin{bmatrix} 0 \\ x_0 \end{bmatrix}\right) = 1$ gilt somit:

$$\begin{aligned}
\nabla_\theta J(\theta) &= \sum_{i=0}^{N-1} \int_{\mathcal{S}} P^{(i)}(\boldsymbol{s}', \boldsymbol{s}_0, u_\theta) \cdot \frac{\partial u_\theta(\boldsymbol{s}')}{\partial \theta} \cdot \nabla_a Q^{u_\theta}(\boldsymbol{s}', a)\Big|_{a=u_\theta(\boldsymbol{s}')} \mathrm{d}\boldsymbol{s}' \\
&= \sum_{i=0}^{N-1} \mathsf{E}_{\boldsymbol{s} \sim P_i^{u_\theta}} \left[\nabla_\theta u_\theta(\boldsymbol{s}) \cdot \nabla_a Q^{u_\theta}(\boldsymbol{s}, a)\Big|_{a=u_\theta(\boldsymbol{s})} \right]
\end{aligned}$$

Hierbei bezeichnet $P_i^{u_\theta}$ die Verteilung des Zustandes $\boldsymbol{s}$ im Zeitpunkt t_i unter der Policy u_θ. □

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A. Herleitungen

A.1. Neural ODEs

Für die Jacobi-Matrix $\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{u}} : \mathbb{R}^m \to \mathbb{R}^n$ gilt:

$$\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{u}} = \left(\underbrace{\frac{\partial \boldsymbol{f}}{\partial W^{(1)}}}_{n\times(h\cdot n)} \quad \underbrace{\frac{\partial \boldsymbol{f}}{\partial b^{(1)}}}_{n\times h} \quad \underbrace{\frac{\partial \boldsymbol{f}}{\partial W^{(N)}}}_{n\times(n\cdot h)} \quad \underbrace{\frac{\partial \boldsymbol{f}}{\partial b^{(N)}}}_{n\times n} \right)$$

Die konkreten Matrizen werden konkret spaltenweise gebaut, d.h. für die $i-$te Zeile von $\frac{\partial \boldsymbol{f}}{\partial W^{(1)}}$:

$$\left(\underbrace{\frac{\partial \boldsymbol{f}_i}{\partial W^{(1)}_{11}} \cdots \frac{\partial \boldsymbol{f}_i}{\partial W^{(1)}_{h1}}}_{\text{1. Spalte von } W^{(1)}} \Bigg| \underbrace{\frac{\partial \boldsymbol{f}_i}{\partial W^{(1)}_{12}} \cdots \frac{\partial \boldsymbol{f}_i}{\partial W^{(1)}_{h2}}}_{\text{2. Spalte von } W^{(1)}} \Bigg| \cdots \Bigg| \underbrace{\frac{\partial \boldsymbol{f}_i}{\partial W^{(1)}_{1n}} \cdots \frac{\partial \boldsymbol{f}_i}{\partial W^{(1)}_{hn}}}_{n \text{ Spalte von } W^{(1)}} \right) \in \mathbb{R}^{1\times(h\cdot n)}$$

Transponieren wir die gesamte Matrix $\left(\frac{\partial \boldsymbol{f}}{\partial W^{(1)}}\right)^T$ so erhalten wir für $\left(\frac{\partial \boldsymbol{f}}{\partial W^{(1)}}\right)^T \cdot \boldsymbol{p}(t)$:

$$\left(\begin{array}{cccc}
\frac{\partial \boldsymbol{f}_1}{\partial W^{(1)}_{11}} & \frac{\partial \boldsymbol{f}_2}{\partial W^{(1)}_{11}} & \cdots & \frac{\partial \boldsymbol{f}_n}{\partial W^{(1)}_{11}} \\
\vdots & \vdots & & \vdots \\
\frac{\partial \boldsymbol{f}_1}{\partial W^{(1)}_{h1}} & \frac{\partial \boldsymbol{f}_2}{\partial W^{(1)}_{h1}} & \cdots & \frac{\partial \boldsymbol{f}_n}{\partial W^{(1)}_{h1}} \\
\hline
\frac{\partial \boldsymbol{f}_1}{\partial W^{(1)}_{12}} & \frac{\partial \boldsymbol{f}_2}{\partial W^{(1)}_{12}} & \cdots & \frac{\partial \boldsymbol{f}_n}{\partial W^{(1)}_{12}} \\
\vdots & \vdots & & \vdots \\
\frac{\partial \boldsymbol{f}_1}{\partial W^{(1)}_{h2}} & \frac{\partial \boldsymbol{f}_2}{\partial W^{(1)}_{h2}} & \cdots & \frac{\partial \boldsymbol{f}_n}{\partial W^{(1)}_{h2}} \\
\hline
\vdots & \vdots & & \vdots \\
\hline
\frac{\partial \boldsymbol{f}_1}{\partial W^{(1)}_{1n}} & \frac{\partial \boldsymbol{f}_2}{\partial W^{(1)}_{1n}} & \cdots & \frac{\partial \boldsymbol{f}_n}{\partial W^{(1)}_{1n}} \\
\vdots & \vdots & & \vdots \\
\frac{\partial \boldsymbol{f}_1}{\partial W^{(1)}_{hn}} & \frac{\partial \boldsymbol{f}_2}{\partial W^{(1)}_{hn}} & \cdots & \frac{\partial \boldsymbol{f}_n}{\partial W^{(1)}_{hn}}
\end{array}\right) \cdot \begin{pmatrix} \boldsymbol{p}_1(t) \\ \boldsymbol{p}_2(t) \\ \vdots \\ \boldsymbol{p}_n(t) \end{pmatrix} = \left(\begin{array}{c}
\sum_{i=1}^n \frac{\partial \boldsymbol{f}_i}{\partial W^{(1)}_{11}} \boldsymbol{p}_i(t) \\
\vdots \\
\sum_{i=1}^n \frac{\partial \boldsymbol{f}_i}{\partial W^{(1)}_{h1}} \boldsymbol{p}_i(t) \\
\hline
\sum_{i=1}^n \frac{\partial \boldsymbol{f}_i}{\partial W^{(1)}_{12}} \boldsymbol{p}_i(t) \\
\vdots \\
\sum_{i=1}^n \frac{\partial \boldsymbol{f}_i}{\partial W^{(1)}_{h2}} \boldsymbol{p}_i(t) \\
\hline
\vdots \\
\hline
\sum_{i=1}^n \frac{\partial \boldsymbol{f}_i}{\partial W^{(1)}_{1n}} \boldsymbol{p}_i(t) \\
\vdots \\
\sum_{i=1}^n \frac{\partial \boldsymbol{f}_i}{\partial W^{(1)}_{hn}} \boldsymbol{p}_i(t)
\end{array}\right) = \left(\begin{array}{c}
\boldsymbol{\alpha}_1 \boldsymbol{\delta}_1 \boldsymbol{x}_1 \\
\vdots \\
\boldsymbol{\alpha}_h \boldsymbol{\delta}_h \boldsymbol{x}_1 \\
\hline
\boldsymbol{\alpha}_1 \boldsymbol{\delta}_1 \boldsymbol{x}_2 \\
\vdots \\
\boldsymbol{\alpha}_h \boldsymbol{\delta}_h \boldsymbol{x}_2 \\
\hline
\vdots \\
\hline
\boldsymbol{\alpha}_1 \boldsymbol{\delta}_1 \boldsymbol{x}_n \\
\vdots \\
\boldsymbol{\alpha}_h \boldsymbol{\delta}_h \boldsymbol{x}_n
\end{array}\right)$$

Für die einzelne Komponente gilt nun:

$$\sum_{i=1}^n \frac{\partial \boldsymbol{f}_i}{\partial W^{(1)}_{jk}} \cdot \boldsymbol{p}_i(t) = \sum_{i=1}^n \boldsymbol{\alpha}_j W^{(N)}_{ij} \boldsymbol{p}_i(t) \boldsymbol{x}_k = \boldsymbol{\alpha}_j \cdot \Bigg(\underbrace{\left(W^{(N)}\right)^T \cdot \boldsymbol{p}(t)}_{:=\boldsymbol{\delta}} \Bigg)_j \cdot \boldsymbol{x}_k = \left(\boldsymbol{\alpha} \odot \boldsymbol{\delta}\right)_j \cdot \boldsymbol{x}_k$$

Für den Beitrag zum Gradienten erhalten wir:

$$\nabla_{W^{(1)}} J(\boldsymbol{u}) = (\boldsymbol{\alpha} \odot \boldsymbol{\delta}) \cdot \boldsymbol{x}^T \in \mathbb{R}^{h\times n}$$

Für die i-te Zeile von $\frac{\partial \boldsymbol{f}}{\partial W^{(N)}}$ gilt:

$$\left(\underbrace{\frac{\partial \boldsymbol{f}_i}{\partial W^{(N)}_{11}} \cdots \frac{\partial \boldsymbol{f}_i}{\partial W^{(N)}_{n1}}}_{\text{1. Spalte von } W^{(N)}} \Bigg| \underbrace{\frac{\partial \boldsymbol{f}_i}{\partial W^{(N)}_{12}} \cdots \frac{\partial \boldsymbol{f}_i}{\partial W^{(N)}_{n2}}}_{\text{2. Spalte von } W^{(N)}} \Bigg| \cdots \Bigg| \underbrace{\frac{\partial \boldsymbol{f}_i}{\partial W^{(N)}_{1h}} \cdots \frac{\partial \boldsymbol{f}_i}{\partial W^{(N)}_{nh}}}_{h \text{ Spalte von } W^{(N)}} \right) \in \mathbb{R}^{1\times(n\cdot h)}$$

Aufstellen der Matrix $\left(\frac{\partial \boldsymbol{f}}{\partial W^{(N)}}\right)^T \boldsymbol{p}$:

$$\left(\frac{\partial \boldsymbol{f}}{\partial W^{(N)}}\right)^T \cdot \boldsymbol{p}(t) = \left(\underbrace{\sum_{i=1}^{n} \frac{\partial \boldsymbol{f}_i}{\partial W_{11}^{(N)}} \cdot \boldsymbol{p}_i \cdots \sum_{i=1}^{n} \frac{\partial \boldsymbol{f}_i}{\partial W_{n1}^{(N)}} \cdot \boldsymbol{p}_i}_{\text{1. Spalte von } W^{(N)}} \Bigg| \cdots \Bigg| \underbrace{\sum_{i=1}^{n} \frac{\partial \boldsymbol{f}_i}{\partial W_{1h}^{(N)}} \cdot \boldsymbol{p}_i \cdots \sum_{i=1}^{n} \frac{\partial \boldsymbol{f}_i}{\partial W_{nh}^{(N)}} \cdot \boldsymbol{p}_i}_{h \text{ Spalte von } W^{(N)}} \right)^T$$

$$= \left(\boldsymbol{a}_1\boldsymbol{p}_1 \cdots \boldsymbol{a}_1\boldsymbol{p}_n \Big| \boldsymbol{a}_2\boldsymbol{p}_1 \cdots \boldsymbol{a}_2\boldsymbol{p}_n \Big| \cdots \Big| \boldsymbol{a}_h\boldsymbol{p}_1 \cdots \boldsymbol{a}_h\boldsymbol{p}_n \right)^T$$

Für die einzelne Komponente gilt:

$$\sum_{i=1}^{n} \frac{\partial \boldsymbol{f}_i}{\partial W_{jk}^{(N)}} \cdot \boldsymbol{p}_i = \sum_{i=1}^{n} \delta_{ij} \cdot \boldsymbol{a}_k \cdot \boldsymbol{p}_i = \boldsymbol{a}_k \cdot \boldsymbol{p}_j$$

Für den Beitrag zum Gradienten gilt:

$$\nabla_{W^{(N)}} J(\boldsymbol{u}) = \boldsymbol{p} \cdot \boldsymbol{a}^T$$

Für die Ableitung $\left(\frac{\partial \boldsymbol{f}}{\partial b^{(1)}}\right)^T \boldsymbol{p}(t)$

$$\begin{pmatrix} \frac{\partial \boldsymbol{f}_1}{\partial b_1^{(1)}} & \cdots & \frac{\partial \boldsymbol{f}_n}{\partial b_1^{(1)}} \\ \vdots & \ddots & \vdots \\ \frac{\partial \boldsymbol{f}_1}{\partial b_h^{(1)}} & \cdots & \frac{\partial \boldsymbol{f}_n}{\partial b_h^{(1)}} \end{pmatrix} \cdot \begin{pmatrix} \boldsymbol{p}_1 \\ \vdots \\ \boldsymbol{p}_n \end{pmatrix} = \begin{pmatrix} \sum_{i=1}^{n} \frac{\partial \boldsymbol{f}_i}{\partial b_1^{(1)}} \cdot \boldsymbol{p}_i \\ \vdots \\ \sum_{i=1}^{n} \frac{\partial \boldsymbol{f}_i}{\partial b_h^{(1)}} \cdot \boldsymbol{p}_i \end{pmatrix} = \begin{pmatrix} \boldsymbol{\alpha}_1\boldsymbol{\delta}_1 \\ \vdots \\ \boldsymbol{\alpha}_h\boldsymbol{\delta}_h \end{pmatrix} = \boldsymbol{\alpha} \odot \boldsymbol{\delta}$$

Für die einzelnen Komponenten gilt:

$$\sum_{i=1}^{n} \frac{\partial \boldsymbol{f}_i}{\partial b_j^{(1)}} \cdot \boldsymbol{p}_i = \sum_{i=1}^{n} W_{ij}^{(N)} \boldsymbol{\alpha}_j \cdot \boldsymbol{p}_i = \boldsymbol{\alpha}_j \boldsymbol{\delta}_j$$

Für die Ableitung $\left(\frac{\partial \boldsymbol{f}}{\partial b^{(N)}}\right)^T \boldsymbol{p}(t)$

$$\begin{pmatrix} \frac{\partial \boldsymbol{f}_1}{\partial b_1^{(N)}} & \cdots & \frac{\partial \boldsymbol{f}_n}{\partial b_1^{(N)}} \\ \vdots & \ddots & \vdots \\ \frac{\partial \boldsymbol{f}_1}{\partial b_n^{(N)}} & \cdots & \frac{\partial \boldsymbol{f}_n}{\partial b_n^{(N)}} \end{pmatrix} \cdot \begin{pmatrix} \boldsymbol{p}_1 \\ \vdots \\ \boldsymbol{p}_n \end{pmatrix} = \begin{pmatrix} \sum_{i=1}^{n} \frac{\partial \boldsymbol{f}_i}{\partial b_1^{(N)}} \cdot \boldsymbol{p}_i \\ \vdots \\ \sum_{i=1}^{n} \frac{\partial \boldsymbol{f}_i}{\partial b_n^{(N)}} \cdot \boldsymbol{p}_i \end{pmatrix} = \begin{pmatrix} \boldsymbol{p}_1 \\ \vdots \\ \boldsymbol{p}_n \end{pmatrix} = \boldsymbol{p}(t)$$

Für die einzelnen Komponenten gilt:

$$\sum_{i=1}^{n} \frac{\partial \boldsymbol{f}_i}{\partial b_j^{(N)}} \cdot \boldsymbol{p}_i = \sum_{i=1}^{n} \delta_{ij} \cdot \boldsymbol{p}_i = \boldsymbol{p}_j$$

Für die Ableitung unserer Zielfunktion gilt:

$$\nabla_{\boldsymbol{u}} J(\boldsymbol{u}) = \int_{t_0}^{t_f} \left(\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{u}}\right)^T \cdot \boldsymbol{p}(t)\,\mathrm{d}t = \begin{bmatrix} \int_{t_0}^{t_f} \left(\frac{\partial \boldsymbol{f}}{\partial W^{(1)}}\right)^T \cdot \boldsymbol{p}(t)\,\mathrm{d}t \\ \int_{t_0}^{t_f} \left(\frac{\partial \boldsymbol{f}}{\partial b^{(1)}}\right)^T \cdot \boldsymbol{p}(t)\,\mathrm{d}t \\ \int_{t_0}^{t_f} \left(\frac{\partial \boldsymbol{f}}{\partial W^{(N)}}\right)^T \cdot \boldsymbol{p}(t)\,\mathrm{d}t \\ \int_{t_0}^{t_f} \left(\frac{\partial \boldsymbol{f}}{\partial b^{(N)}}\right)^T \cdot \boldsymbol{p}(t)\,\mathrm{d}t \end{bmatrix}$$

A.2. LNN Herleitung

Seien M Trejaktorien zu N Zeitpunkten im Zustandsraum $\mathbb{R}^2$ vorhanden, so speichern wir die Information in $\boldsymbol{Q}, \dot{\boldsymbol{Q}}, \boldsymbol{U}, \ddot{\boldsymbol{Q}} \in \mathbb{R}^{N\times M}$.

$$\boldsymbol{Q} = \begin{bmatrix} q_1^{(0)} & \cdots & q_1^{(M-1)} \\ \vdots & \ddots & \vdots \\ q_N^{(0)} & \cdots & q_N^{(M-1)} \end{bmatrix} \quad \boldsymbol{Q}_i = \begin{bmatrix} q_i^{(0)} & \cdots & q_i^{(M-1)} \end{bmatrix}$$

Wir definieren weiter $\boldsymbol{X} \in \mathbb{R}^{N\times 2\times M}$ und die Auswertung in einem Zeitpunkt als $\boldsymbol{X}_i \in \mathbb{R}^{2\times M}$ durch

$$\boldsymbol{X}_i = \begin{bmatrix} \boldsymbol{Q}_i \\ \dot{\boldsymbol{Q}}_i \end{bmatrix} = \begin{bmatrix} q_i^{(0)} & \cdots & q_i^{(M-1)} \\ \dot{q}_i^{(0)} & \cdots & \dot{q}_i^{(M-1)} \end{bmatrix} \quad \left[\boldsymbol{X}_i^{(j)}\right] = \begin{bmatrix} q_i^{(j)} \\ \dot{q}_i^{(j)} \end{bmatrix}$$

Für die Lagrange-Funktion gilt nun mit Aktivierungsfunktion $\boldsymbol{\sigma}$

$$\boldsymbol{L}(\boldsymbol{X}_i) = \underbrace{W^{(N)}}_{1\times h}\boldsymbol{\sigma}\left(\underbrace{W^{(1)}\cdot\boldsymbol{X}_i}_{h\times M} + b^{(1)}\cdot\mathbf{1}_M^T\right) + \underbrace{b^{(N)}\mathbf{1}_M^T}_{1\times M} \in \mathbb{R}^{1\times M}$$

Wir definieren $\boldsymbol{A} \in \mathbb{R}^{N\times h\times M}$ und ausgewertet in einem Zeitpunkt t_i als $\boldsymbol{A}_i \in \mathbb{R}^{h\times M}$ durch:

$$\left[\boldsymbol{A}_i^{(j)}\right]_k := \boldsymbol{\sigma}\left(\sum_{s=1}^{2} W_{ks}^{(1)}\left[\boldsymbol{X}_i^{(j)}\right]_s + b_k^{(1)}\right) \quad \left[\boldsymbol{z}_i^{(j)}\right]_k := \sum_{s=1}^{2} W_{ks}^{(1)}\left[\boldsymbol{X}_i^{(j)}\right]_s + b_k^{(1)}$$

$$\boldsymbol{A}_i := \begin{bmatrix} | & & | \\ \boldsymbol{A}_i^{(0)} & \cdots & \boldsymbol{A}_i^{(M-1)} \\ | & & | \end{bmatrix} = \boldsymbol{\sigma}\left[W^{(1)}\cdot\boldsymbol{X}_i + b^{(1)}\cdot\mathbf{1}_M^T\right]$$

Wir verwenden für die Aktivierungsfunktion tanh und softplus:[12]

	tanh	softplus
$\left[\boldsymbol{A}_i^{(j)}\right]_k := \boldsymbol{\sigma}\left(\left[\boldsymbol{z}_i^{(j)}\right]_k\right)$	$\tanh\left(\left[\boldsymbol{z}_i^{(j)}\right]_k\right)$	$\log\left(1+\exp\left[\boldsymbol{z}_i^{(j)}\right]_k\right)$
$\left[\boldsymbol{\alpha}_i^{(j)}\right]_k := \left.\frac{\partial\boldsymbol{\sigma}(\boldsymbol{z})}{\partial\boldsymbol{z}}\right\|_{\boldsymbol{z}=\left[\boldsymbol{z}_i^{(j)}\right]_k}$	$1-\left[\boldsymbol{A}_i^{(j)}\right]_k^2$	$\frac{\exp\left[\boldsymbol{z}_i^{(j)}\right]_k}{1+\exp\left[\boldsymbol{z}_i^{(j)}\right]_k}$
$\left[\boldsymbol{\beta}_i^{(j)}\right]_k := \left.\frac{\partial\boldsymbol{\alpha}(\boldsymbol{z})}{\partial\boldsymbol{z}}\right\|_{\boldsymbol{z}=\left[\boldsymbol{z}_i^{(j)}\right]_k}$	$(-2)\cdot\left[\boldsymbol{A}_i^{(j)}\right]_k\cdot\left[\boldsymbol{\alpha}_i^{(j)}\right]_k$	$\left[\boldsymbol{\alpha}_i^{(j)}\right]_k\cdot\left(1-\left[\boldsymbol{\alpha}_i^{(j)}\right]_k\right)$
$\left[\boldsymbol{\gamma}_i^{(j)}\right]_k := \left.\frac{\partial\boldsymbol{\beta}(\boldsymbol{z})}{\partial\boldsymbol{z}}\right\|_{\boldsymbol{z}=\left[\boldsymbol{z}_i^{(j)}\right]_k}$	$2\cdot\left[\boldsymbol{\alpha}_i^{(j)}\right]_k\cdot\left(3\cdot\left[\boldsymbol{A}_i^{(j)}\right]_k^2-1\right)$	$\left[\boldsymbol{\beta}_i^{(j)}\right]_k\cdot\left(1-2\cdot\left[\boldsymbol{\alpha}_i^{(j)}\right]_k\right)$

Wir definieren $\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma} \in \mathbb{R}^{N\times h\times M}$ und für einen festen Zeitpunkt als $\boldsymbol{\alpha}_i, \boldsymbol{\beta}_i, \boldsymbol{\gamma}_i \in \mathbb{R}^{h\times M}$.

[12] Let f, g be defined as $f(z) := \tanh(z)$ and $g(z) = \log(1+e^z)$, the derivatives are given by

$$f'(z) = 1-\tanh(z)^2 \quad f''(z) = (-2)\cdot f(z)\cdot f'(z) \quad f^{(3)}(z) = (-2)\cdot f'(z)\left(1-f(z)^2\right)+4\cdot f(z)^2 f'(z) = 6f'(z)f(z)^2-2f'(z)$$

$$g'(z) = \frac{e^z}{1+e^z} \quad g''(z) = \frac{e^z}{(1+e^z)^2} = g'(z)\cdot(1-g'(z)) \quad g^{(3)}(z) = g''(z)\cdot[1-2\cdot g'(z)]$$

Sei h gegeben dann gilt mit $\boldsymbol{\theta} \in \mathbb{R}^{4h+1}$ für die ersten Ableitungen der Lagrange-Funktion: [13]

$$\frac{\partial \boldsymbol{L}(\boldsymbol{X}_i)}{\partial q} = W^{(N)} \cdot \left[\boldsymbol{\alpha}_i \odot \left(\left[W^{(1)}\right]^{(1)} \cdot \mathbf{1}_M^T\right)\right] \in \mathbb{R}^{1\times M}$$
$$\frac{\partial \boldsymbol{L}(\boldsymbol{X}_i)}{\partial \dot{q}} = W^{(N)} \cdot \left(\boldsymbol{\alpha}_i \odot \left(\left[W^{(1)}\right]^{(2)} \cdot \mathbf{1}_M^T\right)\right) \in \mathbb{R}^{1\times M}$$

Für die Ableitung von $\partial_q \boldsymbol{L}$ koponentweise nach den Parametern gilt:

$$\frac{\partial}{\partial W_{k1}^{(1)}} \frac{\partial \boldsymbol{L}(\boldsymbol{X}_i^{(j)})}{\partial q} = W_{1k}^{(N)} \cdot \left(\left[\boldsymbol{\beta}_i^{(j)}\right]_k \cdot \left[\boldsymbol{X}_i^{(j)}\right]_1 \cdot W_{k1}^{(1)} + \left[\boldsymbol{\alpha}_i^{(j)}\right]_k\right)$$
$$\frac{\partial}{\partial W_{k2}^{(1)}} \frac{\partial \boldsymbol{L}(\boldsymbol{X}_i^{(j)})}{\partial q} = W_{1k}^{(N)} \cdot \left[\boldsymbol{\beta}_i^{(j)}\right]_k \cdot \left[\boldsymbol{X}_i^{(j)}\right]_2 \cdot W_{k1}^{(1)}$$
$$\frac{\partial}{\partial b_k^{(1)}} \frac{\partial \boldsymbol{L}(\boldsymbol{X}_i^{(j)})}{\partial q} = W_{1k}^{(N)} \cdot \left[\boldsymbol{\beta}_i^{(j)}\right]_k \cdot W_{k1}^{(1)}$$
$$\frac{\partial}{\partial W_{1k}^{(N)}} \frac{\partial \boldsymbol{L}(\boldsymbol{X}_i^{(j)})}{\partial q} = \left[\boldsymbol{\alpha}_i^{(j)}\right]_k \cdot W_{k1}^{(1)}$$

Für alle Trajektorien parallel gilt im Batch-Format:

$$\frac{\partial}{\partial \left[W^{(1)}\right]^{(1)}} \frac{\partial \boldsymbol{L}(\boldsymbol{X}_i)}{\partial q} = \left[\left(W^{(N)}\right)^T \cdot \mathbf{1}_M^T\right] \odot \left(\boldsymbol{\beta}_i \odot [\mathbf{1}_h \cdot \boldsymbol{Q}_i] \odot \left[\left(W^{(1)}\right)^{(1)} \cdot \mathbf{1}_M^T\right] + \boldsymbol{\alpha}_i\right)$$
$$\frac{\partial}{\partial \left[W^{(1)}\right]^{(2)}} \frac{\partial \boldsymbol{L}(\boldsymbol{X}_i)}{\partial q} = \left[\left(W^{(N)}\right)^T \cdot \mathbf{1}_M^T\right] \odot \boldsymbol{\beta}_i \odot \left[\mathbf{1}_h \cdot \dot{\boldsymbol{Q}}_i\right] \odot \left[\left(W^{(1)}\right)^{(1)} \cdot \mathbf{1}_M^T\right]$$
$$\frac{\partial}{\partial b^{(1)}} \frac{\partial \boldsymbol{L}(\boldsymbol{X}_i)}{\partial q} = \left[\left(W^{(N)}\right)^T \cdot \mathbf{1}_M^T\right] \odot \boldsymbol{\beta}_i \odot \left[\left(W^{(1)}\right)^{(1)} \cdot \mathbf{1}_M^T\right]$$
$$\frac{\partial}{\partial W^{(N)}} \frac{\partial \boldsymbol{L}(\boldsymbol{X}_i)}{\partial q} = \boldsymbol{\alpha}_i \odot \left[\left(W^{(1)}\right)^{(1)} \cdot \mathbf{1}_M^T\right]$$

Für die zweiten Ableitungen der Lagrange-Funktion gilt: [14]

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[13] For a fixed state $\left[\boldsymbol{X}_i^{(j)}\right] = \left[q_i^{(j)} \quad \dot{q}_i^{(j)}\right]^T$ and time the Lagrangian is a scalar

$$\boldsymbol{L}(\boldsymbol{X}_i^{(j)}) = \sum_{k=1}^{h} W_{1k}^{(N)} \cdot \sigma\left(\sum_{s=1}^{2} W_{ks}^{(1)} \left[\boldsymbol{X}_i^{(j)}\right]_s + b_k^{(1)}\right) + b^{(N)}$$

hence the partial derivatives with respect to q and $\dot{q}$ are given by

$$\frac{\partial \boldsymbol{L}(\boldsymbol{X}_i^{(j)})}{\partial q} = \sum_{k=1}^{h} W_{1k}^{(N)} \cdot \frac{\partial}{\partial z}\sigma\left(z = \sum_{s=1}^{2} W_{ks}^{(1)} \left[\boldsymbol{X}_i^{(j)}\right]_s + b_k^{(1)}\right) \cdot W_{k1}^{(1)} = W^{(N)} \cdot \left(\left[\boldsymbol{\alpha}_i^{(j)}\right] \odot \left[W^{(1)}\right]^{(1)}\right)$$
$$\frac{\partial \boldsymbol{L}(\boldsymbol{X}_i^{(j)})}{\partial \dot{q}} = \sum_{k=1}^{h} W_{1k}^{(N)} \cdot \frac{\partial}{\partial z}\sigma\left(z = \sum_{s=1}^{2} W_{ks}^{(1)} \left[\boldsymbol{X}_i^{(j)}\right]_s + b_k^{(1)}\right) \cdot W_{k2}^{(1)} = W^{(N)} \cdot \left(\left[\boldsymbol{\alpha}_i^{(j)}\right] \odot \left[W^{(1)}\right]^{(2)}\right)$$

where $\left[W^{(1)}\right]^{(s)}$ denotes the s-th column of the Matrix $W^{(1)}$.

[14] For a fixed state $\left[\boldsymbol{X}_i^{(j)}\right] = \left[q_i^{(j)} \quad \dot{q}_i^{(j)}\right]^T$ the second order partial derivatives are given by

$$C_i^{(j)} = \frac{\partial^2 \boldsymbol{L}(\boldsymbol{X}_i^{(j)})}{\partial \dot{q} \partial q} = \sum_{k=1}^{h} W_{1k}^{(N)} \cdot \left[\boldsymbol{\beta}_i^{(j)}\right]_k \cdot W_{k1}^{(1)} \cdot W_{k2}^{(1)} = W^{(N)} \cdot \left(\boldsymbol{\beta}_i^{(j)} \odot \left[W^{(1)}\right]^{(1)} \odot \left[W^{(1)}\right]^{(2)}\right) \in \mathbb{R}$$
$$H_i^{(j)} = \frac{\partial^2 \boldsymbol{L}(\boldsymbol{X}_i^{(j)})}{\partial \dot{q}^2} = \sum_{k=1}^{h} W_{1k}^{(N)} \cdot \left[\boldsymbol{\beta}_i^{(j)}\right]_k \cdot \left[W_{k2}^{(1)}\right]^2 = W^{(N)} \cdot \left(\boldsymbol{\beta}_i^{(j)} \odot \left[W^{(1)}\right]^{(2)} \odot \left[W^{(1)}\right]^{(2)}\right) \in \mathbb{R}$$

$$\boldsymbol{C}_i = \frac{\partial^2 \boldsymbol{L}(\boldsymbol{X}_i)}{\partial \dot{q} \partial q} = W^{(N)} \cdot \left[\boldsymbol{\beta}_i \odot \left(\left[W^{(1)} \right]^{(1)} \cdot \mathbf{1}_M^T \right) \odot \left(\left[W^{(1)} \right]^{(2)} \cdot \mathbf{1}_M^T \right) \right] \in \mathbb{R}^{1 \times M}$$
$$\boldsymbol{H}_i = \frac{\partial^2 \boldsymbol{L}(\boldsymbol{X}_i)}{\partial \dot{q}^2} = W^{(N)} \cdot \left[\boldsymbol{\beta}_i \odot \left(\left[W^{(1)} \right]^{(2)} \cdot \mathbf{1}_M^T \right) \odot \left(\left[W^{(1)} \right]^{(2)} \cdot \mathbf{1}_M^T \right) \right] \in \mathbb{R}^{1 \times M}$$

Für die Ableitungen von $\boldsymbol{H}_i$ nach den Parametern gilt:

$$\frac{\partial \boldsymbol{H}_i}{\partial \left[W^{(1)} \right]^{(1)}} = \left(\mathbf{1}_M \cdot W^{(N)} \right)^T \odot \left[\boldsymbol{\gamma}_i \odot \left(\mathbf{1}_h \cdot \boldsymbol{Q}_i \right) \odot \left(\left[W^{(1)} \right]^{(2)} \cdot \mathbf{1}_M^T \right) \odot \left(\left[W^{(1)} \right]^{(2)} \cdot \mathbf{1}_M^T \right) \right] \in \mathbb{R}^{h \times M}$$
$$\frac{\partial \boldsymbol{H}_i}{\partial \left[W^{(1)} \right]^{(2)}} = \left(\mathbf{1}_M \cdot W^{(N)} \right)^T \odot \left(\left[W^{(1)} \right]^{(2)} \cdot \mathbf{1}_M^T \right) \odot \left[\left(\left[W^{(1)} \right]^{(2)} \cdot \mathbf{1}_M^T \right) \odot \boldsymbol{\gamma}_i \odot \left(\mathbf{1}_h \cdot \dot{\boldsymbol{Q}}_i \right) + 2\boldsymbol{\beta}_i \right]$$
$$\frac{\partial \boldsymbol{H}_i}{\partial b^{(1)}} = \left(\mathbf{1}_M \cdot W^{(N)} \right)^T \odot \boldsymbol{\gamma}_i \odot \left(\left[W^{(1)} \right]^{(2)} \cdot \mathbf{1}_M^T \right) \odot \left(\left[W^{(1)} \right]^{(2)} \cdot \mathbf{1}_M^T \right)$$
$$\frac{\partial}{\partial W^{(N)}} = \boldsymbol{\beta}_i \odot \left(\left[W^{(1)} \right]^{(2)} \cdot \mathbf{1}_M^T \right) \odot \left(\left[W^{(1)} \right]^{(2)} \cdot \mathbf{1}_M^T \right)$$

Für die Ableitungen von $\boldsymbol{C}_i$ nach den Parametern gilt:

$$\frac{\partial \boldsymbol{C}_i}{\partial \left[W^{(1)} \right]^{(1)}} = \left(\mathbf{1}_M \cdot W^{(N)} \right)^T \odot \left(\left[W^{(1)} \right]^{(2)} \cdot \mathbf{1}_M^T \right) \odot \left[\boldsymbol{\gamma}_i \odot \left(\mathbf{1}_h \cdot \boldsymbol{Q}_i \right) \odot \left(\left[W^{(1)} \right]^{(1)} \cdot \mathbf{1}_M^T \right) + \boldsymbol{\beta}_i \right]$$
$$\frac{\partial \boldsymbol{C}_i}{\partial \left[W^{(1)} \right]^{(2)}} = \left(\mathbf{1}_M \cdot W^{(N)} \right)^T \odot \left(\left[W^{(1)} \right]^{(1)} \cdot \mathbf{1}_M^T \right) \odot \left[\boldsymbol{\gamma}_i \odot \left(\mathbf{1}_h \cdot \dot{\boldsymbol{Q}}_i \right) \odot \left(\left[W^{(1)} \right]^{(2)} \cdot \mathbf{1}_M^T \right) + \boldsymbol{\beta}_i \right]$$
$$\frac{\partial \boldsymbol{C}_i}{\partial b^{(1)}} = \left(\mathbf{1}_M \cdot W^{(N)} \right)^T \odot \left(\left[W^{(1)} \right]^{(1)} \cdot \mathbf{1}_M^T \right) \odot \left(\left[W^{(1)} \right]^{(2)} \cdot \mathbf{1}_M^T \right) \odot \boldsymbol{\gamma}_i$$
$$\frac{\partial \boldsymbol{C}_i}{\partial W^{(N)}} = \boldsymbol{\beta}_i \odot \left(\left[W^{(1)} \right]^{(1)} \cdot \mathbf{1}_M^T \right) \odot \left(\left[W^{(1)} \right]^{(2)} \cdot \mathbf{1}_M^T \right)$$

Für $\boldsymbol{N}_i := \frac{\partial \boldsymbol{L}_i}{\partial q} - \boldsymbol{C}_i \cdot \dot{q} + u$ erhalten wir: $\frac{\partial \boldsymbol{N}_i}{\partial \boldsymbol{\theta}} = \frac{\partial}{\partial \boldsymbol{\theta}} \frac{\partial \boldsymbol{L}_i}{\partial q} - \frac{\partial \boldsymbol{C}_i}{\partial \boldsymbol{\theta}} \dot{q}$

$$\begin{aligned}
\frac{\partial \boldsymbol{N}_i}{\partial \left[W^{(1)} \right]^{(1)}} &= \left[\left(W^{(N)} \right)^T \cdot \mathbf{1}_M^T \right] \odot \left(\boldsymbol{\beta}_i \odot \left[\mathbf{1}_h \cdot \boldsymbol{Q}_i \right] \odot \left[\left(W^{(1)} \right)^{(1)} \cdot \mathbf{1}_M^T \right] + \boldsymbol{\alpha}_i \right) \\
&\quad - \left[\mathbf{1}_h \cdot \dot{\boldsymbol{Q}}_i \right] \odot \left(\mathbf{1}_M \cdot W^{(N)} \right)^T \odot \left(\left[W^{(1)} \right]^{(2)} \cdot \mathbf{1}_M^T \right) \odot \left[\boldsymbol{\gamma}_i \odot \left(\mathbf{1}_h \cdot \boldsymbol{Q}_i \right) \odot \left(\left[W^{(1)} \right]^{(1)} \cdot \mathbf{1}_M^T \right) + \boldsymbol{\beta}_i \right] \\
\frac{\partial \boldsymbol{N}_i}{\partial \left[W^{(1)} \right]^{(2)}} &= \left[\left(W^{(N)} \right)^T \cdot \mathbf{1}_M^T \right] \odot \boldsymbol{\beta}_i \odot \left[\mathbf{1}_h \cdot \dot{\boldsymbol{Q}}_i \right] \odot \left[\left(W^{(1)} \right)^{(1)} \cdot \mathbf{1}_M^T \right] \\
&\quad - \left[\mathbf{1}_h \cdot \dot{\boldsymbol{Q}}_i \right] \odot \left(\mathbf{1}_M \cdot W^{(N)} \right)^T \odot \left(\left[W^{(1)} \right]^{(1)} \cdot \mathbf{1}_M^T \right) \odot \left[\boldsymbol{\gamma}_i \odot \left(\mathbf{1}_h \cdot \dot{\boldsymbol{Q}}_i \right) \odot \left(\left[W^{(1)} \right]^{(2)} \cdot \mathbf{1}_M^T \right) + \boldsymbol{\beta}_i \right] \\
\frac{\partial \boldsymbol{N}_i}{\partial b_j^{(1)}} &= \left[\left(W^{(N)} \right)^T \cdot \mathbf{1}_M^T \right] \odot \boldsymbol{\beta}_i \odot \left[\left(W^{(1)} \right)^{(1)} \cdot \mathbf{1}_M^T \right] \\
&\quad - \left[\mathbf{1}_h \cdot \dot{\boldsymbol{Q}}_i \right] \odot \left(\mathbf{1}_M \cdot W^{(N)} \right)^T \odot \left(\left[W^{(1)} \right]^{(1)} \cdot \mathbf{1}_M^T \right) \odot \left(\left[W^{(1)} \right]^{(2)} \cdot \mathbf{1}_M^T \right) \odot \boldsymbol{\gamma}_i \\
\frac{\partial \boldsymbol{N}_i}{\partial W_{1j}^{(N)}} &= \boldsymbol{\alpha}_i \odot \left[\left(W^{(1)} \right)^{(1)} \cdot \mathbf{1}_M^T \right] \\
&\quad - \left[\mathbf{1}_h \cdot \dot{\boldsymbol{Q}}_i \right] \odot \boldsymbol{\beta}_i \odot \left(\left[W^{(1)} \right]^{(1)} \cdot \mathbf{1}_M^T \right) \odot \left(\left[W^{(1)} \right]^{(2)} \cdot \mathbf{1}_M^T \right)
\end{aligned}$$

Wir erhalten für den Gradienten:

$$\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) = \frac{1}{N \cdot M} \sum_{j=0}^{M-1} \sum_{i=0}^{N-1} \boldsymbol{p}_i^{(j)} \cdot \left[-\frac{\partial H_{\boldsymbol{\theta}}}{\partial \boldsymbol{\theta}} \ddot{\boldsymbol{q}}_{\boldsymbol{\theta}} + \frac{\partial N_{\boldsymbol{\theta}}}{\partial \boldsymbol{\theta}} \right] \quad \boldsymbol{p}_i^{(j)} := 2 H_{\boldsymbol{\theta}}^{-1} \left[\ddot{q} \left(q_i^{(j)}, \dot{q}_i^{(j)}, u_i^{(j)} \right) - \ddot{q}_i^{(j)} \right]$$